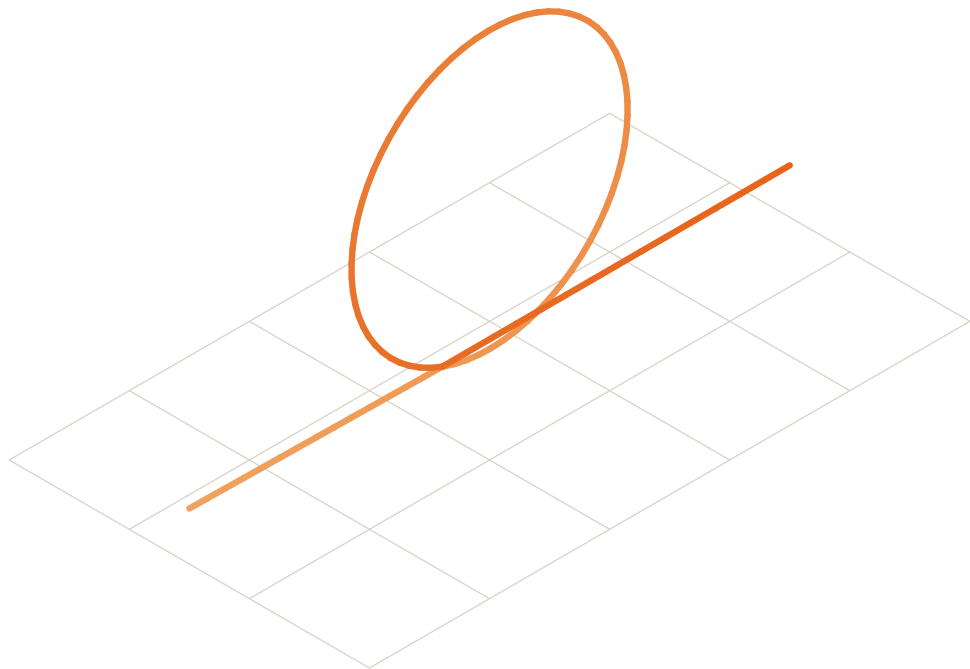




THE TRICKTIONARY

A visual field guide to FPV freestyle tricks

Flight paths · control inputs · how-to



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How to use this book

Every trick gets one page. Here's how to read it, the anatomy is the same for all of them.

- **The header**

Each trick opens with its name, a difficulty chip, and its motion-family category. Where known, the originator is credited beside them.

- **The illustration**

A 3D isometric view of the flight path. You're looking down at the scene: the ground grid sits flat, obstacles are solid, and the orange path runs light-to-dark from entry to exit with an arrow at the finish. Numbered markers tie the path to the steps.

- **How to**

A plain-language walkthrough of the maneuver, drawn from source tutorials and credited beneath the text.

- **Mastery levels**

Three star levels for each trick: one star for landing it at all, two for clean form, three for perfect form. Use them to gauge how solid a trick really is, not just whether you have done it once.

- **Control inputs**

Each step shows a twin-stick diagram (see Controls & Conventions) so you can read the throttle, yaw, pitch, and roll for that moment alongside the description.

- **The notes**

Prerequisites & where to practice, common mistakes, variations, and related tricks, everything you need to decide if you're ready and what to try next.

- **Sources**

Every entry links its sources at the foot of the page. Nothing here is invented; it's compiled and credited.

Controls & conventions

READING THE STICKS (MODE 2)

Control inputs are shown as a pair of sticks. Throughout this book the stick diagrams carry no text labels, so learn the four axes here once. An arrow shows a stick moving; a filled dot shows an input held in place; an empty center means the stick is centered.

LEFT STICK

Up / down Throttle. Up adds power and lift; down cuts it.

Left / right Yaw. Rotates the drone left or right in place.

RIGHT STICK

Up / down Pitch. Up drops the nose to fly forward; down raises it.

Left / right Roll. Tips the drone onto its left or right side.

EXAMPLES



Throttle up, held



Full pitch down



Roll left



Yaw right, pitch up

READING THE ILLUSTRATION

Viewpoint

You look down at the scene in isometric 3D. The ground grid is flat; altitude is vertical. Dotted drop-lines connect points on the path to the ground.

The path

The flight path runs orange, light at entry and saturated at the exit, ending in an arrowhead. A dashed line on the ground is its shadow.

Obstacles

Walls, bars, and buildings are drawn as solid objects and correctly hide the parts of the path that pass behind them.

Markers & rotation

Numbered markers match the how-to steps. Orange curved arrows indicate rotation (roll, yaw, or pitch).

The difficulty scale

Beginner	First tricks, clean flips, rolls, orbits. Learn these before anything else.	25 tricks
Novice	Building blocks combined: powerloops, matty flips, split-S. Comfortable acro required.	29 tricks
Intermediate	Multi-input tricks and obstacle work: rewinds, taps, trippy spins.	44 tricks
Advanced	Fast, committed, low-margin lines and chained rotations.	34 tricks
Master	The hardest named tricks, precise, high-consequence, rarely landed clean.	11 tricks

143 tricks in all.

Safety & where to fly

- **Know the rules**

In the U.S., recreational flight follows the FAA's rules for recreational flyers; flying for any non-recreational purpose requires a Part 107 certificate. Pass the free TRUST test, register your drone if it's over 250 g, and check for a Remote ID requirement.

- **Altitude & airspace**

Stay at or below 400 ft AGL and clear of controlled airspace unless you have authorization (LAANC). Use a current airspace map before you fly somewhere new.

- **Line of sight & spotter**

Keep the aircraft in visual line of sight, with a visual observer if you're in goggles. Many parks and events require this.

- **Where to fly**

Pick open spaces away from people, roads, and airports: fields, sanctioned FPV spots, and clubs (AMA fields). Some state parks require a permit or ban drones outright, check before you go, and get landowner permission on private property.

- **Fly within your skill**

Give yourself altitude to recover, drop your rates when learning, and practice new tricks in a simulator first. Every sim crash is a free lesson.

Rules change and vary by location. This page is a starting orientation, not legal advice, always confirm current regulations for where you fly.

Glossary

Acro / Rate mode

Manual flight mode with no self-leveling; the sticks command rotation rate. Required for nearly every freestyle trick.

Arm

Enabling the motors so the quad will spin up. You arm before takeoff (usually on a switch) and disarm to cut power.

AUX / switch

An auxiliary channel on the radio, used to arm, change modes, or trigger features mid-flight.

Bando

An abandoned building or structure that pilots treat as a freestyle playground. Short for 'abandoned'.

Betaflight

The most common open-source flight-control firmware, where rates, flight modes, and tuning are set up.

Blip

A brief tap of throttle, used to nudge altitude or kick a stalled quad back into motion.

BNF / RTF

Bind-N-Fly (needs your own radio) / Ready-To-Fly (includes everything).

CLM

Camera-Locked Motion, keeping a subject fixed in frame throughout a trick, a scoring requirement in competition.

Cinewhoop

A small ducted quad built for smooth cinematic flying in tight spaces.

Disarm

Cutting motor power. A 'slide disarm' or 'crash and disarm' ends a line deliberately.

Failsafe

The programmed response when the quad loses its radio or video link, usually cutting the motors or holding position to prevent a flyaway.

FPV

First-Person View, flying via a live camera feed to goggles or a screen. Is not a crime.

Freestyle

Expressive, trick-oriented flying (as opposed to racing).

Horizon mode

A self-leveling mode that still lets you flip and roll, then eases back toward level when you release the sticks. Angle mode is fully self-leveling; Acro has no leveling at all.

Invert

Flipping the quad upside-down; many tricks pass through or hold an inverted state.

LiPo

Lithium-polymer battery, the high-discharge packs that power most FPV quads. Store and charge them carefully.

LOS

Line-Of-Sight, keeping visual contact with the aircraft, often a legal requirement.

Powerloop

A loop up-and-over an obstacle using throttle through the top; also a whole trick family.

Prop wash

Turbulent air thrown off the quad's own props, felt as a jittery wobble when you descend through it or stop hard.

Punch / punch-out

A sharp burst of throttle for a fast vertical climb.

Rates

Radio settings that map stick travel to rotation speed. Lower rates = gentler, higher = snappier.

Rewind

Reversing a motion back along its entry path, often out of a stall.

Stall

Chopping throttle so the quad momentarily stops flying and hangs or drops before recovery.

Throttle

Motor power, the left stick's vertical axis in Mode 2.

VTX

Video transmitter, the part that beams the camera feed from the quad to your goggles.

Whoop

A micro ducted FPV drone (typically 65 to 85mm); safe, agile, great for learning.

Yaw

Rotation about the vertical axis, the drone turning left/right in place.

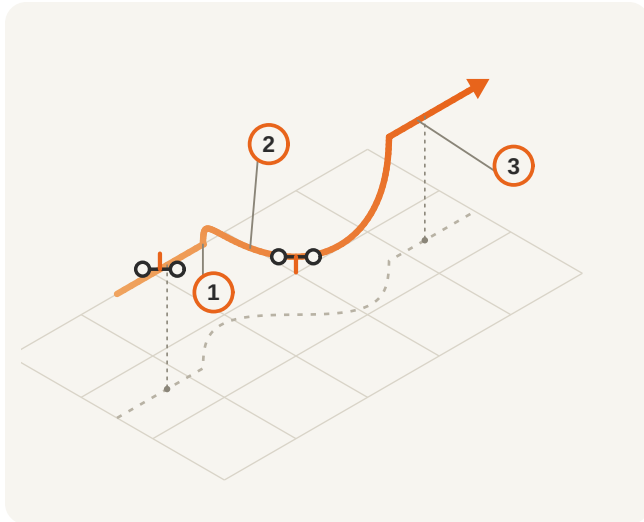
5-inch

A common freestyle quad class with 5-inch props; more power and range than a whoop.

Barrel Roll

Beginner

Open Air



HOW TO

A roll flown as a corkscrew rather than on the axis, the quad spiraling around its line of travel. Settle onto a line, corkscrew through the barrel, and catch level on the same track. The wider, looping path is what separates it from a tight axial roll.

MASTERY LEVELS

- ★☆☆ Corkscrew through a barrel roll and recover.
- ★★★ Smooth, round barrel roll holding the line.
- ★★★★ Clean barrel roll at any size, both directions.

STEP BY STEP CONTROL INPUTS



Settle onto a line



Corkscrew through a barrel roll



Catch level on the same track

PREREQUISITES & WHERE TO PRACTICE

- A clean axial roll
- Coordinated roll and pitch

Open air. Fly the roll as a wide corkscrew, coordinating a little pitch with the roll so the path loops out and back, and catch level on your track. Both directions.

COMMON MISTAKES

- ✗ Rolling on the axis instead of corkscrewing
- ✗ An uneven, wobbly barrel
- ✗ Drifting off the line

VARIATIONS

Roll

A tight axial roll.

Double Roll

Two rolls in a row.

RELATED TRICKS

Roll

The axial version.

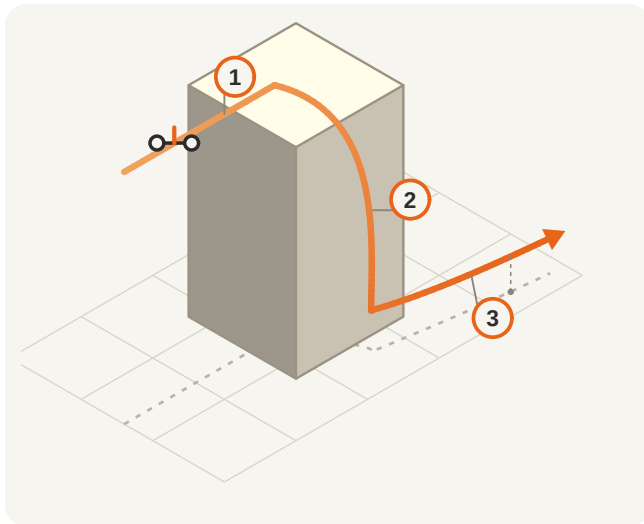
Power Drill

A vertical corkscrew.

Building Dive

Beginner

Wall Tricks



HOW TO

A committed descent down the face of a tall structure.

Cross the roofline with your speed under control, then drop the nose and track straight down the face, staying close but never clipping it. As the ground rushes up, flare the nose back to level and feed in throttle so you pull out smoothly along the base.

MASTERY LEVELS

- ★☆☆ Dive the face and pull out level before the ground.
- ★★★ Tight, controlled descent close to the face with a smooth flare.
- ★★★★ Fast committed dive hugging the building, flared level at the base.

STEP BY STEP CONTROL INPUTS



1 Cross the roofline with your speed in check



2 Commit the nose down and track the face



3 Flare the nose up and feed throttle to level out

PREREQUISITES & WHERE TO PRACTICE

- Confident fast descents with a clean pull-out
- Good speed judgement near a surface

Start on a modest drop with plenty of run-out at the bottom before trying anything tall. Scout the base for obstacles, you need clean air to flare into. Keep some throttle in reserve for the pull-out.

COMMON MISTAKES

- ✗ Diving too fast to arrest before the ground
- ✗ Drifting into the face on the way down
- ✗ Flaring too late, so you pancake at the base

VARIATIONS

Roof gap dive

Drop through a gap in the roofline before the face.

Spiral dive

Add rotation on the way down.

RELATED TRICKS

Wall Ride

Tracking a vertical surface, but alongside it.

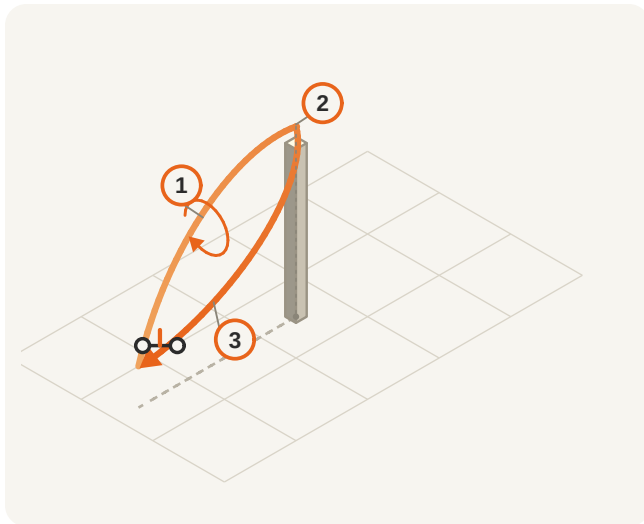
Split S

Another way to trade altitude fast.

Cinnamon Roll

Beginner

Jump Ropes



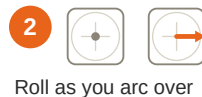
HOW TO

A jump rope with a roll baked in. Swing up toward the top of an object as you would a jump rope, but roll the quad as you arc over, then drop back down the same side. The roll adds a twist of style to the up-and-over rhythm.

MASTERY LEVELS

- ★☆☆ Add a roll to the jump-rope swing.
- ★★★☆☆ Smooth roll folded into an even swing.
- ★★★★★ Rhythmic swings with a clean roll, over and back.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- A solid jump rope
- A controllable roll

Get the jump-rope rhythm down first, then add the roll as you go over the top. Keep it smooth and even so the roll flows with the swing.

COMMON MISTAKES

- ✗ Rolling too fast, breaking the rhythm
- ✗ A lopsided swing
- ✗ Losing the object as you roll

VARIATIONS

Jump Rope

Without the roll.

Burrito Roll

Roll the other way and go under.

RELATED TRICKS

Jump Rope

The base swing.

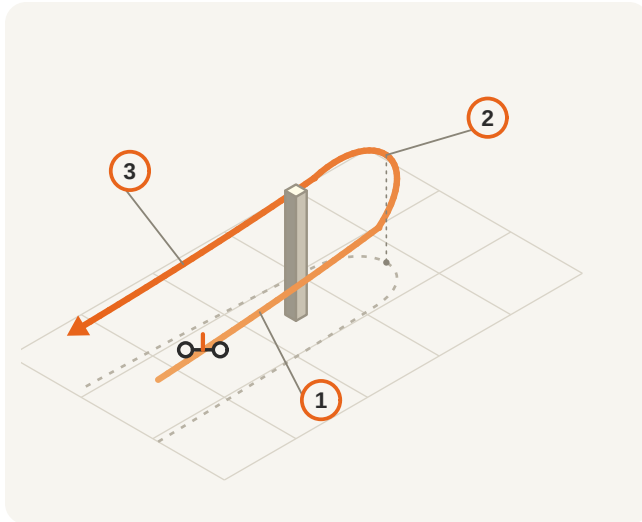
Burrito Roll

A jump-rope roll that finishes under.

Cradle

Beginner

Orbitals



HOW TO

A tracking pass that ends flying backward. Fly past an object, then pitch backward to track it and keep it in view. As you pick up backward momentum, level out by pitching forward again and continue flying backward until you have completely cleared the object. It is mostly pitch, leaning the quad back then forward, with just enough roll or yaw to shift smoothly around the object, and you keep it in view the whole time.

MASTERY LEVELS

- ★☆☆ Fly past the object and back up, keeping it in view.
- ★★★ Smooth pitch-back and level backward pass, object framed.
- ★★★★ Seamless forward-to-backward pass, object locked in view.

STEP BY STEP CONTROL INPUTS



1 Fly forward past the object



2 Pitch back to track it and pick up backward momentum



3 Pitch forward to level, drift backward until cleared

PREREQUISITES & WHERE TO PRACTICE

- Confident backward flight
- Tracking an object while maneuvering

Pick an isolated post or object. Fly past it, then pitch back to slow and reverse while keeping it framed, and drift backward until you have cleared it. It is all in smooth pitch control, with only a little roll or yaw to shift around the object.

COMMON MISTAKES

- ✗ Losing the object out of frame
- ✗ Pitching back too hard and climbing away
- ✗ Too much roll or yaw, swinging off the track

VARIATIONS

Inverted Cradle

Carry the pitch-back all the way to inverted.

Cradle Rewind

Pitch forward again and rewind past the object.

RELATED TRICKS

Orbits

Another object-tracking maneuver.

Side-Lock Rewind

A tracking rewind past an object.

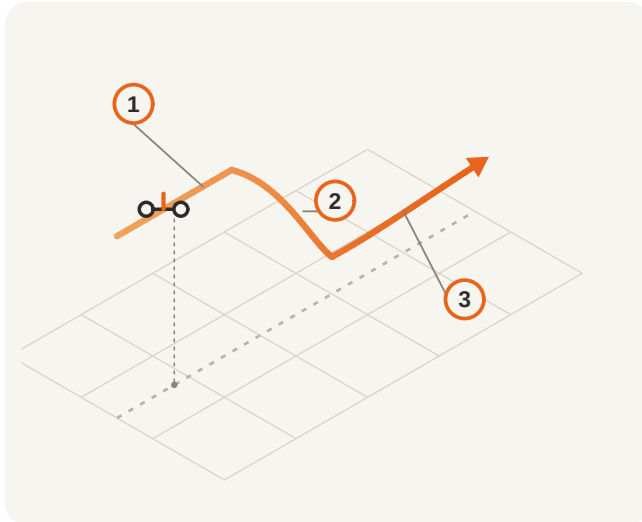
Trippy Spin

Builds on the cradle, taken inverted.

Dive

Beginner

Open Air



HOW TO

The basic descent: drop the nose and dive, then pull out. From level flight cross into the dive with your speed in check, commit the nose down to track the line down, and flare to level as the ground comes up. It is the foundation for building dives and every steep descent.

MASTERY LEVELS

- ★☆☆ Dive the line and pull out before the ground.
- ★★★ Controlled dive with a smooth level pull-out.
- ★★★★ Fast, committed dive flared clean at the bottom.

STEP BY STEP CONTROL INPUTS

1  
Cross with speed in check

2  
Drop the nose and dive

3  
Flare and pull out low

PREREQUISITES & WHERE TO PRACTICE

- Confident forward flight
- A feel for pulling out of a descent

Start with shallow dives and plenty of run-out, steepening as your pull-out gets clean. Keep some throttle in reserve to flare. Watch your altitude.

COMMON MISTAKES

- ✗ Diving too steep to arrest in time
- ✗ Pulling out too late
- ✗ Ballooning instead of tracking the line down

VARIATIONS

Building Dive

A dive down a structure's face.

Backwards Dive

The same dive flown tail-first.

RELATED TRICKS

Building Dive

A committed dive down a face.

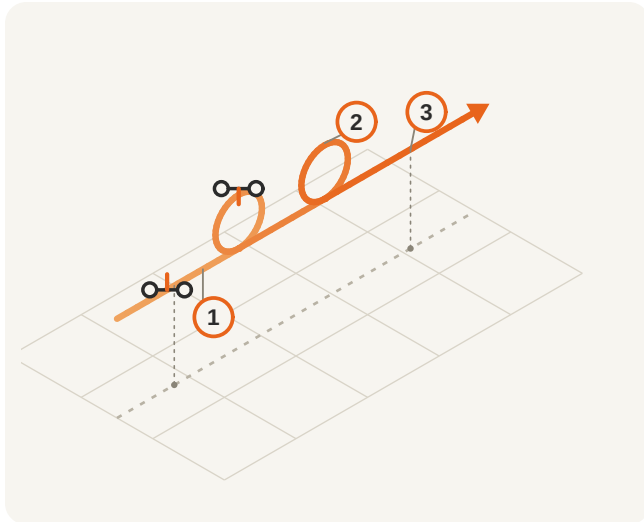
Split S

Another way to trade altitude.

Double Flip

Beginner

Open Air



HOW TO

Two flips in a row. From level flight punch full forward pitch and let the quad rotate through two complete loops on the pitch axis in one motion, keeping the throttle low so you hold your height, then catch level and carry on. Pace it so you do not over-rotate the second flip.

MASTERY LEVELS

- ★☆☆ Two flips and recover to level.
- ★★★☆☆ Even flips with no altitude gain, clean catch.
- ★★★★ Two tight, on-axis flips holding the line, both directions.

STEP BY STEP CONTROL INPUTS



Cruise level with room above



Two flips back to back



Catch level and continue

PREREQUISITES & WHERE TO PRACTICE

- A clean single flip
- Enough altitude for two rotations

High over open grass. Link the two flips without a pause and keep throttle low so you do not climb. Catch level the moment the second flip comes around.

COMMON MISTAKES

- ✗ Ballooning up through the flips
- ✗ Pausing between rotations
- ✗ Catching late on the second flip

VARIATIONS

Triple Flip

Three loops linked in one motion.

Double Roll

The doubled idea on the roll axis.

RELATED TRICKS

Flip

The single version.

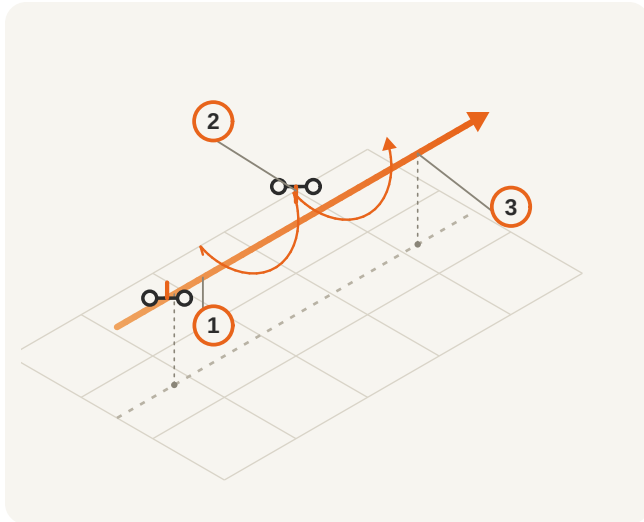
Double Roll

Two rolls instead of flips.

Double Roll

Beginner

Open Air



HOW TO

Two back-to-back rolls. Hold a straight, level line and roll the quad through two full rotations in one smooth, continuous motion, keeping the throttle steady so the line stays flat, then catch level on the same track. It is the single roll doubled, so pace the rate that the second rotation is as clean as the first.

MASTERY LEVELS

- ★☆☆ Two full rolls and recover on the same line.
- ★★★☆☆ Smooth, even rolls with no heading drift.
- ★★★★★ Two crisp rolls at any rate, both directions, dead on line.

STEP BY STEP CONTROL INPUTS

1 Settle onto a straight, level line

2 Two full rolls in a row

3 Catch level on the same track

PREREQUISITES & WHERE TO PRACTICE

- A clean single roll
- Steady throttle through rotation

Plenty of height. Link two rolls in one motion rather than pausing between them, and keep the rate even so both come out clean.

COMMON MISTAKES

- ✗ Pausing between the two rolls
- ✗ Throttle sag on the second rotation
- ✗ Drifting off the line

VARIATIONS

Juicy Double Roll

The double roll flown out of a juicy flick.

Triple Roll

Three rotations linked in one pass.

Vanny Roll

A roll carried through a turn around an obstacle.

RELATED TRICKS

Roll

The single-rotation version.

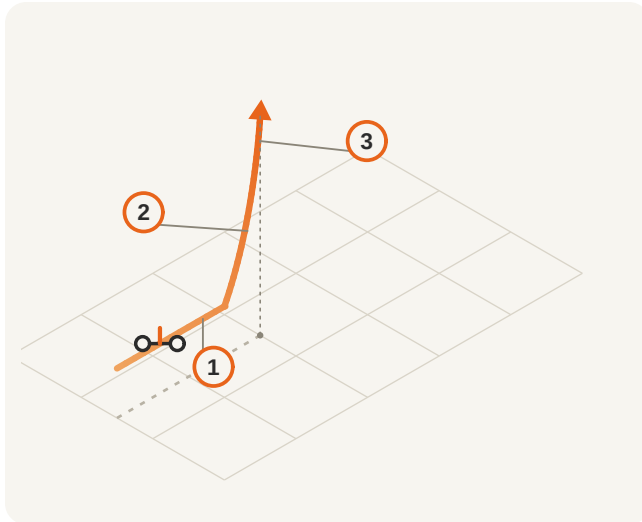
Double Flip

The same idea on the pitch axis.

Face Punch

Beginner

Miscellaneous



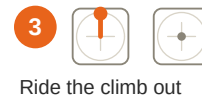
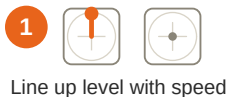
HOW TO

A hard vertical punch-out. From a level line, slam full throttle to rocket straight up, riding the acceleration as high as you dare before you level off. It is the simplest show of power in the book, all about a clean, committed, dead-vertical line.

MASTERY LEVELS

- ★☆☆ Punch out into a fast vertical climb.
- ★★★☆☆ Clean, straight punch with a controlled top.
- ★★★★ Explosive, dead-vertical punch, held straight all the way up.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- Confident, aggressive throttle
- A feel for when to level off

Open air with plenty of ceiling. Punch straight up and hold the vertical, then level off with room to spare. Watch your battery, a hard punch draws a lot of current.

COMMON MISTAKES

- ✗ Drifting off vertical under power
- ✗ Punching so hard you lose orientation
- ✗ Leaving no room to level off

VARIATIONS

Power Flip

Add a flip on the way up.

Power Roll

Add a roll on the way up.

RELATED TRICKS

Power Flip

A punch with a flip in it.

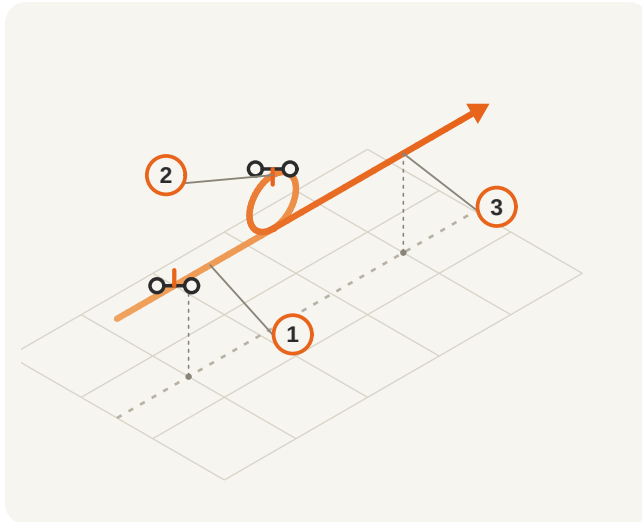
Powerloop

Another power-throttle move.

Flip

Beginner

Open Air



HOW TO

The building-block rotation. From level flight, feed in a touch of throttle for a cushion of altitude, then punch full forward pitch and let the quad whip all the way around the pitch axis. Keep the throttle low through the rotation so you climb as little as possible, and catch the stick back to level the instant the horizon comes back around. Feed throttle in again on the exit and carry the same line.

MASTERY LEVELS

- ★☆☆ Complete the rotation and recover to level without hitting the ground.
- ★★★ Consistent height and a clean level catch, both directions.
- ★★★★ Tight, on-axis flip that holds its line with no altitude gain.

STEP BY STEP CONTROL INPUTS



1 Cruise level, blip a little altitude



2 Punch full forward pitch, let it rotate a full 360



3 Catch level as the horizon returns, feed throttle back

PREREQUISITES & WHERE TO PRACTICE

- Stable hover and level cruise in acro (rate) mode
- Enough altitude that a late catch still recovers

Do them high over open grass first. A flip only needs airspace, no obstacle, which is why it is usually the very first trick. Drop the entry height as your catch gets consistent.

COMMON MISTAKES

- ✗ Not enough altitude, so a slightly late catch ends in the ground
- ✗ Carrying too much throttle and ballooning up through the rotation
- ✗ Catching late and over-rotating past level

VARIATIONS

Double Flip

Two full pitch rotations back to back before catching level.

Flip Rewind

Flip, then reverse back along the entry line out of a stall.

RELATED TRICKS

Roll

The same idea on the roll axis instead of pitch.

Powerloop

A flip translated up and over an obstacle with throttle.

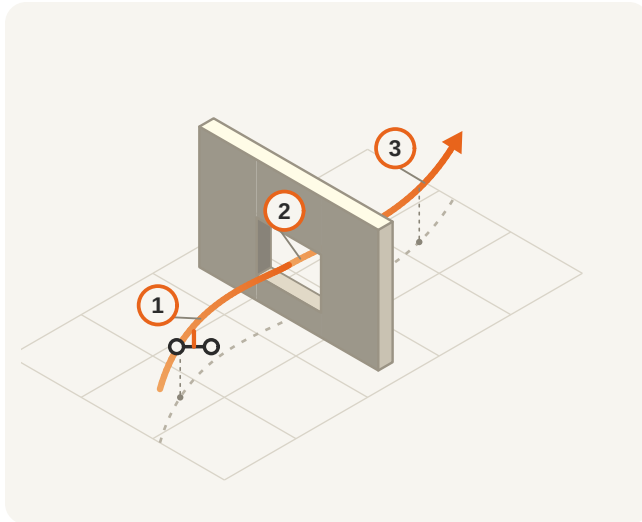
Matty Flip

A partial flip over a gap while tracking a subject.

Gaps

Beginner

Gaps



HOW TO

Threading a gap is about commitment and a clean line, not a rotation. Pick the opening early and line the quad up pointed through it well before you arrive, then hold that line all the way through, do not grab for corrections in the gap itself. Look through the opening to where you want to exit, not at the frame of the gap. Speed is your friend once the line is set.

MASTERY LEVELS

- ★☆☆ Clear the gap cleanly on a committed line.
- ★★★☆☆ Smooth line with no last-second corrections, repeatable.
- ★★★★ Fast, tight gaps taken on any chosen line without hesitation.

STEP BY STEP CONTROL INPUTS



Spot the opening and line up early



Commit and thread the gap on a fixed line



Hold the line out the far side

PREREQUISITES & WHERE TO PRACTICE

- Confident, stable forward flight
- Good depth perception through the goggles

Start with wide, forgiving gaps, a soccer goal, a wide doorway, the space between two trees, and shrink them as your line gets repeatable. Always scout the gap and its approach before committing.

COMMON MISTAKES

- ✗ Staring at the frame of the gap instead of through it
- ✗ Making late stick corrections inside the opening
- ✗ Approaching at an angle instead of squared up

VARIATIONS

Low gap

An opening threaded fast and close to the ground.

Angled gap

A gap taken on a banked, non-level line.

RELATED TRICKS

Wall Ride

Proximity flying along a surface rather than through it.

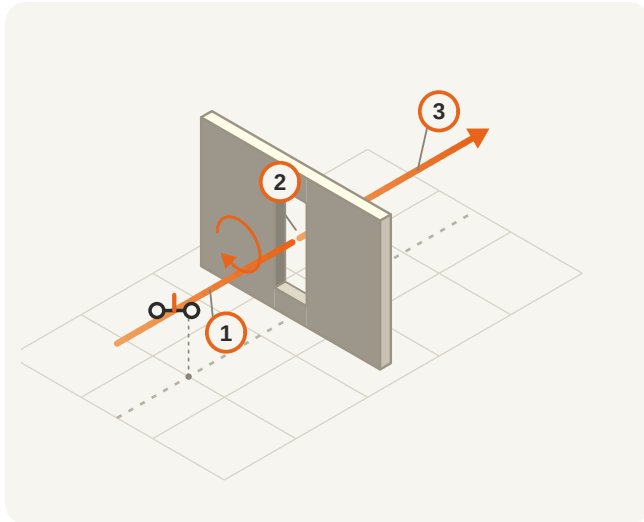
Powerloop

Often set up by threading a gap first.

Knife Edge

Beginner

Gaps



HOW TO

Threading a slot too narrow to fly through flat. Line up the opening early, then roll the quad up onto its side, knife-edge, so its narrow profile fits the gap, hold that bank through the opening, and roll level again on the far side. Commit to the line before the gap and look through it, not at it.

MASTERY LEVELS

- ★☆☆ Fit through the narrow gap on the roll axis.
- ★★★☆☆ Clean knife-edge with a level exit.
- ★★★★★ Tight gaps taken knife-edge on a committed line.

STEP BY STEP CONTROL INPUTS



1 Pick a comfortable gap



2 Roll 90° onto the drone's side



3 Fly through sideways, watching exit momentum

PREREQUISITES & WHERE TO PRACTICE

- Confident, stable forward flight
- A controllable 90 degree roll and hold

Start with taller, wider slots, a doorway or a gap between close trees, and narrow them as your knife-edge holds steady. Set the bank before the gap, not in it.

COMMON MISTAKES

- ✗ Rolling in the gap instead of before it
- ✗ Not enough bank to clear the sides
- ✗ Grabbing corrections mid-gap

VARIATIONS

Reverse Knife Edge

Knife-edge banked the other way.

Ninja Star

A knife-edge gap taken as part of a combo.

RELATED TRICKS

Gaps

Threading an opening on a flat line.

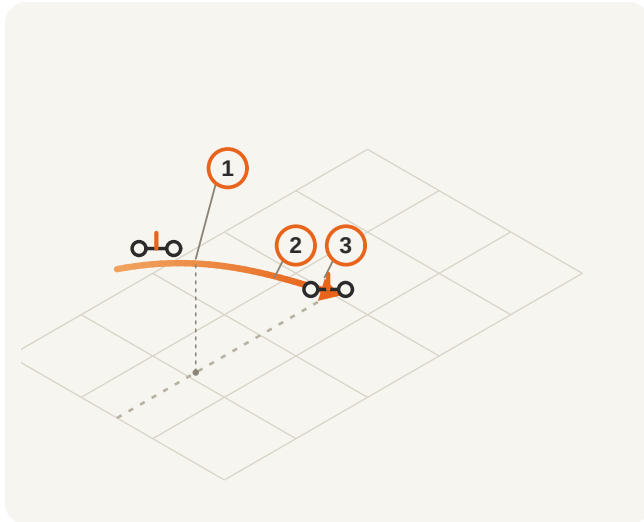
Roll

The roll that sets the knife-edge.

Land

Beginner

Miscellaneous



HOW TO

The clean way to end a flight. Line up your descent, ease the throttle down to sink toward the ground, and feather it to a soft touch. A controlled landing protects your quad and looks a lot better than a fall out of the sky.

MASTERY LEVELS

- ★☆☆ Bring it down and land without a bounce.
- ★★★ Smooth, controlled descent to a soft touch.
- ★★★★ Precise landing on a chosen spot, feather-soft.

STEP BY STEP CONTROL INPUTS



Line up your descent



Ease down toward the ground



Touch down softly

PREREQUISITES & WHERE TO PRACTICE

- Smooth throttle control near the ground
- Patience on the descent

Practise slow, controlled descents to a soft touch on a flat, clear spot. The last few inches are all throttle finesse, so take them slow.

COMMON MISTAKES

- ✗ Descending too fast and thumping down
- ✗ Cutting throttle too high and dropping
- ✗ Landing on uneven ground or debris

VARIATIONS

Slide Disarm

Slide to a stop instead.

Perch

Land on top of an object.

RELATED TRICKS

Slide Disarm

A sliding finish.

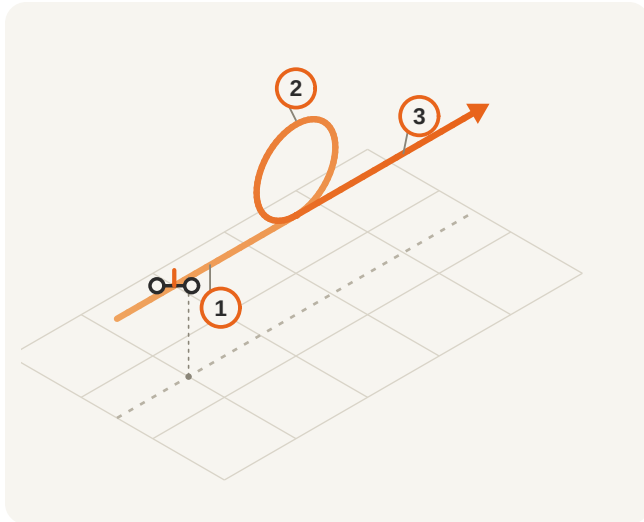
Perch

Landing on top of an object.

Loop

Beginner

Open Air



HOW TO

The loop, or flip: a full 360 rotation on the pitch axis, the drone tumbling nose-over-tail through a vertical circle. Carry a level line, pull a full loop, and catch level on the same line. It is one of the first big tricks every pilot learns, and the root of half the tricks in this book.

MASTERY LEVELS

- ★☆☆ Carry a full loop and recover.
- ★★★ Round, even loop that holds the line.
- ★★★★ Clean loops at any size, both directions.

STEP BY STEP CONTROL INPUTS



Carry a level line



Pull a full loop



Catch level on the line

PREREQUISITES & WHERE TO PRACTICE

- Confident throttle and pitch
- Enough altitude for a loop

Give yourself height, carry some speed, and pull a full loop, catching level as it comes back round. Keep it round and even, and try both forward and back.

COMMON MISTAKES

- ✗ Not enough speed or height to carry the loop
- ✗ An egg-shaped, uneven loop
- ✗ Dropping altitude through the top

VARIATIONS

Double Flip

Two loops in a row.

Laser Flip

A tight, fast loop that holds the line.

RELATED TRICKS

Flip

A tight version of the same rotation.

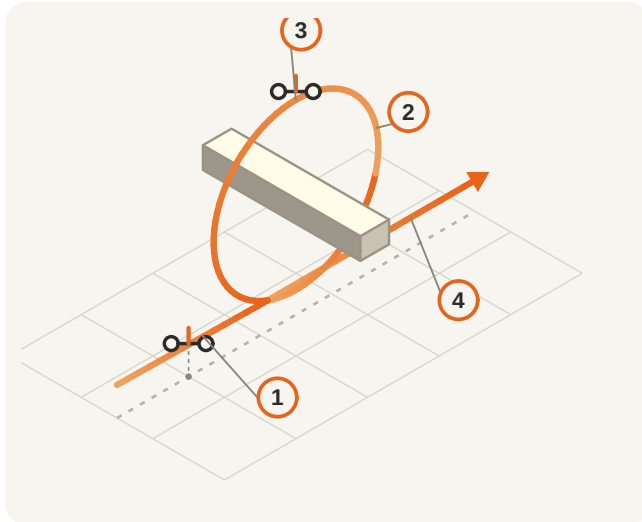
Powerloop

A loop flown around an object.

Maverick Loop

Beginner

Maverick Loops



HOW TO

A powerloop you never invert for: instead of pitching forward through inverted, you pitch back and carry the loop around the object, staying forward-facing the whole time. Approach under the object, pitch back and climb above it, hold the back-pitch over the top moving backwards, then drop throttle to return and pass under again. The quad stays upright throughout, which is what sets it apart from a powerloop.

MASTERY LEVELS

- ★☆☆ Loop around the object without inverting.
- ★★★☆☆ Round loop, forward-facing the whole way.
- ★★★★ Tight, committed loop, upright and dead on line.

STEP BY STEP CONTROL INPUTS



Approach under the obstacle



Pitch back and climb above the object



Over the top, then drop throttle to return



Pass back under the object

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- Comfortable carrying a hard back-pitch

Find a horizontal object with clear air around it, a beam, branch, or bar. Approach under it, pitch back to climb above, carry the loop over the top moving backwards, then drop throttle to return under. Keep the quad upright the whole way, never letting it pitch through inverted.

COMMON MISTAKES

- ✗ Letting it pitch through inverted like a powerloop
- ✗ Not enough height to clear the top
- ✗ Ballooning into the object on the way over

VARIATIONS

Half Mavvy

Only the bottom half of the loop.

Mavvelmann

A Maverick loop into an Immelmann exit.

RELATED TRICKS

Powerloop

The inverted, forward-pitching cousin.

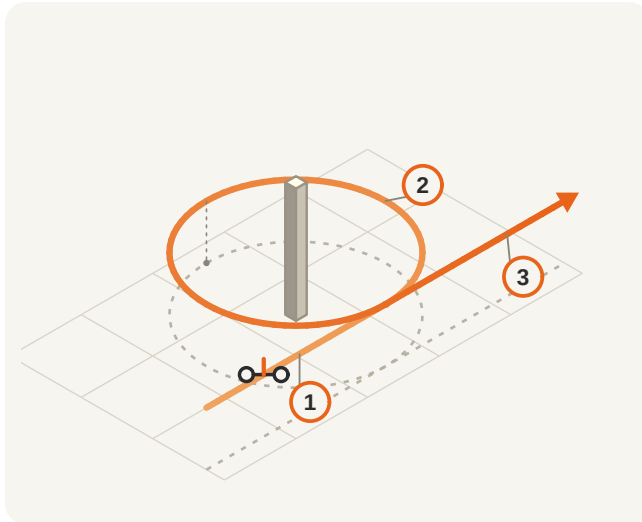
Immelmann

Shares the looping mechanic.

Orbits

Beginner

Orbitals



HOW TO

Approach the object head-on but slightly to the side. Add roll toward the object plus a bit of yaw the same way and a tiny amount of pitch: the roll tilts the drone, the mixed yaw counterbalances the tilt to keep it level, and the pitch supplies forward momentum around the object. Keep the object slightly off-center to hold your orbit distance and speed.

MASTERY LEVELS

- ★☆☆ Fly a full circle around the subject keeping it in frame.
- ★★★ Round, constant-radius orbit with the subject held centered.
- ★★★★ Smooth cinematic orbits at any radius, both directions, subject locked.

STEP BY STEP CONTROL INPUTS



Approach with the subject off one side



Bank and yaw together, camera locked on the center



Release on a tangent when you have the shot

PREREQUISITES & WHERE TO PRACTICE

- Coordinated turns, banking and yawing together
- Keeping a subject framed while moving

Pick an isolated subject with room around it, a lone tree, a cone, a friend, and fly complete circles at a fixed radius. Both directions. Holding the radius and height constant is the whole skill.

COMMON MISTAKES

- ✗ Yaw and roll out of sync, so the circle turns egg-shaped
- ✗ Letting the nose drift off the subject
- ✗ Changing radius or height mid-orbit

VARIATIONS

Trippy Spin

An inverted orbit around the top of a pole.

Cinematic orbit

A slow, wide orbit flown for smooth video.

RELATED TRICKS

Yaw Spin

Spinning in place instead of circling a subject.

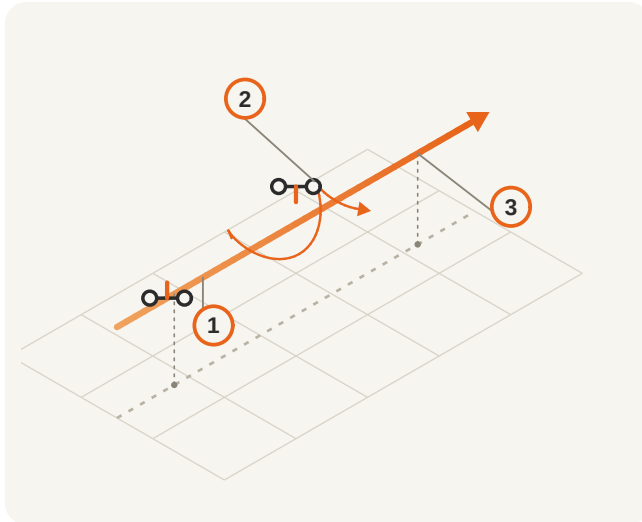
Trippy Spin

The inverted, elevated cousin.

Roll

Beginner

Open Air



HOW TO

The roll is the flip's sideways twin. Hold a straight, level line and roll the quad a full 360 around its roll axis in one smooth motion. Because a roll does not pull the quad off its line the way a flip does, you barely need to touch throttle, just keep it steady and let the roll carry through. Start slow and floaty; speed and style come later.

MASTERY LEVELS

- ★☆☆ Full 360 roll and recover upright on the same line.
- ★★★ Smooth, steady roll with no throttle sag or heading drift.
- ★★★★ Slow, controlled roll you can stop on any point, both directions.

STEP BY STEP CONTROL INPUTS



Settle onto a straight, level line



Full 360 on the roll axis in one motion



Catch level on the same track

PREREQUISITES & WHERE TO PRACTICE

- Stable level cruise
- A horizon or ground reference to catch level against

Open airspace, plenty of height. Pick a direction (most pilots start rolling to their dominant side) and keep the throttle dead steady so you learn the roll in isolation.

COMMON MISTAKES

- ✗ Chopping or pumping throttle mid-roll and dropping out
- ✗ Stopping short of a full rotation and exiting off-axis
- ✗ Mixing in accidental pitch, so the line bends

VARIATIONS

Double Roll

Two full roll rotations before catching level.

Vanny Roll

A slow, floaty roll carried through a turn.

RELATED TRICKS

Flip

The same rotation on the pitch axis.

Split S

A half roll to inverted, then a descending half loop.

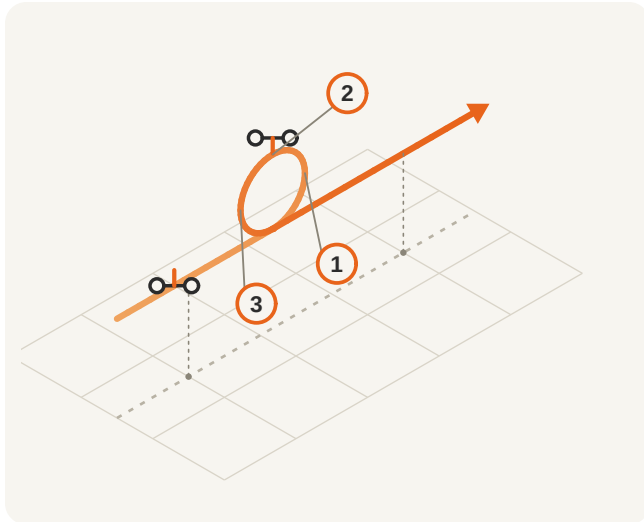
Vanny Roll

The stylish slow-roll variation.

Segmented Flips/Rolls

Beginner

Open Air



HOW TO

A flip or roll broken into stop-and-go segments instead of one smooth rotation. Rotate a quarter turn and stop, hold, then step to the next, working around in clean stages until you are level again. It trains precise rotational control, the ability to stop the rotation exactly where you want.

MASTERY LEVELS

- ★☆☆ Take a flip or roll around in clean segments.
- ★★★ Even, controlled stops at each quarter.
- ★★★★ Crisp segmented rotation, any number of stops.

STEP BY STEP CONTROL INPUTS

1  

Rotate a quarter and stop

2  

Step through the halfway point

3  

Finish the rotation to level

PREREQUISITES & WHERE TO PRACTICE

- A clean flip and roll
- Precise control of the rotation rate

Start with quarter-turn stops on the pitch axis, holding each cleanly, then try the roll axis. The skill is stopping the rotation dead at each stage, so keep the inputs small and deliberate.

COMMON MISTAKES

- ✗ Rotation drifting past each stop
- ✗ Uneven segments
- ✗ Losing altitude through the stops

VARIATIONS

Segmented Roll

Stop-and-go on the roll axis.

Double Flip

The same rotation, done smoothly.

RELATED TRICKS

Flip

The smooth version.

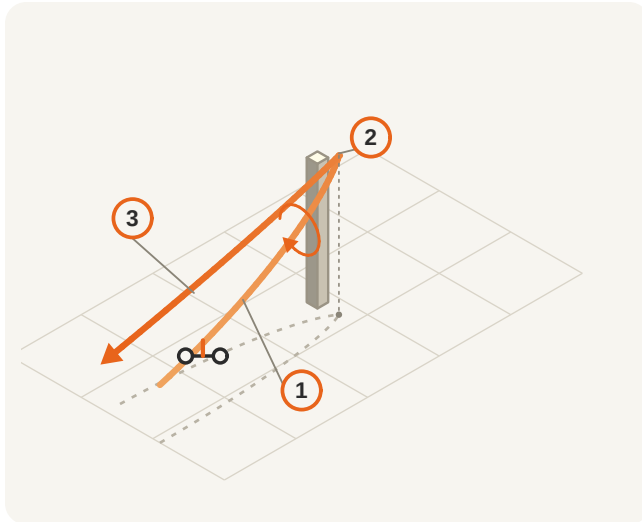
Roll

The roll-axis version.

Side-Lock Rewind

Beginner

Orbitals



HOW TO

Fly past an object, then roll 90° toward it while pitching back to keep it in view. Hold the pitch back until you reach a stall point, then blip throttle to rewind back the way you came. Finally roll away from the object to level out.

MASTERY LEVELS

- ★☆☆ Rewind back the way you came past the object.
- ★★★ Clean stall and a smooth rewind holding the object in view.
- ★★★★ Crisp stall, tidy rewind, and a level roll-away.

STEP BY STEP CONTROL INPUTS



1 Fly past the object and roll toward it



2 Pitch back to a stall, keeping it in view



3 Blip to rewind back, then roll away level

PREREQUISITES & WHERE TO PRACTICE

- Comfortable reaching and holding a stall
- Tracking an object while maneuvering

Pick an isolated pole or post. Fly past it and work the roll-and-pitch to keep it framed as you climb to the stall, then blip back down the same line. Rolling away cleanly at the end is the finishing touch.

COMMON MISTAKES

- ✗ Rolling away before the stall, so the rewind never sets up
- ✗ Losing the object out of frame at the stall point
- ✗ A harsh blip that throws the rewind off its line

VARIATIONS

Side-Lock Rewind (both sides)

Mirror it on the opposite side of the object.

Stall Rewind

The same rewind without the object to track.

RELATED TRICKS

Orbits

Another object-locked maneuver.

Snapback

A quicker stop-and-reverse.

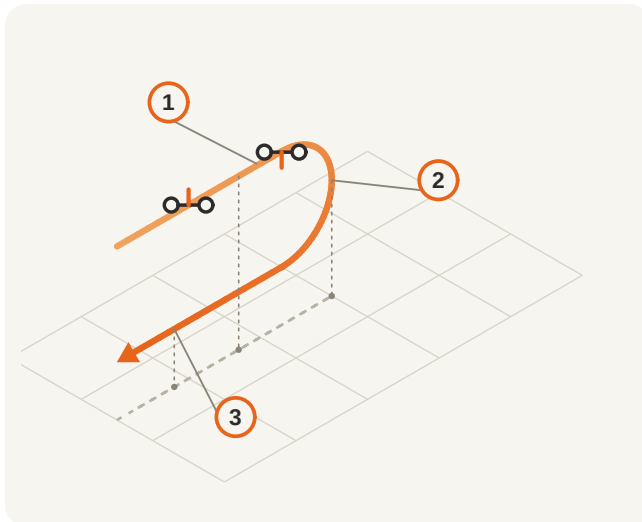
Wall Ride

More object-and-surface proximity work.

Split S

Beginner

Split-S



HOW TO

A fast way to reverse direction and shed altitude. From level flight, roll a half turn to inverted, then pull back-pitch through a descending half loop. You come out flying the opposite direction and lower than you started, essentially the bottom half of a loop entered upside down. Keep the pull smooth so the exit is level, not a dive.

MASTERY LEVELS

- ★☆☆ Roll inverted, pull through, and exit the opposite direction upright.
- ★★★☆☆ Level exit with no dive and controlled altitude loss.
- ★★★★★ Tight, symmetric half-loop that exits exactly on the reciprocal line.

STEP BY STEP CONTROL INPUTS

1 Approach object and roll inverted

2 Dive down

3 Catch and continue

PREREQUISITES & WHERE TO PRACTICE

- A clean half roll to inverted
- Comfortable applying pitch while inverted

Give yourself altitude to spend, since the whole trick descends. A split-S is the standard way to turn around and drop into a lower line, so practice it as a transition, not just in isolation.

COMMON MISTAKES

- ✗ Pulling too hard and diving out instead of leveling
- ✗ Not fully inverting before you pull, so the exit corkscrews
- ✗ Running out of altitude, since the trick always loses height

VARIATIONS

Power Split

Enter from the top of a powerloop instead of level flight.

Barani

A split-S with a half twist that exits facing forward.

RELATED TRICKS

Immelmann

The reverse: a climbing half loop that gains altitude.

Roll

The half roll that starts the trick.

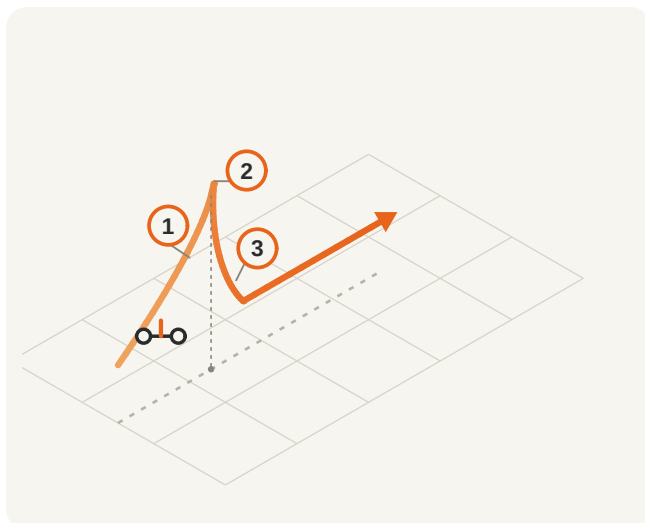
Powerloop

Shares the loop mechanic, entered over an obstacle.

Stall

Beginner

Open Air



HOW TO

Letting the quad run out of lift and hang, then catching it. Climb until you bleed off speed, chop the throttle so the quad stalls and drops, then feed throttle back in to recover before the ground. The stall is the pause that sets up rewinds and drops.

MASTERY LEVELS

- ★☆☆ Stall the quad and recover cleanly.
- ★★★ Controlled stall with a smooth drop-out.
- ★★★★ Deliberate stall held on the spot, tidy recovery.

STEP BY STEP CONTROL INPUTS



Climb and bleed off speed



Chop throttle to a stall



Drop, then feed throttle to recover

PREREQUISITES & WHERE TO PRACTICE

- Smooth throttle control
- A feel for the point of stall

Climb, chop, and catch, high up with room below. Learn where the stall point is and how much throttle it takes to recover. It is the building block of every stall trick.

COMMON MISTAKES

- ✗ Chopping too late, no room to recover
- ✗ Feeding throttle too hard on recovery
- ✗ Drifting off the spot in the stall

VARIATIONS

Stall Rewind

Rewind out of the stall.

Front Swing

A stall flown off the side of an object.

RELATED TRICKS

Stall Rewind

A stall into a rewind.

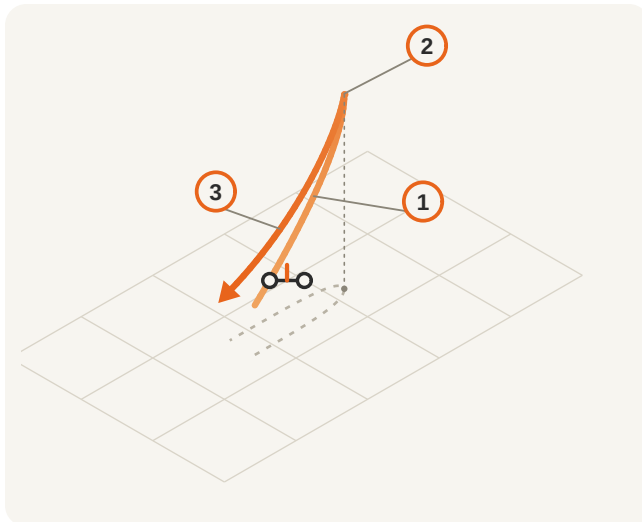
Front Swing

A side-on stall.

Stall Rewind

Beginner

Rewinds



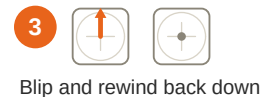
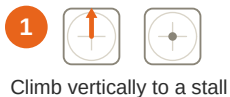
HOW TO

A vertical stall and reverse. Climb straight up until the quad runs out of speed and stalls at the top, hang there for a beat, then blip the throttle to kick it into a rewind straight back down the line it climbed. The cleaner the vertical, the tidier the rewind.

MASTERY LEVELS

- ★☆☆ Climb to a stall and rewind back down.
- ★★★☆☆ Clean vertical stall with a smooth rewind.
- ★★★★★ Crisp stall and a tight rewind back down the same line.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- Comfortable with a vertical climb and stall
- A feel for the blip that starts a rewind

Open air with height. Climb dead vertical, let it stall, and practise the little blip that sends it back down the same line. Keep the climb straight so the rewind tracks true.

COMMON MISTAKES

- ✗ Drifting off vertical, so the rewind wanders
- ✗ Blipping too hard and overshooting
- ✗ Not letting it stall before the rewind

VARIATIONS

Stall Flip Rewind

Add a flip at the stall before rewinding.

Stall 360 Rewind

Add a yaw 360 at the stall.

RELATED TRICKS

Snapback

A horizontal stop-and-reverse.

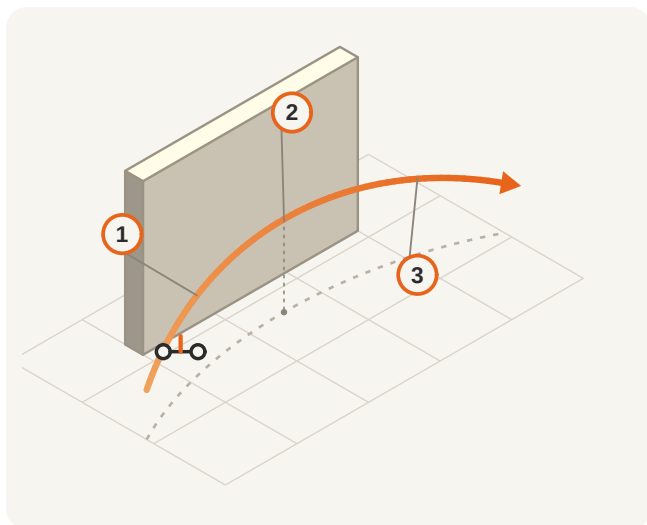
Side-Lock Rewind

A rewind flown beside an object.

Wall Ride

Beginner

Wall Tricks



HOW TO

Fly towards a wall. A few inches away, perform a 90° roll away from it and immediately cut throttle to glide smoothly alongside the wall.

MASTERY LEVELS

- ☆☆☆ Track alongside the wall for a beat without clipping it.
- ☆☆☆ Hold a steady gap and glide the length of the wall smoothly.
- ☆☆☆ Long, close, controlled ride with a clean peel-away at the end.

STEP BY STEP CONTROL INPUTS



Approach the wall on a shallow line



Roll ~90 onto the wall, ease off throttle to glide



Roll back level and peel away at the end

PREREQUISITES & WHERE TO PRACTICE

- Confident proximity flying near a surface
- Smooth throttle control at low margins

Find a clean, tall wall with nothing to snag, a barn, a bando, a big retaining wall. Start with a wide gap off the face and creep closer as your control improves. Never ride a wall you have not inspected.

COMMON MISTAKES

- ✗ Rolling too late and climbing away from the wall
- ✗ Carrying too much throttle, so you drift off the face
- ✗ Fixating on the wall instead of looking down its length

VARIATIONS

Reverse Wall Ride

Ride the wall flying backward.

Wall Kiss

A quick tap and peel instead of a full ride.

RELATED TRICKS

Gaps

Line-and-commit proximity flying through, not along.

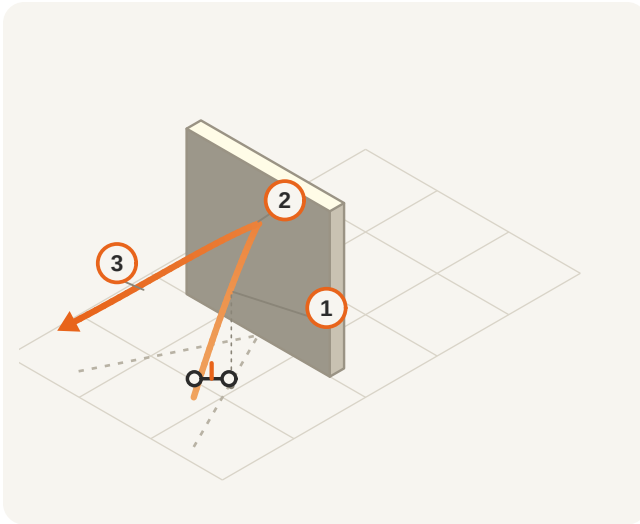
Building Dive

Tracking a vertical surface downward.

Wall Tap

Beginner

Wall Tricks



HOW TO

The basic wall touch. Approach a flat wall on a shallow line, ease up to tap the face lightly, and peel away smoothly. It is all about control and a soft touch, the foundation for the tap tricks that add rotations.

MASTERY LEVELS

- ★☆☆ Tap the wall and fly away under control.
- ★★★ Clean, light tap with a smooth peel-away.
- ★★★★ Precise taps on a chosen spot, no bounce.

STEP BY STEP CONTROL INPUTS



Approach the wall on a shallow line



Tap the flat face



Peel away from the wall

PREREQUISITES & WHERE TO PRACTICE

- Confident proximity flying near a surface
- A soft, controlled touch

Find a clean, flat wall and start with the gentlest taps, building confidence before you add speed. Scout the face for anything that could snag.

COMMON MISTAKES

- ✗ Coming in too steep or too fast
- ✗ Tapping too hard and bouncing off
- ✗ Drifting into the wall after the tap

VARIATIONS

Loop Tap

Add a roll and pitch into the tap.

Ceiling Tap

Tap an overhang instead of a wall.

RELATED TRICKS

Loop Tap

The tap with a rotation added.

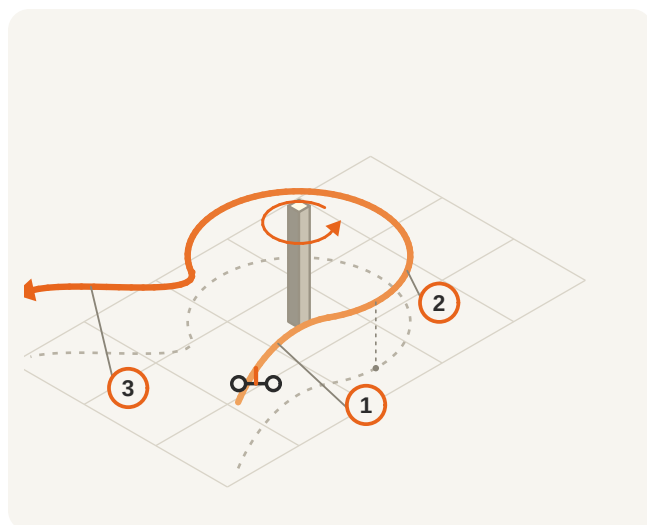
Wall Ride

Riding along the wall instead of tapping.

Whiplash

Beginner

Orbitals



HOW TO

Fly at an object, then pitch back and yaw around it so you swing around the back with the nose locked on it, and pitch forward to level out. Fly towards the object, pitch back to keep the top of it in view while yawing around it, then pitch forward until you level out. The object never leaves the frame, and the whip comes from the yaw carrying you around the back of it.

MASTERY LEVELS

- ★☆☆ Swing around the object keeping it in view.
- ★★★ Smooth swing around the back, object held framed.
- ★★★★ Tight, controlled swing, object locked in view the whole way.

STEP BY STEP CONTROL INPUTS



Fly towards the object



Pitch back and yaw around it, object in view



Pitch forward to level out, object in view

PREREQUISITES & WHERE TO PRACTICE

- Coordinated yaw and pitch
- Keeping a subject framed while maneuvering

Pick an object with clear room around it. Fly at it, then pitch back and yaw to swing around the back, keeping it centred in frame, and pitch forward to level out. Work both directions and keep it framed the whole way.

COMMON MISTAKES

- ✗ Losing the object out of frame through the swing
- ✗ Not enough pitch-back to carry around the back
- ✗ A jerky level-out that drops the frame

VARIATIONS

Pole Dance

Wrap the swing into a spiral around a pole.

Orbits

A full circle around the subject.

RELATED TRICKS

Orbits

Circling a subject at a fixed radius.

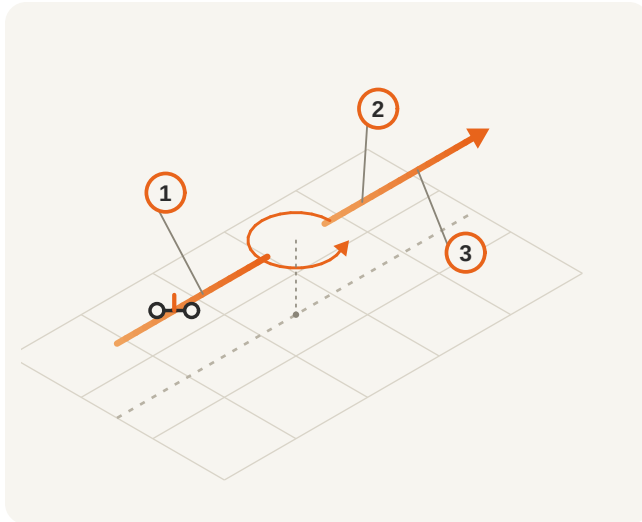
Pole Dance

Spiralling around a pole.

Yaw Spin

Beginner

Open Air



HOW TO

A flat 360 spin. Carry a little momentum into your spot, then spin the quad a full turn on the yaw axis while holding altitude, like a top spinning in place. Keep the throttle steady through the rotation so the quad neither climbs nor drops, and roll out on the heading you came in on. It is the foundation for every spin trick that follows.

MASTERY LEVELS

- ★☆☆ Complete a full flat 360 and end roughly on your heading.
- ★★★☆☆ Pinned on the spot at one height, exit on the entry heading.
- ★★★★★ Crisp, centered spin at any speed, both directions, dead on heading.

STEP BY STEP CONTROL INPUTS



Carry momentum into the spot



Full flat 360 on yaw, throttle held steady



Roll out on the original heading

PREREQUISITES & WHERE TO PRACTICE

- Comfortable holding a stable hover and heading
- Smooth throttle control to hold altitude while yawing

Over open grass at a comfortable height. Practice both directions. Watch that the quad stays on the spot and at one height through the whole turn.

COMMON MISTAKES

- ✗ Letting throttle sag, so the quad drops through the spin
- ✗ Drifting off the spot instead of pinning it
- ✗ Coming out on a different heading than you entered

VARIATIONS

Inverted Yaw Spin

The same flat spin held while inverted.

Trippy Spin

An inverted yaw orbit around the top of a pole.

RELATED TRICKS

Orbits

Circling a subject rather than spinning in place.

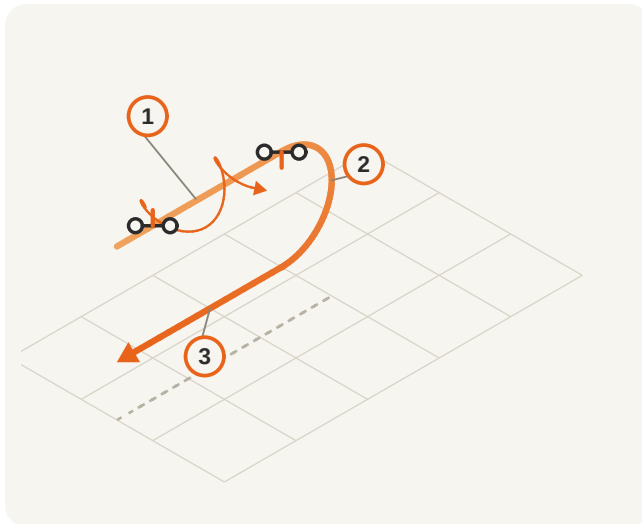
Inverted Yaw Spin

The inverted, advanced counterpart.

540 Split S

Novice

Split-S



HOW TO

A 540 roll flowing into a Split-S. Roll one and a half turns on the entry so you finish inverted, then pull through the descending half loop of a Split-S, exiting the opposite direction and lower. The roll count is what makes it read clean.

MASTERY LEVELS

- ★☆☆ Land the 540 roll and split-S out.
- ★★★ Clean 1.5 rolls into a smooth Split-S.
- ★★★★ Precise 540 straight into a tidy Split-S exit.

STEP BY STEP CONTROL INPUTS



540 roll on the entry



Pull the descending half loop



Exit the opposite direction, lower

PREREQUISITES & WHERE TO PRACTICE

- A precise, countable 540 roll
- A solid Split-S

Nail the 540 roll so you can stop it exactly inverted, then chain it into the Split-S descent. Precision on the roll is the whole trick.

COMMON MISTAKES

- ✗ Over or under rotating the 540
- ✗ A gap between the roll and the split
- ✗ Diving out instead of leveling

VARIATIONS

Split S

Without the leading 540.

540 Half Matty

A 540 into a Half Matty instead.

RELATED TRICKS

Split S

The reversal it finishes with.

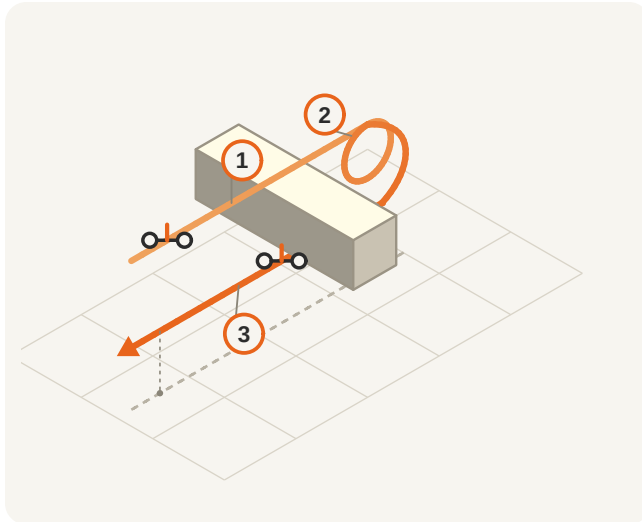
Double Roll

Shares the multi-rotation roll.

Anti Matty

Novice

Matty Flips



HOW TO

The Matty flipped the other way, a full backflip instead of a half front-flip. Fly over the object, then cut throttle and pull a full backflip, over-rotating 15 to 30 degrees to line yourself up, dropping down behind the object flying backward. Fly backward under the object, feeding in throttle and pitching forward to level out as you pass beneath it. Because the flip is a full rotation, you come out upright rather than inverted.

MASTERY LEVELS

- ★☆☆ Backflip over the object and fly back under it.
- ★★★☆☆ Clean full backflip, tidy backward pass underneath.
- ★★★★★ Crisp backflip, aligned exit, smooth level-out under the object.

STEP BY STEP CONTROL INPUTS



Fly over the object



Cut throttle, full backflip, over-flip to align



Fly backward under, throttle up and level out

PREREQUISITES & WHERE TO PRACTICE

- A confident full backflip
- Comfortable flying backward and keeping an object framed

Find an object with clear air above and below. Fly over it, cut throttle and commit to a full backflip, over-rotating a touch to line up, then fly backward under it and feed in throttle to level out. Give yourself height: the backflip drops you as you go.

COMMON MISTAKES

- ✗ Under-rotating the backflip, so you cannot line up
- ✗ Not enough height, so you drop too low
- ✗ Losing the object out of frame

VARIATIONS

Anti Matty Rewind

Rewind back under out of the exit.

Power Anti Matty

Carry it over a taller object with more throttle.

RELATED TRICKS

Matty Flip

The front-flip version, over then under.

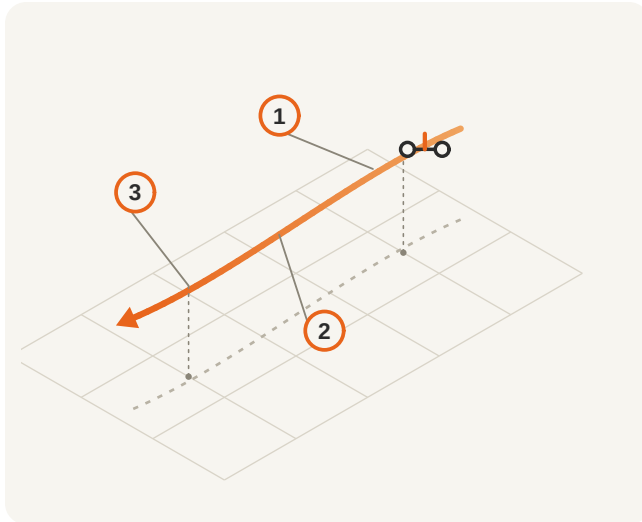
Loop

The full rotation at its core.

Backward Flying

Novice

Open Air



HOW TO

The essential skill of flying in reverse. Pitch back to get the quad moving tail-first, then hold a steady backward line, keeping it tracking true. Backward flight underpins every reversal and rewind, so it is worth drilling until it feels natural.

MASTERY LEVELS

- ★☆☆ Fly a straight line backward under control.
- ★★★ Smooth, steady backward flight, no wander.
- ★★★★ Confident backward flight on any line, both directions.

STEP BY STEP CONTROL INPUTS

1

Get the quad moving backward

2

Hold a steady backward line

3

Keep it tracking true

PREREQUISITES & WHERE TO PRACTICE

- Stable forward flight
- Comfortable reading a backward line

Open ground. Get the quad moving backward and hold a straight line, resisting the urge to over-correct. Practise until backward feels as natural as forward.

COMMON MISTAKES

- ✗ Over-correcting and weaving
- ✗ Drifting off the line
- ✗ Losing orientation flying tail-first

VARIATIONS

Backwards Dive

A dive flown backward.

Beginner Switch

Snap around to backward.

RELATED TRICKS

Beginner Switch

Snapping around to backward.

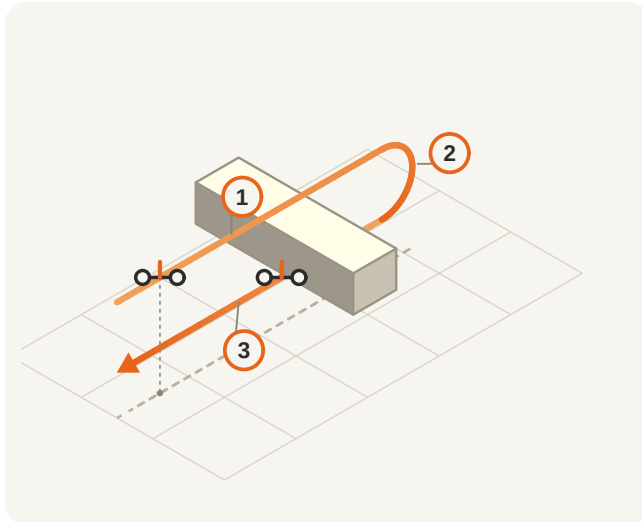
Cradle

A backward tracking pass.

Beginner Matty

Novice

Matty Flips



HOW TO

The Matty in its gentlest form, over a low, forgiving object. Fly over it with the camera tracking, throw a small flip at the far edge, and drop back under. Low and slow, it is the place to learn the Matty motion before you take it to bigger obstacles.

MASTERY LEVELS

- ★☆☆ Clear the low object and pass back under it.
- ★★★ Smooth, small Matty with a clean under-pass.
- ★★★★ Tidy, controlled Matty over a low obstacle, both ways.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- A clean flip
- Comfortable flying under and over an object

Pick a low bar or rail with clear air below. Keep it small and slow, focusing on the timing of the flip and keeping the object framed, then build up to taller obstacles.

COMMON MISTAKES

- ✗ Flipping too early
- ✗ Losing the object out of frame
- ✗ Dropping onto the object underneath

VARIATIONS

Matty Flip

The full-height version.

Power Matty

Over a much taller obstacle.

RELATED TRICKS

Matty Flip

The standard version.

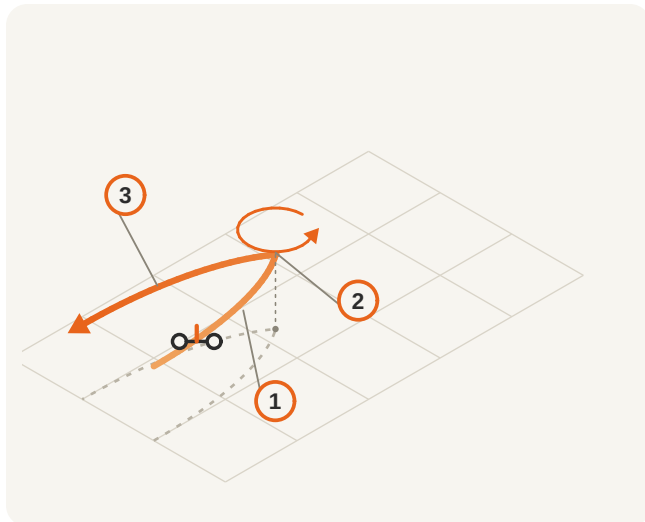
Flip

The rotation at its core.

Beginner Switch

Novice

Powerloops



HOW TO

The building block of every switch trick: a quick 180 yaw that turns you around. Fly forward on your line, snap a half yaw so the nose whips around, and continue flying backward. Master the clean switch here before you fold it into loops and taps.

MASTERY LEVELS

- ★☆☆ Switch 180 and fly out backward.
- ★★★☆☆ Clean, quick switch with a steady backward exit.
- ★★★★★ Snappy 180 switch, smooth into controlled backward flight.

STEP BY STEP CONTROL INPUTS



1 Fly forward on your line



2 Snap a 180 yaw to switch



3 Fly out backward

PREREQUISITES & WHERE TO PRACTICE

- A stable forward line
- Comfortable flying backward

Open air. Fly a straight line, snap the 180 yaw, and hold a clean backward line out. Both directions. Keep the switch quick and the exit steady.

COMMON MISTAKES

- ✗ A slow switch that drifts
- ✗ Over-rotating past 180
- ✗ Wobbling into the backward flight

VARIATIONS

Power Switch

Fold the switch into a powerloop.

Whiplash

A fast yaw whip around an object.

RELATED TRICKS

Whiplash

A fast yaw whip.

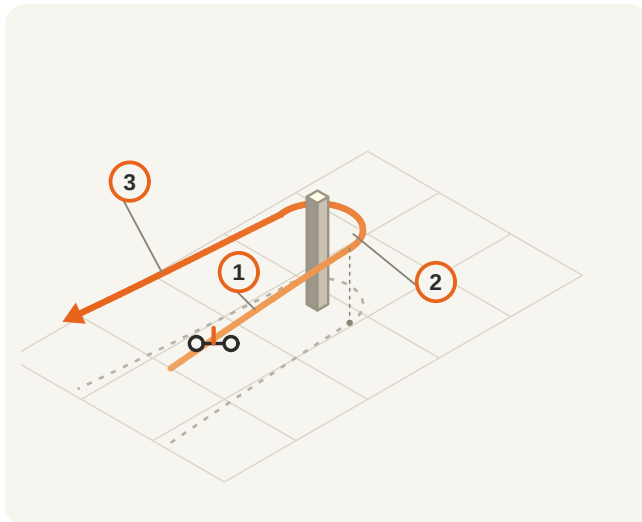
Yaw Spin

A full flat 360.

Boomerang

Novice

Open Air



HOW TO

An out-and-back curve wrapped around an object like a pole, the flight path of a thrown boomerang. Approach the obstacle, then pitch back and roll slightly right while yawing a touch left to carry the turn around it, and feed in throttle and pitch forward to exit. The object sits at the heart of the turn: you sweep around it and back out the way you came.

MASTERY LEVELS

- ★☆☆ Curve out around the object and back.
- ★★★ Smooth, rounded turn around the object.
- ★★★★ Tight, controlled boomerang around the object, both directions.

STEP BY STEP CONTROL INPUTS



Approach the obstacle



Pitch back, roll right, yaw left around it



Throttle and pitch forward to exit

PREREQUISITES & WHERE TO PRACTICE

- Coordinated pitch, roll, and yaw
- A feel for a rounded turn

Pick a pole or post with room around it. Approach it, then pitch back and coordinate a little roll and yaw to carry the turn around it and back out. Keep it smooth and rounded, both directions.

COMMON MISTAKES

- ✗ A flat, skidding turn instead of a rounded one
- ✗ Clipping the object on the turn
- ✗ Not coordinating the roll and yaw

VARIATIONS

Orbits

A full circle around the object.

Pendulum

An out-and-back swing.

RELATED TRICKS

Orbits

A full circle around a subject.

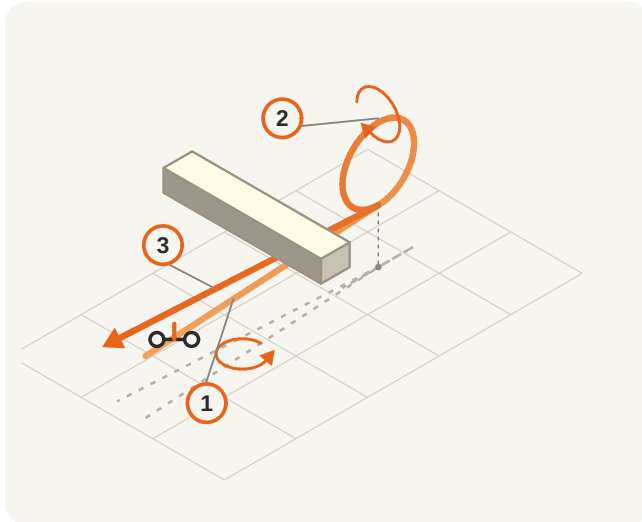
Whiplash

A tighter swing around an object.

Burrito Roll

Novice

Jump Ropes



HOW TO

A full roll flown under an object while facing sideways the whole time. Yaw 90 degrees to face sideways and enter beneath the object, throttle up and roll a full 360 staying sideways, then roll back under the object to the point you entered, the way you came in. It wraps a full roll into an under-and-back pass, keeping the quad sideways throughout.

MASTERY LEVELS

- ★☆☆ Roll under the object and back to the entry.
- ★★★ Smooth 360 roll, clean back-under to where you started.
- ★★★★ Tight, sideways burrito roll, dead on the entry line, both directions.

STEP BY STEP CONTROL INPUTS



90° flat yaw spin, enter under the object sideways



Facing sideways, throttle up and roll a full 360



Facing sideways, roll left back under, the way you came

PREREQUISITES & WHERE TO PRACTICE

- A clean 360 roll
- Comfortable flying sideways (yawed 90°) under an object

Find a horizontal object like the jump-rope bar. Yaw sideways, drop under it, throttle up and roll a full 360 staying sideways, then roll back under to where you entered. Keep it low and on the same line.

COMMON MISTAKES

- ✗ Losing the sideways heading through the roll
- ✗ Drifting off the entry line
- ✗ Clipping the object on the way under

VARIATIONS

Jump Rope

The loop-around-the-object version.

Double Burrito

Two burrito rolls linked.

RELATED TRICKS

Jump Rope

The loop-around cousin, also flown sideways.

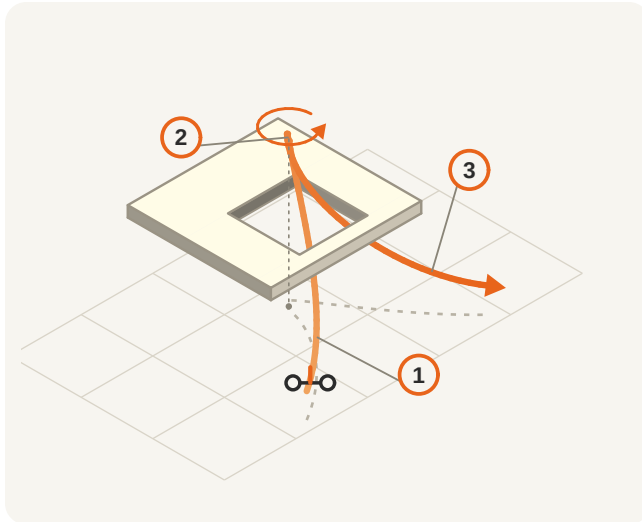
Barrel Roll

The 360 roll on its own.

Eject Roll

Novice

Miscellaneous



HOW TO

A vertical gap pass with a roll pinned at the top.

Throttle-punch up through a horizontal gap and pitch forward so you are facing straight down, pin a stationary 360 roll while looking down at the opening, then stop the roll and drop back down through the gap and out. It is a gap flown up-and-down instead of side-to-side, with a roll on top.

MASTERY LEVELS

- ★☆☆ Punch up through the gap, roll, and drop back through.
- ★★★ Clean up-and-through, tidy 360 roll, smooth drop back down.
- ★★★★ Crisp eject roll, pinned 360, dead through the gap both ways.

STEP BY STEP CONTROL INPUTS



Punch up through the gap, pitch to face down



Facing the gap from above, pin a 360 roll



Stop the roll, drop back down through and out

PREREQUISITES & WHERE TO PRACTICE

- A clean 360 roll
- Confident vertical punch-outs and gap flying

Find a horizontal gap, a doorway laid flat or the space between two beams. Punch up through it, pitch to face down, pin a full roll, then drop back through and out. Give yourself height above the gap for the roll.

COMMON MISTAKES

- ✗ Not enough punch to clear the gap for the roll
- ✗ Drifting off the gap during the roll
- ✗ Clipping the edge on the way back down

VARIATIONS

Juicy Eject

A floatier eject.

Stellar Eject Roll

An eject roll taken further.

RELATED TRICKS

Gaps

The gap pass, flown side-to-side instead.

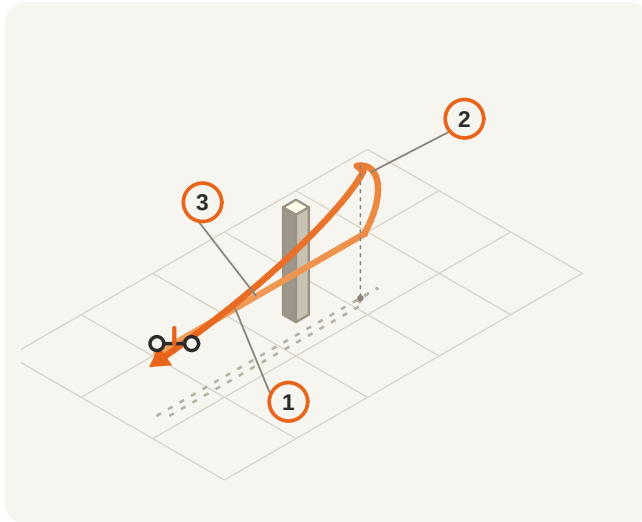
Barrel Roll

The 360 roll on its own.

Front Swing

Novice

Orbitals



HOW TO

Approach an object from the side, slip just past it, pitch up into a stall, and come backwards out the way you came. Approach the side of the object, and as you pass it slightly, pitch up to stall the quad, then blip the throttle and pitch back to slide backwards out along your approach line. It is a controlled stall-and-reverse played out beside an object.

MASTERY LEVELS

- ★☆☆ Approach, stall past the object, and reverse out.
- ★★★ Clean approach into a tidy stall and a smooth reverse.
- ★★★★ Crisp stall just past the object, dead-straight reverse out.

STEP BY STEP CONTROL INPUTS



1 Approach the side of the object



2 Just past it, pitch up to a stall



3 Blip and pitch back, backwards out the way you came

PREREQUISITES & WHERE TO PRACTICE

- A confident stall
- Comfortable flying backward on a line

Pick an object with room around it. Approach from the side, slip just past it, pitch up to a clean stall, then blip and slide backwards out the way you came. Keep the reverse straight on your approach line.

COMMON MISTAKES

- ✗ Stalling too early, before passing the object
- ✗ Drifting off the approach line on the reverse
- ✗ A harsh blip that overshoots

VARIATIONS

Pendulum

A swinging out-and-back.

Boomerang

An out-and-back around an object.

RELATED TRICKS

Stall Rewind

A stall reversed, flown vertically.

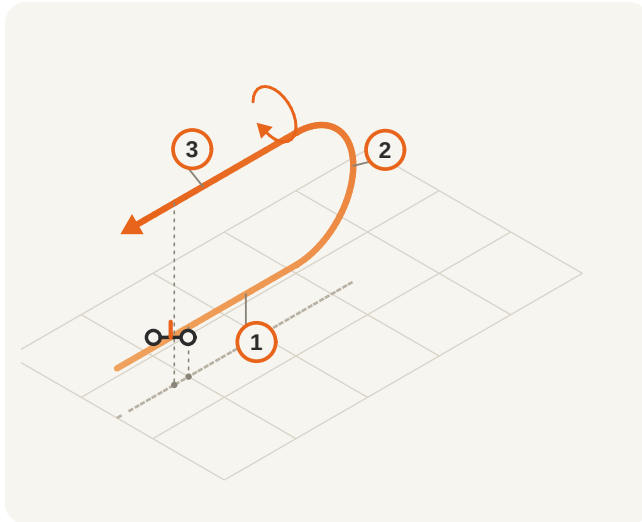
Boomerang

An out-and-back around an object.

Immelmann

Novice

Powerloops



HOW TO

Begin a Powerloop, but at the peak of the loop execute a rapid 180° roll to level out and complete the trick.

MASTERY LEVELS

- ★☆☆ Complete the half loop and roll out upright the opposite direction.
- ★★★☆☆ Smooth loop and a clean level roll-out with altitude gained.
- ★★★★ Round loop, crisp 180 roll right at the peak, dead-level exit.

STEP BY STEP CONTROL INPUTS



Pull up into a half loop with throttle



Half roll upright at the top of the loop



Exit level the opposite way, higher

PREREQUISITES & WHERE TO PRACTICE

- Comfortable looping on the pitch axis (powerloop)
- A reliable half roll to level

Give yourself altitude and open space. Practice it as the mirror of a Split-S: where the Split-S drops and reverses, the Immelmann climbs and reverses. Enter with enough speed to carry the loop.

COMMON MISTAKES

- ✗ Rolling before you reach the top, so the exit corkscrews
- ✗ Running out of speed halfway up the loop
- ✗ Exiting off-heading instead of a clean reciprocal

VARIATIONS

Immelloop

A powerloop with a 180 roll at the apex.

Immelmatt

Blend an Immelmann into a Matty line.

RELATED TRICKS

Split S

The descending mirror image.

Powerloop

Shares the loop; the Immelmann is a half loop plus a roll.

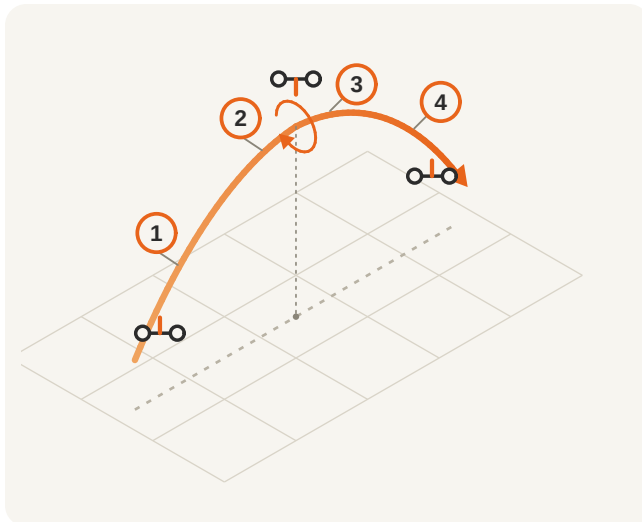
Matty Flip

Another over-the-top reversal, done over a gap.

Inverted Rewind

Novice

Open Air



HOW TO

A stall rewind flown through inverted. Blip the throttle to float up to a stall point, then roll a full 180 to the right so you pass through inverted at the top, and immediately roll 180 back to the left to flatten out, rewinding the roll. Blip the throttle again to catch the drop, coming back down at the same rate you climbed.

MASTERY LEVELS

- ★☆☆ Blip up, double-roll through inverted, and catch the drop.
- ★★★☆☆ Smooth stall with a clean roll and rewind at the top.
- ★★★★★ Crisp stall, tight through inverted, floated back down on the line.

STEP BY STEP CONTROL INPUTS

1



Blip throttle to a stall out point

2



180 roll to the right

3



180 roll to the left to flatten out

4



Blip throttle to catch the drop

PREREQUISITES & WHERE TO PRACTICE

- A confident stall out
- A crisp, controllable roll

Get the stall timing dialed first, then add the double roll at the top. Everything feels reversed as you pass through inverted, so start high and take it slow. Keep the line straight so the rewind tracks true.

COMMON MISTAKES

- ✗ Stalling the roll halfway and dropping inverted
- ✗ Drifting off the line through the peak
- ✗ A harsh blip that throws the drop

VARIATIONS

Stall Rewind

The same, flown upright.

Inverted Yaw Spin

Another inverted hold-and-move.

RELATED TRICKS

Stall Rewind

The upright version.

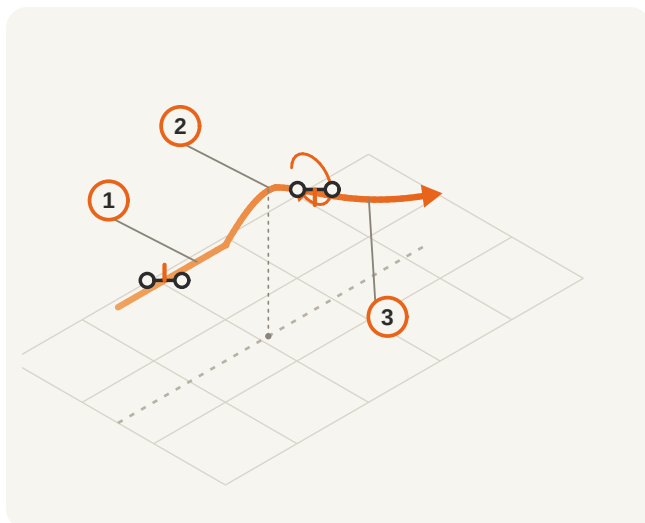
Inverted Yaw Spin

Another inverted maneuver.

Juicy Eject

Novice

Miscellaneous



HOW TO

A floaty, stylish way to bail off a line. Blip the throttle to unload the quad, then let it drift into a soft, slow roll away and down, floating out clear of the line rather than diving off it. It is the eject roll made juicy, all about float and finesse.

MASTERY LEVELS

- ★☆☆ Unload and float a roll away off the line.
- ★★★ Smooth, floaty roll into a soft descent.
- ★★★★ Effortless, juicy eject that drifts out clean.

STEP BY STEP CONTROL INPUTS



Slightly roll left



Blip throttle for eject and soft roll away



Float out clear of the line

PREREQUISITES & WHERE TO PRACTICE

- A clean roll
- A feel for unloading with throttle

Practise the blip-and-drift, keeping the roll slow and floaty rather than snappy. It doubles as a graceful bail-out. Film it back to check it reads smooth.

COMMON MISTAKES

- ✗ Rolling too fast, losing the float
- ✗ Diving instead of drifting
- ✗ Not unloading enough

VARIATIONS

Eject Roll

A snappier eject.

Juicy Flick

A juicy pop to inverted.

RELATED TRICKS

Eject Roll

A snappier bail.

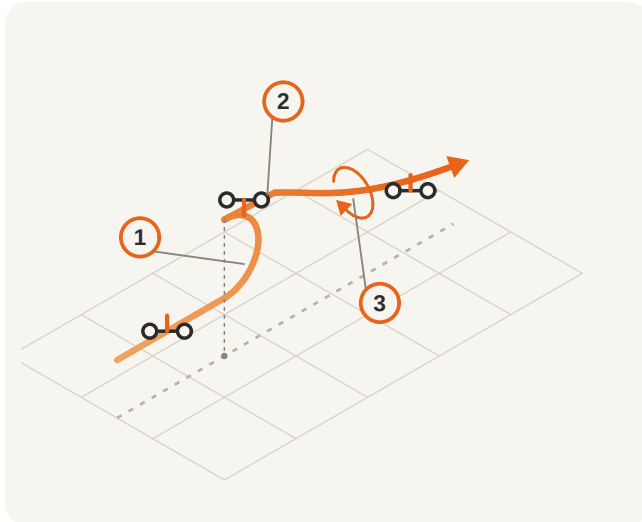
Juicy Flick

Another juicy, floaty move.

Juicy Flick

Novice

Open Air



HOW TO

From a cruising line, blip the throttle to unload the quad and snap a 180 pitch to flick it up to inverted, hang there floaty for a beat, then roll out slowly and smoothly to recover forward on the same line. It is small and floaty by design, more style than power, which is exactly what makes it read as juicy.

MASTERY LEVELS

- ★☆☆ Pop to inverted and roll out upright on the same line.
- ★★★☆☆ Floaty, unhurried flick with a smooth slow roll-out.
- ★★★★★ Effortless, stylish flick that stays perfectly on line, both ways.

STEP BY STEP CONTROL INPUTS



Blip the throttle and snap a 180 pitch to inverted



Hang inverted, floaty, for a beat



Slow roll out and recover forward

PREREQUISITES & WHERE TO PRACTICE

- A clean flip and confident inverted recovery
- A feel for unloading the quad with throttle

Plenty of height at first. The trick is in the timing of the blip and the patience of the roll-out, so keep it slow and floaty rather than forcing it. Film it back to check it reads as smooth.

COMMON MISTAKES

- ✗ Rolling out too fast, killing the floaty look
- ✗ Not unloading enough, so the flick is stiff
- ✗ Losing the line, so you recover off-track

VARIATIONS

Juicy Double Roll

Finished with a quick double (540) roll.

Double Juicy

Two flicks linked before recovering.

Juicy Flick Rewind

Flick, then reverse back along the line.

RELATED TRICKS

Juicy Double Roll

The same flick, double-rolled out.

Flip

The full rotation the flick is a stylised piece of.

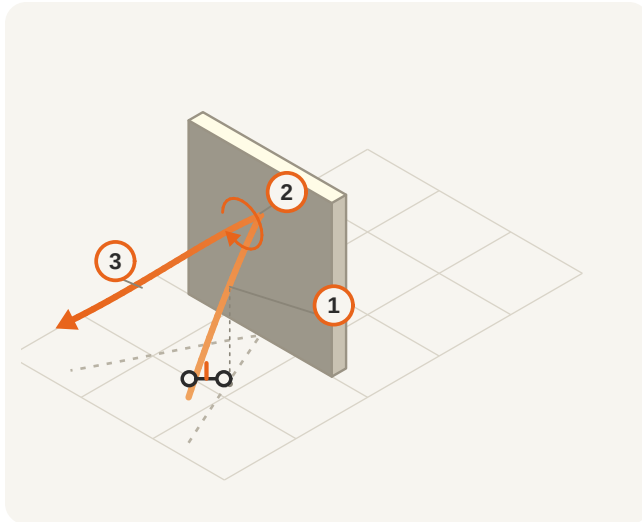
Roll

The slow roll-out that finishes it.

Loop Tap

Novice

Wall Tricks



HOW TO

Fly towards an object at 45° or less. Apply a bit of throttle and simultaneously perform a 90° roll and a 45° pitch back to execute a wall tap while facing backward. Immediately follow with a 180° pitch back and roll away from the wall to finish.

MASTERY LEVELS

- ★☆☆ Tap the wall and peel away under control.
- ★★★ Clean tap facing backward with a smooth roll-away.
- ★★★★ Precise tap and an immediate, tidy reverse off the wall.

STEP BY STEP CONTROL INPUTS



Approach the wall on a shallow line



Roll and pitch back to tap the face, facing backward



Pitch back and roll away from the wall

PREREQUISITES & WHERE TO PRACTICE

- Confident proximity flying near a surface
- Comfortable tapping a wall and reversing

Find a clean, tall wall and start with gentle taps, building up the roll-and-pitch as your confidence grows. Scout the face first, you want a flat, snag-free surface to tap.

COMMON MISTAKES

- ✗ Coming in too steep instead of a shallow line
- ✗ Tapping too hard and knocking yourself off line
- ✗ Slow to reverse away, so you drift into the wall

VARIATIONS

Loop Tap Rewind

Rewind straight back off the tap.

Double Tap

Two taps linked along the same wall.

RELATED TRICKS

Wall Ride

Riding along the wall rather than tapping it.

Building Dive

Another committed wall maneuver.

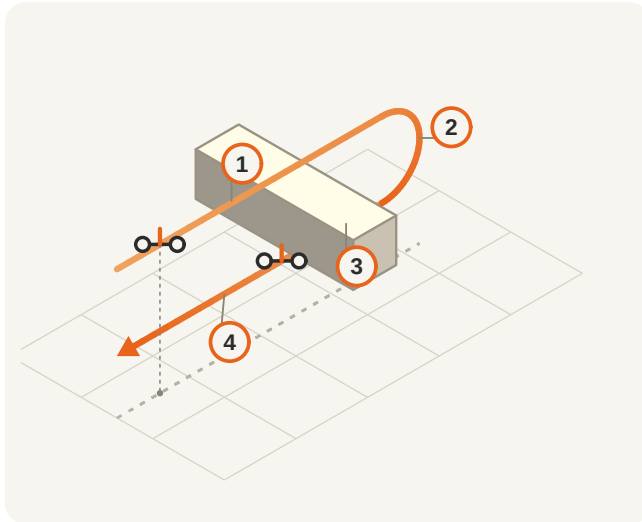
Snapback

Shares the tap-and-reverse feel.

Matty Flip

Novice

Matty Flips



HOW TO

Fly over an object and execute a partial front flip over it while keeping the object in view as you descend. Gradually increase throttle and adjust pitch to level out and fly backwards beneath the object.

MASTERY LEVELS

- ★☆☆ Pass over the object, flip, and return under it in any form.
- ★★★☆☆ Keep the object framed through the flip with a smooth, controlled return. Good form.
- ★★★★★ Object stays centered the entire time with a seamless over then under line. Perfect form.

STEP BY STEP CONTROL INPUTS

1

Fly over the object, camera tracking it (light forward pitch)

2

Partial front flip over the far edge (forward pitch)

3

Gradually increase throttle to arrest and level inverted

4

Adjust pitch to fly backward beneath the object

PREREQUISITES & WHERE TO PRACTICE

- Solid Forward Flip with clean level exits
- Comfortable flying backward and tracking a subject
- Basic Gap/proximity confidence

Needs a gap or elevated obstacle you can pass over and then back under, a railing, low wall, fallen log, or playground structure. Pick a subject to track (a tree, a friend, a car) so you can practice keeping it framed through the flip.

COMMON MISTAKES

- ✗ Losing the subject during the flip, rushing the rotation
- ✗ Drifting off the line so the return doesn't clear the gap
- ✗ Flipping too early before fully clearing the far edge
- ✗ Not enough separation, so the exit clips the obstacle

VARIATIONS

Half Matty

Stop at inverted, a reverse Split-S that reverses direction with altitude.

540 Half Matty

Add a full flip for 540° of total pitch rotation.

Matty Twister

Add a yaw twist during the flip.

Power Matty

Fly it with powerloop throttle to carry it over a taller obstacle.

RELATED TRICKS

Anti Matty

Forward-flip variant toward the direction of travel.

Matty 360

Matty with an added 360° element.

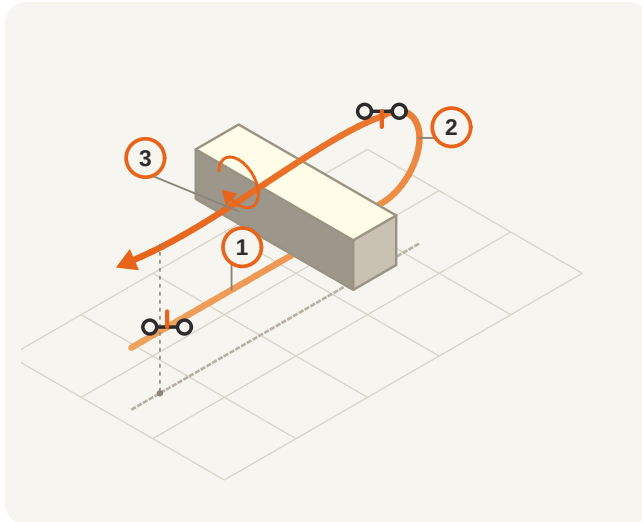
Snapback

Half-flip reversal without the gap pass-under.

Mavvelmann

Novice

Maverick Loops



HOW TO

Pass forward under the object, then pitch up into a half loop to almost inverted. Once above the object, rock back and pitch the drone down, then level out with a 180 roll over the object, coming back the way you started. It stacks a half loop and a rolling reversal, adding both roll and pitch on the pass back over the top.

MASTERY LEVELS

- ★☆☆ Loop up over the object and roll out backward.
- ★★★☆☆ Clean half-loop up into a tidy roll back over.
- ★★★★★ Seamless Mavvelmann, roll and pitch out over the object, dead on line.

STEP BY STEP CONTROL INPUTS



1 Pass under, pitch up to almost inverted



2 Once above it, rock back and pitch down



3 180 roll to level, back over toward start

PREREQUISITES & WHERE TO PRACTICE

- A confident half loop
- A clean 180 roll to level

Pass under the object and pitch up hard to almost inverted, then as you come above it, rock back and roll 180 to level out heading back the way you came. Give yourself vertical room.

COMMON MISTAKES

- ✗ Not enough pitch to reach inverted above the object
- ✗ Rolling before you are above the object
- ✗ Losing the line on the reversal

VARIATIONS

Immelmann

A half-loop and roll in the open.

Maverick Loop

A loop under the object.

RELATED TRICKS

Immelmann

The half-loop-and-roll it echoes.

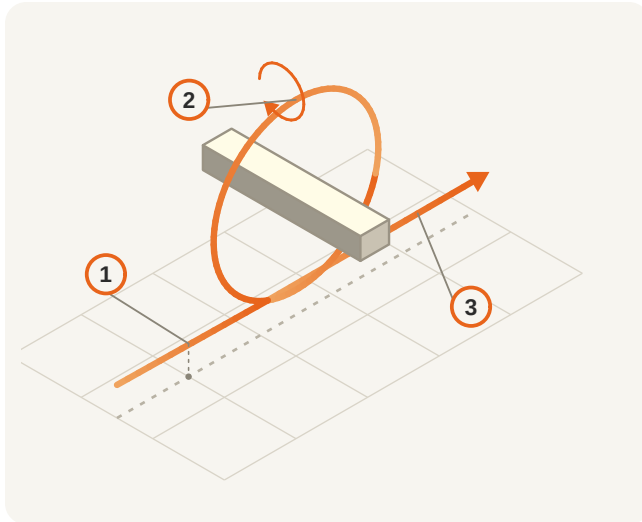
Maverick Loop

A loop under the object.

Mavvy Roll

Novice

Maverick Loops



HOW TO

A powerloop with a full 360 roll pinned at the top of the loop. Pass under the object and punch vertical into the loop, and at the peak roll a full 360, then level the roll and carry the loop back down and under the object. It caps the powerloop with a roll right at the top.

MASTERY LEVELS

- ★☆☆ Powerloop with a 360 roll at the top.
- ★★★ Clean powerloop, tidy roll pinned at the peak.
- ★★★★ Seamless powerloop and roll, dead on line both ways.

STEP BY STEP CONTROL INPUTS

1



Pass under the object, punch vertical

2



At the peak, pin a 360 roll

3



Level the roll, complete the loop back under

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A controllable 360 roll

Get the powerloop solid, then add the roll at the very top of the loop where you hang inverted. Give yourself height and commit to the loop before the roll.

COMMON MISTAKES

- ✗ Rolling before you reach the peak
- ✗ Not enough speed to carry the loop
- ✗ Losing the loop line on the roll

VARIATIONS

Powerloop

Without the roll at the top.

Inverted 360 Powerloop

A yaw 360 at the top instead of a roll.

RELATED TRICKS

Powerloop

The loop it is built on.

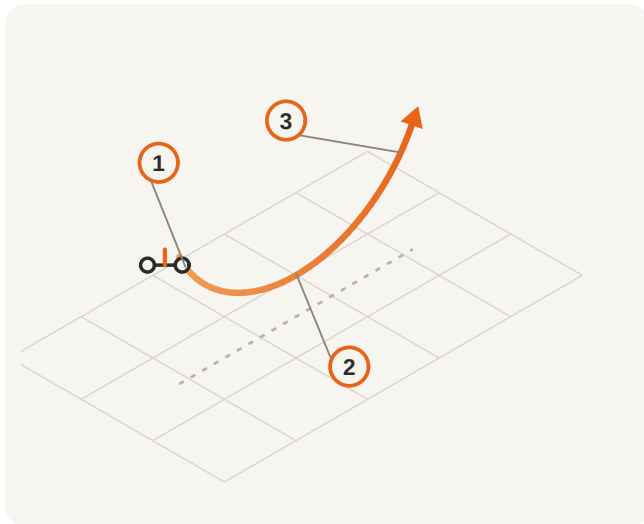
Barrel Roll

The 360 roll on its own.

Pendulum

Novice

Orbitals



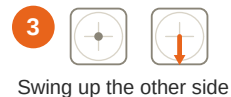
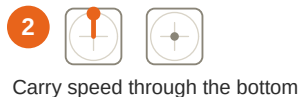
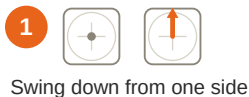
HOW TO

A swinging arc like a pendulum on a string. Swing down from one side, carry your speed through the bottom, and let it swing up the other side, matching the height on each swing. It is a rhythm move that trades altitude for speed and back again.

MASTERY LEVELS

- ★☆☆ Swing down through and up the other side.
- ★★★ Smooth, even swing like a pendulum.
- ★★★★ Rhythmic swings, matched on both sides.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- Smooth pitch and throttle
- A feel for trading height for speed

Find an open arc and swing down and up, keeping both sides even. Let momentum do the work through the bottom. Build the rhythm so it flows like a real pendulum.

COMMON MISTAKES

- ✗ Uneven swings, higher on one side
- ✗ Throttling through instead of coasting
- ✗ Losing the rhythm

VARIATIONS

Jump Rope

An up-and-over swing over an object.

Boomerang

An out-and-back curve.

RELATED TRICKS

Jump Rope

An over-an-object swing.

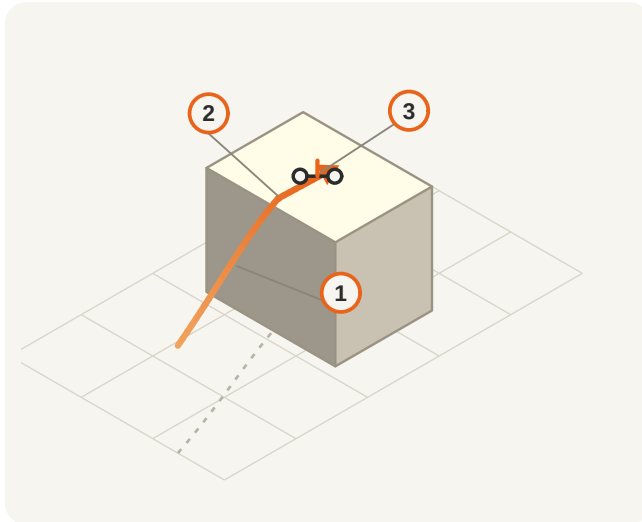
Orbits

Another arcing maneuver.

Perch

Novice

Miscellaneous



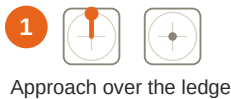
HOW TO

Landing on top of an object instead of the ground. Approach over a ledge, wall, or box, flare to bleed off your speed, and settle gently onto the top surface, holding it there. It is a precision landing that turns a piece of the scenery into a perch.

MASTERY LEVELS

- ★☆☆ Set down on top of the object and hold.
- ★★★ Smooth approach and a controlled settle.
- ★★★★ Precise, gentle perch on a chosen spot, no bounce.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- Precise low-speed control
- A gentle touch on the throttle

Start with a wide, flat top, a low wall or a table, and practise settling on it softly. Approach slow, flare, and set down light. Tighten your target as your control improves.

COMMON MISTAKES

- ✗ Coming in too fast and overshooting the top
- ✗ Not flaring, so you clip the edge
- ✗ Setting down hard and bouncing off

VARIATIONS

Slide to Perch

Slide across the top before stopping.

Wall Ride

Track a vertical face instead of the top.

RELATED TRICKS

Slide Disarm

A ground-level settle-and-stop.

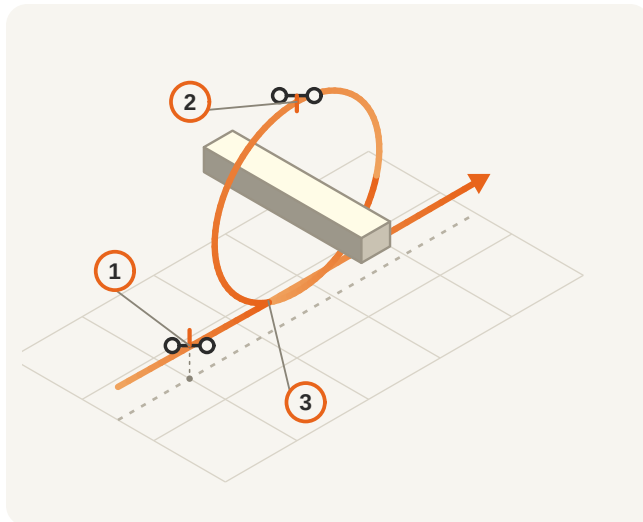
Wall Tap

Touching a surface rather than resting on it.

Powerloop

Novice

Powerloops



HOW TO

Fly under and past the obstacle with speed, then apply throttle and gradually pitch back to arc up into the loop until you're fully inverted directly over or slightly behind it. Carry throttle across the top, then relax pitch as gravity brings the nose down, increasing throttle again as you approach the ground so you exit level on your entry line.

MASTERY LEVELS

- ★☆☆ Fly under the obstacle, around the top, and back through in any form.
- ★★★☆☆ Keep a smooth flight path with no jerking through the whole trick. Good form.
- ★★★★★ Smooth, with consistent distance maintained through the whole path. Perfect form.

STEP BY STEP CONTROL INPUTS



Approach under object, pitching drone up



At the peak of the loop, float through



Catch at the bottom of the loop

PREREQUISITES & WHERE TO PRACTICE

- Comfortable with basic Flip and inverted recovery
- Consistent throttle control through a full rotation

Needs a horizontal obstacle to loop over, a tree branch, gap, banner bar, or doorway. Start with a high, forgiving obstacle and plenty of vertical space; drop the height as it gets consistent.

COMMON MISTAKES

- ✗ Too little throttle over the top, the quad drops out inverted
- ✗ Flipping in place instead of translating up and over the obstacle
- ✗ Chopping throttle early, killing the exit and dropping the nose

VARIATIONS

Sideways Powerloop

Flown on the roll axis, loop the obstacle knife-edge style.

Power Split

Finish the top with a Split-S roll-out instead of continuing the loop.

Immelloop

A 180 roll at the loop's apex, out backwards.

RELATED TRICKS

Immelmann

Half-loop-and-roll that reverses direction with altitude.

Maverick Loop

The inside-out inverse, dive in and loop underneath.

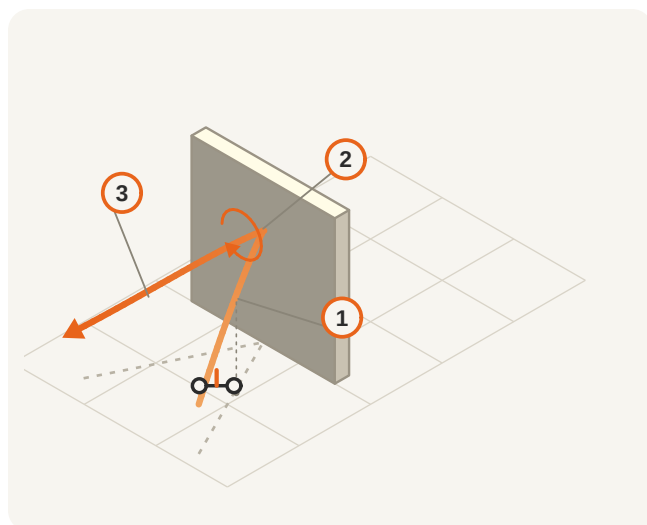
Split S

The downward half-loop counterpart.

Roll Tap

Novice

Wall Tricks



HOW TO

A wall tap with a roll baked in. Approach a flat wall on a shallow line, roll the quad as you reach out to tap the face, then peel away. The roll turns a plain tap into a stylish one, a step up from the basic wall tap.

MASTERY LEVELS

- ★☆☆ Roll into a wall tap and peel away.
- ★★★☆☆ Clean roll into a light tap and smooth exit.
- ★★★★★ Crisp roll-tap on a chosen spot, no bounce.

STEP BY STEP CONTROL INPUTS



Approach the wall on a shallow line



Roll into the tap



Peel away from the wall

PREREQUISITES & WHERE TO PRACTICE

- A confident wall tap
- A controllable roll near a surface

Get the basic wall tap solid first, then add the roll as you touch. Start gentle and build up. Scout the wall for anything that could snag.

COMMON MISTAKES

- ✗ Rolling too early, missing the tap
- ✗ Tapping too hard and bouncing
- ✗ Losing your line on the roll

VARIATIONS

Wall Tap

Without the roll.

Loop Tap

A roll and pitch into the tap.

RELATED TRICKS

Wall Tap

The plain tap.

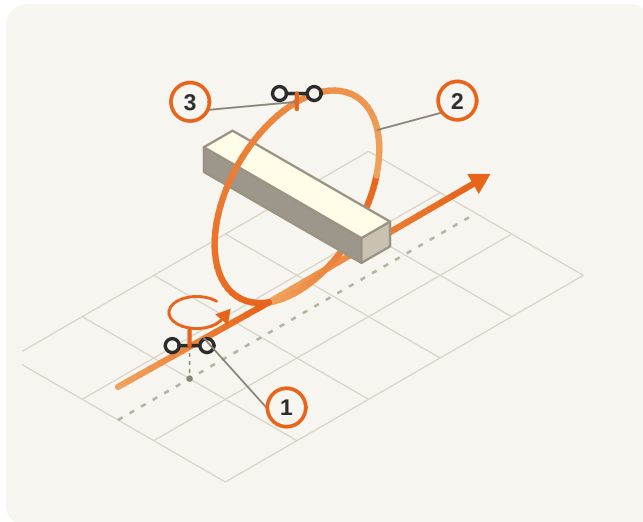
Loop Tap

A tap with roll and pitch.

Side Loop

Novice

Jump Ropes



HOW TO

The same as a jump rope, a loop around a horizontal object entered facing sideways, but instead of staying level you go inverted over the top. Yaw 90 degrees to face sideways and enter beneath the object, throttle up to carry the loop around it, and pass over the top inverted before dropping back down and out. Unlike the jump rope, there is no roll to keep it level, so you invert at the peak.

MASTERY LEVELS

- ★☆☆ Loop around the object, inverted over the top.
- ★★★ Smooth sideways loop, clean inversion at the peak.
- ★★★★ Tight side loop, dead sideways, crisp inverted top.

STEP BY STEP CONTROL INPUTS



1 90° flat yaw spin, enter under the object sideways



2 Throttle up to loop around



3 Pass over the top inverted, cut throttle

PREREQUISITES & WHERE TO PRACTICE

- A confident jump rope or powerloop
- Comfortable flying sideways (yawed 90°)

Find a horizontal object like the jump-rope bar. Yaw sideways, drop under it, and throttle up to carry the loop around, going inverted over the top rather than rolling to stay level. Build the loop up slowly.

COMMON MISTAKES

- ✗ Rolling to stay level like a jump rope
- ✗ Drifting off the sideways heading
- ✗ Not enough throttle to clear the top inverted

VARIATIONS

Jump Rope

The level version, rolling to stay upright.

Burrito Roll

A sideways roll under the object.

RELATED TRICKS

Jump Rope

The level, non-inverted version.

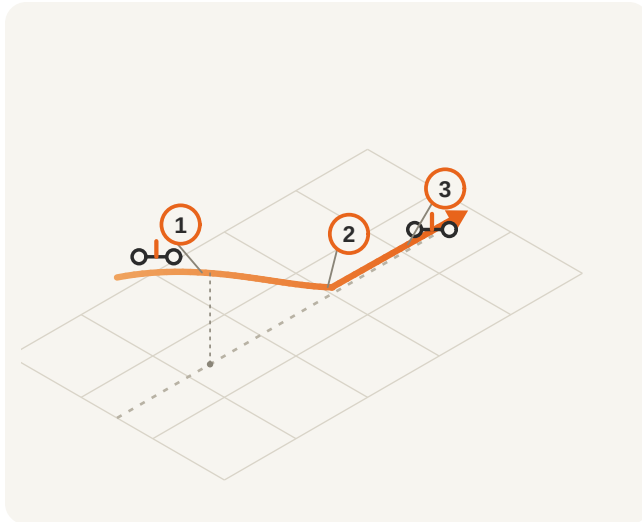
Powerloop

A loop around an object, forward-facing.

Slide Disarm

Novice

Miscellaneous



HOW TO

A stylish way to end a line. Bring the quad down onto a shallow landing line, let it touch and slide along the ground carrying its momentum, and cut the motors mid-slide so it coasts to a stop. It reads as a deliberate finish rather than a landing, the slide disarm every freestyle line wants to end on.

MASTERY LEVELS

- ★☆☆ Set it down and slide to a stop without tumbling.
- ★★★ Smooth approach and a controlled slide.
- ★★★★ Low, fast slide onto a chosen spot, clean disarm.

STEP BY STEP CONTROL INPUTS



Descend onto your landing line



Let it touch and slide along the ground



Cut the motors mid-slide to a stop

PREREQUISITES & WHERE TO PRACTICE

- Smooth throttle control near the ground
- A feel for your quad's momentum

A flat, smooth surface with room to slide. Come in shallow and low, let it touch, and practise timing the disarm so it coasts to rest. Bind your disarm to a switch you can hit cleanly.

COMMON MISTAKES

- ✗ Coming in too steep and tumbling
- ✗ Disarming too early, dropping it short
- ✗ Sliding into something you did not scout

VARIATIONS

Slide to Perch

Slide up onto a low ledge and stop.

Power Slide

Carry more speed into a longer slide.

RELATED TRICKS

Perch

Setting down on top of an object.

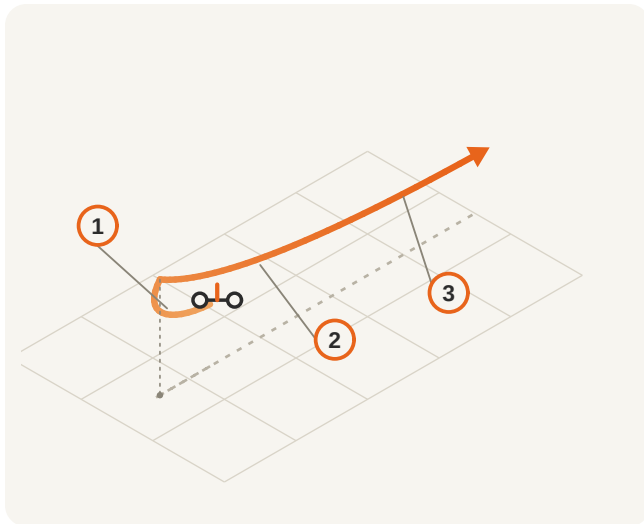
Gaps

Another committed, line-based move.

Slingshot

Novice

Open Air



HOW TO

Load up and fire forward, like drawing and releasing a slingshot. Pull back and up to load, coiling your energy, then release it all into a fast forward line and ride the acceleration out. The bigger the load, the harder the launch.

MASTERY LEVELS

- ★☆☆ Load and release into a fast forward line.
- ★★★☆☆ Smooth load into a snappy release.
- ★★★★★ Explosive slingshot, all the load turned to speed.

STEP BY STEP CONTROL INPUTS

1  
Load by pulling back and up

2  
Release into a fast forward line

3  
Ride the acceleration out

PREREQUISITES & WHERE TO PRACTICE

- Confident pitch and throttle
- A feel for coiling and releasing energy

Practise the load, a smooth pull back and up, then release cleanly into a forward punch. Build the load gradually. It is about turning stored energy into a clean launch.

COMMON MISTAKES

- ✗ A messy load that wastes energy
- ✗ Releasing off-line
- ✗ Not committing to the launch

VARIATIONS

Face Punch

A pure vertical punch-out.

Power Flip

A flip with a power exit.

RELATED TRICKS

Face Punch

A vertical punch-out.

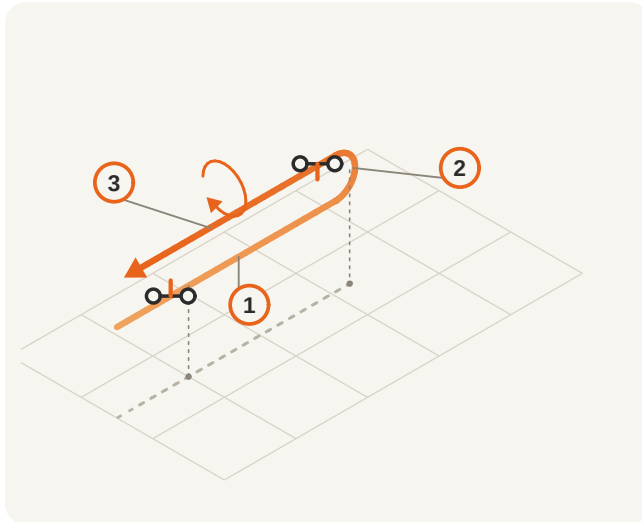
Power Flip

A flip with a power climb.

Snapback

Novice

Open Air



HOW TO

A quick turn-around that reverses your direction. Carry speed into a straight line, snap a 180 pitch, a half flip up and over, so you pitch back on yourself and end up inverted flying the other way, then immediately roll 180 to come upright, now tracking back the way you came. The pitch reverses you and the roll sets you level.

MASTERY LEVELS

- ★☆☆ Reverse with a half flip and roll upright, flying back.
- ★★★ Crisp pitch-and-roll turn-around with a clean exit.
- ★★★★ Sharp 180 pitch into a snappy 180 roll, dead on the line back.

STEP BY STEP CONTROL INPUTS



Commit to a straight line with speed



Snap a 180 pitch, a half flip over the top



Roll 180 to come upright, flying back

PREREQUISITES & WHERE TO PRACTICE

- A confident half flip and a clean half roll
- Comfort chaining a pitch straight into a roll

Open airspace with height to spare. Commit to the line, then snap the 180 pitch and roll upright in one motion. Practise the pitch and the roll separately first, then blend them so there is no pause between.

COMMON MISTAKES

- ✗ Rolling before the pitch flip has fully reversed you
- ✗ A half-hearted pitch, so you do not turn all the way around
- ✗ A sloppy roll that leaves you off-level or off the line

VARIATIONS

Snapback Rewind

Continue the reverse into a full rewind.

Power Snap

Add throttle to snap over a taller line.

RELATED TRICKS

Matty Flip

A half-flip reversal, done over a gap.

Flip

The rotation a snapback stops halfway.

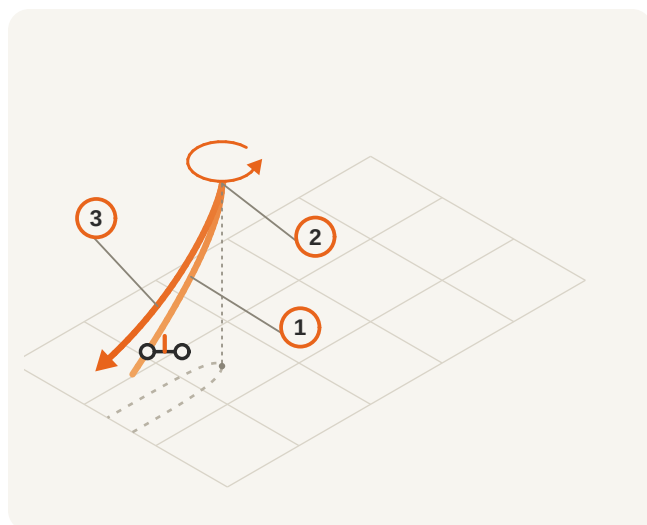
Split S

Another quick reversal, with a roll entry.

Stall 360 Rewind

Novice

Rewinds



HOW TO

A stall rewind with a flat spin at the top, flown facing up. Climb straight up and pitch to a stall facing up, then roll a full 360 while you hang there nose-up, and blip the throttle to rewind back down the line you climbed. Because you are facing straight up, the 360 is a roll rather than a yaw, and it reads as a flat spin from the ground.

MASTERY LEVELS

- ★☆☆ Stall facing up, roll a 360, and rewind down.
- ★★★☆☆ Clean nose-up stall into a flat 360 roll and rewind.
- ★★★★★ Crisp vertical stall, pinned 360 roll, tight rewind.

STEP BY STEP CONTROL INPUTS



1 Climb vertically to a stall facing up



2 Spin a flat 360 roll at the top



3 Blip and rewind back down

PREREQUISITES & WHERE TO PRACTICE

- A solid stall rewind
- A flat 360 roll while facing up

Get the vertical stall and rewind dialed, then add the 360 roll at the top while you hang facing up. Keep it pinned on the spot and the climb straight so the rewind tracks true.

COMMON MISTAKES

- ✗ Drifting off the spot during the roll
- ✗ Not fully stalling before the 360
- ✗ A harsh blip that throws the rewind

VARIATIONS

Stall Rewind

Without the 360.

Stall Flip Rewind

A flip at the stall instead.

RELATED TRICKS

Stall Rewind

The base stall-and-rewind.

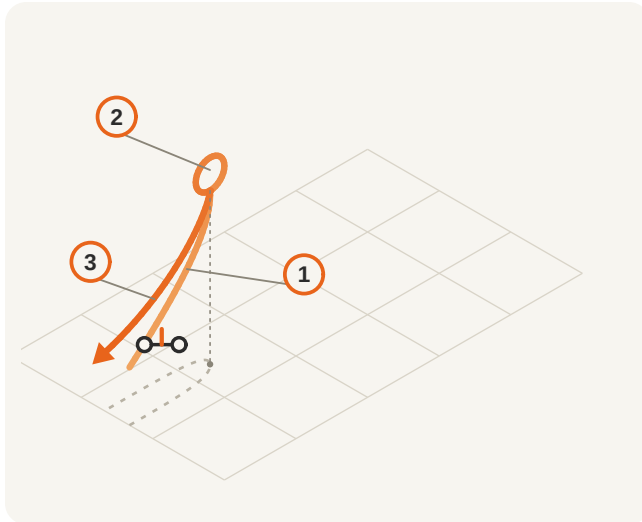
Barrel Roll

The 360 roll at the top.

Stall Flip Rewind

Novice

Rewinds



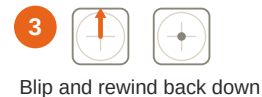
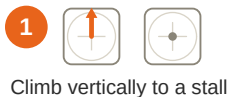
HOW TO

A stall rewind with a flip at the top. Climb straight up to a stall, throw a quick flip as you hang at the peak, then blip the throttle to rewind back down the line you climbed. The flip adds a flourish to the reversal.

MASTERY LEVELS

- ★☆☆ Stall, flip, and rewind back down.
- ★★★☆☆ Clean stall into a tidy flip and rewind.
- ★★★★★ Crisp vertical stall, sharp flip, tight rewind.

STEP BY STEP CONTROL INPUTS



PREREQUISITES & WHERE TO PRACTICE

- A solid stall rewind
- A quick flip at low speed

Get the vertical stall and rewind dialed, then add the flip at the top once you are comfortable hanging there. Keep the climb straight so the rewind tracks.

COMMON MISTAKES

- ✗ Flipping before you reach the stall
- ✗ Drifting off vertical
- ✗ A harsh blip that throws the rewind

VARIATIONS

Stall Rewind

Without the flip.

Stall 360 Rewind

A yaw 360 at the stall instead.

RELATED TRICKS

Stall Rewind

The base stall-and-rewind.

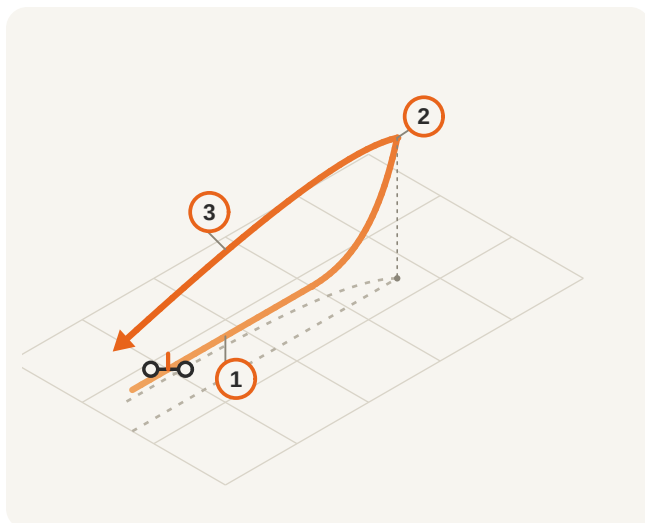
Flip

The rotation at the top.

Throwback

Novice

Rewinds



HOW TO

A throwback: throw the quad out and away, then snap it back on a rewind, as if it were on a bungee. Come in level near the ground and pitch back so it climbs out, and once you are facing the sky blip the throttle, then quickly drop it and pitch down to look back at where you punched out of. Float back with the momentum. The illusion is in the sudden reversal at the far end.

MASTERY LEVELS

- ★☆☆ Throw it out and snap it back.
- ★★★☆☆ Clean throw to the far point, tidy rewind back.
- ★★★★★ Big throw, crisp snap-back, dead on line.

STEP BY STEP CONTROL INPUTS

1



Approach level, then pitch the drone back

2



Facing the sky, blip throttle, then drop it and pitch down to look back

3



Float backwards with momentum

PREREQUISITES & WHERE TO PRACTICE

- A confident stall
- A feel for the stall-and-rewind

Come in level and low, pitch back to climb out, and time the blip so it reads as a sudden reversal at the top. Keep the line straight so it floats back the way it came.

COMMON MISTAKES

- ✗ A soft reversal that reads as a turn, not a snap
- ✗ Drifting off the throw line
- ✗ Blipping too early, before the far point

VARIATIONS

Stall Rewind

A vertical rewind.

Sideways Rewind

A rewind thrown to the side.

RELATED TRICKS

Stall Rewind

The core rewind.

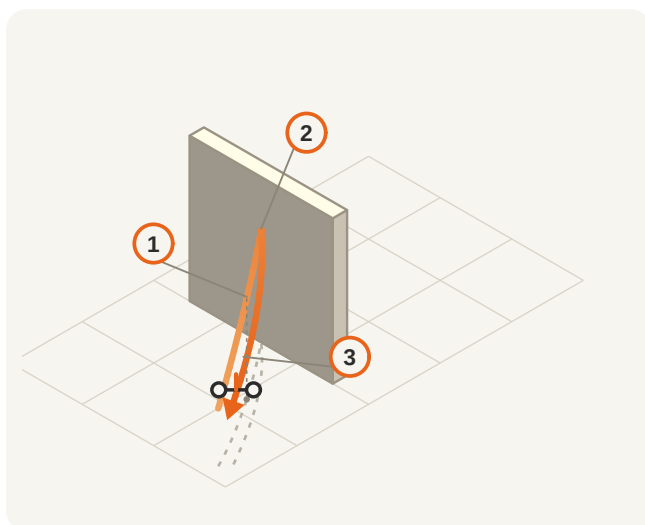
Boomerang

Another out-and-back.

Wall Bonk

Novice

Wall Tricks



HOW TO

A deliberate, harder tap, bonking the wall head-on and bouncing straight back off it. Line up square with the face, commit into the bonk, and ride the bounce back the way you came. It is a bolder cousin of the wall tap, more contact and more commitment.

MASTERY LEVELS

- ★☆☆ Bonk the wall and bounce back under control.
- ★★★☆☆ Clean, controlled bonk with a steady bounce-back.
- ★★★★★ Confident bonk on a chosen spot, tidy recovery.

STEP BY STEP CONTROL INPUTS



Line up head-on with the wall



Bonk the face



Bounce straight back off

PREREQUISITES & WHERE TO PRACTICE

- A confident wall tap
- Nerve to commit into a surface

Start with light bonks on a clean, flat wall and build up the contact as your control improves. Square up before you commit, and be ready to catch the bounce.

COMMON MISTAKES

- ✗ Coming in at an angle and glancing off
- ✗ Bonking too hard and tumbling
- ✗ Not catching the bounce-back

VARIATIONS

Wall Tap

A lighter touch and peel.

Downtown Tap

A roll-and-pitch tap with a flip out.

RELATED TRICKS

Wall Tap

The lighter tap.

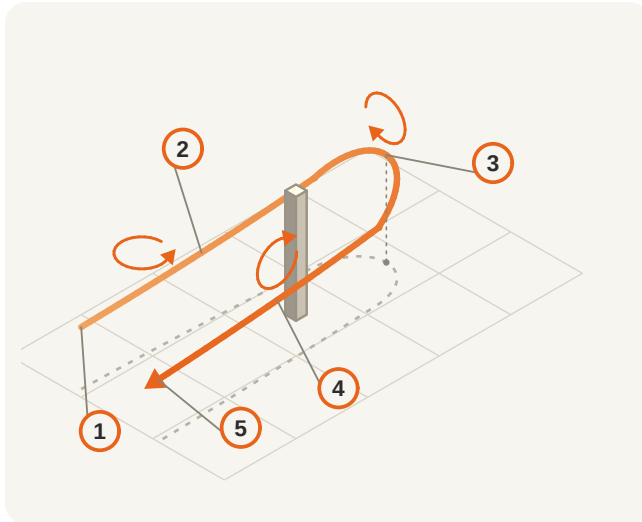
Wall Ride

Riding the face instead.

Alley-Oop

Intermediate

Open Air



HOW TO

A matty flip set up off the side of an object. Approach along the side and, just past it, yaw and roll the quad up until it is facing straight up. Settle it level, then start a matty, rolling as you flip up and over the top, and ride it out flying backwards down the far side. It flows a stall-style pop-up straight into a matty for a big, showy exit.

MASTERY LEVELS

- ★☆☆ Pop up beside the object and matty out backwards.
- ★★★☆☆ Clean pop-up to face up, smooth matty over the top.
- ★★★★★ Crisp pop-up and a tight matty, floated out backwards.

STEP BY STEP CONTROL INPUTS

- | | | | | |
|---|---|---------------------------------------|---|---|
| 1
Approach the side; just past it, yaw left and roll up | 2
Once facing up, pop the sticks back | 3
Once level, start a matty | 4
Roll and continue the matty, then pitch level | 5
Level out and fly backwards |
|---|---|---------------------------------------|---|---|

PREREQUISITES & WHERE TO PRACTICE

- A confident matty flip
- A feel for popping up to face up

Dial the pop-up to facing up off the side of an object first, then add the matty over the top once it settles level. Give yourself height and room, and keep it out to the side so you never drift into the object.

COMMON MISTAKES

- ✗ Rushing the matty before the quad is level
- ✗ Not popping up far enough to face straight up
- ✗ Drifting into the object on the way up

VARIATIONS

Front Swing

The side pop-up without the matty.

Matty Flip

The matty on its own.

RELATED TRICKS

Matty Flip

The flip at the heart of it.

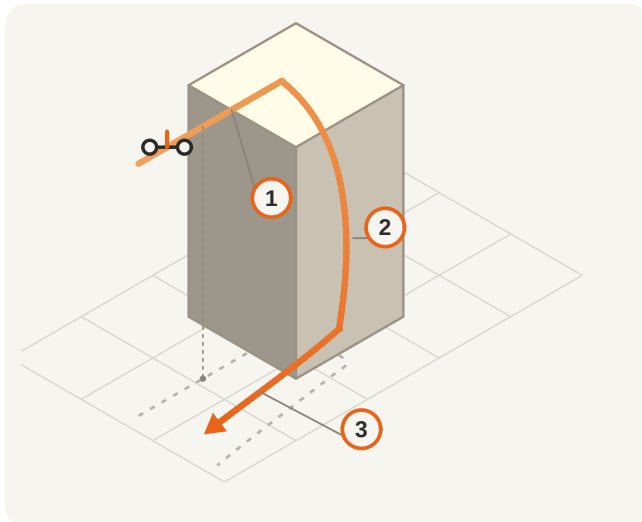
Front Swing

The side pop-up it builds on.

Backwards Dive

Intermediate

Open Air



HOW TO

A building dive flown tail-first. Approach the top of the building head-on and, as you reach the edge, cut all your speed and drop straight down the face, pitching up to look back up the structure so you fall tail-first. Catch the fall near the bottom and keep flying backwards, away from the building the way you came in.

MASTERY LEVELS

- ★☆☆ Drop the face tail-first and back away clean.
- ★★★ Controlled backward dive down the face, smooth catch.
- ★★★★ Committed tail-first dive tight to the face, floated out backwards.

STEP BY STEP CONTROL INPUTS



1 Approach the top of the object to dive



2 Nearing the edge, cut all speed and drop facing up



3 Catch the fall and continue backwards, away from the object

PREREQUISITES & WHERE TO PRACTICE

- A confident building dive
- A feel for backward flight

Dial a forward building dive and backward flight on their own first, then join them: approach the top, cut all speed, and let it fall down the face while you keep it pointed up the structure. Start on a shorter building with plenty of room at the bottom to catch the fall.

COMMON MISTAKES

- ✗ Carrying speed into the drop instead of cutting it
- ✗ Losing the face and drifting off the building
- ✗ Catching the fall too late

VARIATIONS

Building Dive

The forward version down the face.

Backward Flying

The backward line you ride out on.

RELATED TRICKS

Building Dive

The forward version down a building.

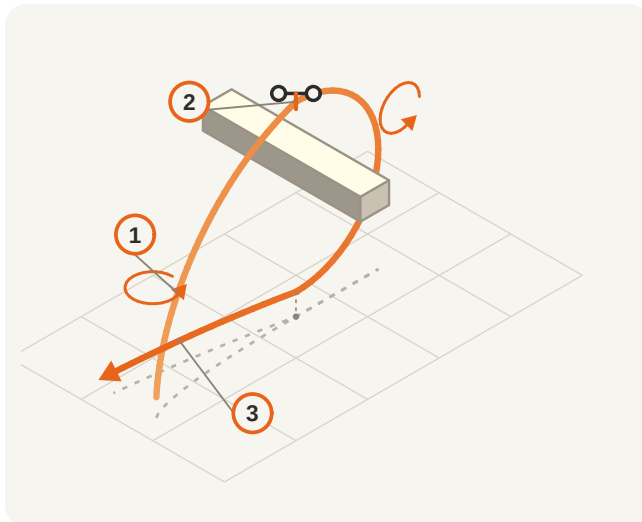
Backward Flying

The backward flight it rides out on.

Blind Loop

Intermediate

Miscellaneous



HOW TO

A loop around an object that closes on itself, flown with a 180 yaw so you spend the loop facing backwards, blind to where you are going. Start low and climb toward the top, spinning a 180 to face back the way you came. Near the top, still backwards, throttle up and pitch up to carry the loop over and down to the bottom, coming out facing where you started. Ride the last of the line back to your start point, finishing exactly where you began.

MASTERY LEVELS

- ★☆☆ Loop the object backwards and finish where you started.
- ★★★☆☆ Smooth blind loop, closing on the start point.
- ★★★★★ Tight, controlled loop flown backwards, dead on the start.

STEP BY STEP CONTROL INPUTS



1 Start low, approach the top with a 180 yaw to face backwards



2 Near the top, throttle up and pitch up to complete the loop



3 Continue the path back to the start point

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- Comfort flying backwards, on feel

Get a powerloop dialed first, then add the 180 yaw so you carry the loop facing backwards. Fly it on feel and aim to finish right where you began. Give yourself room and a clear object to loop around.

COMMON MISTAKES

- ✗ Losing the yaw and drifting off the loop
- ✗ Ballooning the loop and missing the start point
- ✗ Panicking once you are facing backwards

VARIATIONS

Powerloop

A loop over an obstacle.

Maverick Loop

A loop under an obstacle.

RELATED TRICKS

Powerloop

A loop over an obstacle.

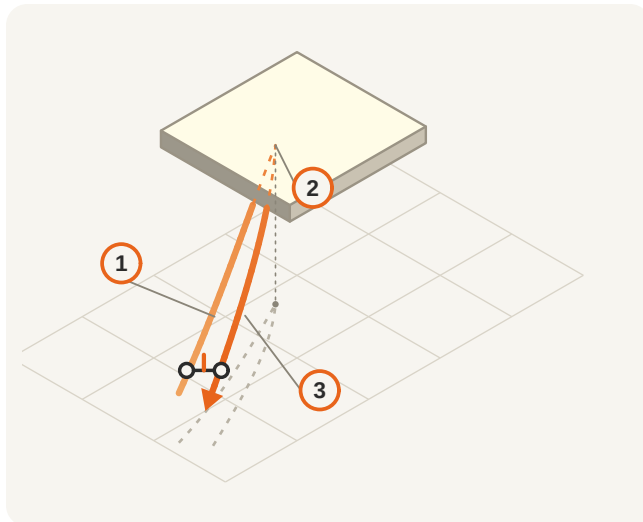
Flip

The loop it is built on.

Ceiling Tap

Intermediate

Wall Tricks



HOW TO

Tapping an overhead surface from below. Approach up from one side to kiss the underside of a ceiling, overhang, or beam with a light touch, then cut throttle and return back down the way you came. Like a wall tap, it is all about a soft touch and a controlled exit, just pointed up instead of across.

MASTERY LEVELS

- ★☆☆ Tap the ceiling and drop away under control.
- ★★★☆☆ Light, clean tap with a smooth drop-down.
- ★★★★★ Precise taps on an overhead surface, no bounce.

STEP BY STEP CONTROL INPUTS



Approach up from the left to the object



Tap the underside



Return back down the way you came

PREREQUISITES & WHERE TO PRACTICE

- Confident vertical control
- A soft, controlled touch

Find a solid overhang or ceiling with clear air below. Start with the gentlest taps and a controlled drop, building up as your vertical control improves. Scout the surface first.

COMMON MISTAKES

- ✗ Punching up too hard and bouncing off
- ✗ Drifting sideways into the surface
- ✗ Dropping out of control after the tap

VARIATIONS

Wall Tap

Tap a vertical wall instead.

Downtown Tap

A tap with a roll and a flip out.

RELATED TRICKS

Wall Tap

The same touch on a vertical wall.

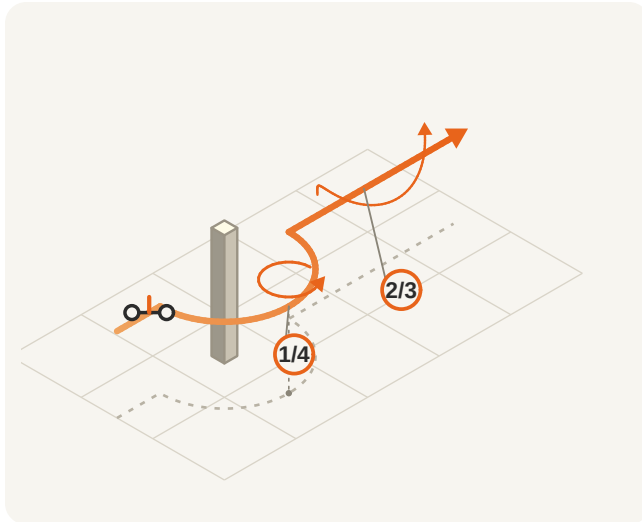
Loop Tap

A tap with a rotation added.

Coin Drop

Intermediate

Miscellaneous



HOW TO

A Vanny Roll flown out and then rewound, like winding a coin back up. Approach the obstacle yaw-tracking it through 180, blip the throttle and cross with a coordinated 360 roll, then rewind the roll and rewind the yaw to come back the way you came. It is a Vanny Roll and its mirror image, back to back.

MASTERY LEVELS

- ★☆☆ Track, roll across, and rewind back clean.
- ★★★☆☆ Smooth 360 roll and a tidy rewind of roll and yaw.
- ★★★★ Coordinated track, roll, and full rewind, dead on line.

STEP BY STEP CONTROL INPUTS



Approach the obstacle, yaw-spin tracking it 180



Throttle blip and cross with a coordinated 360 roll



Rewind the roll



Rewind the yaw

PREREQUISITES & WHERE TO PRACTICE

- A confident Vanny Roll
- Coordinated roll and yaw

Get the Vanny Roll dialed first, then practise rewinding it: after the roll and cross, unroll and unyaw back to where you started. Keep the roll and yaw coordinated in both directions.

COMMON MISTAKES

- ✗ Losing coordination on the rewind
- ✗ Not fully rewinding the roll or the yaw
- ✗ Drifting off the line coming back

VARIATIONS

Vanny Roll

The one-way version without the rewind.

Barrel Roll

The coordinated roll on its own.

RELATED TRICKS

Vanny Roll

The one-way version this rewinds.

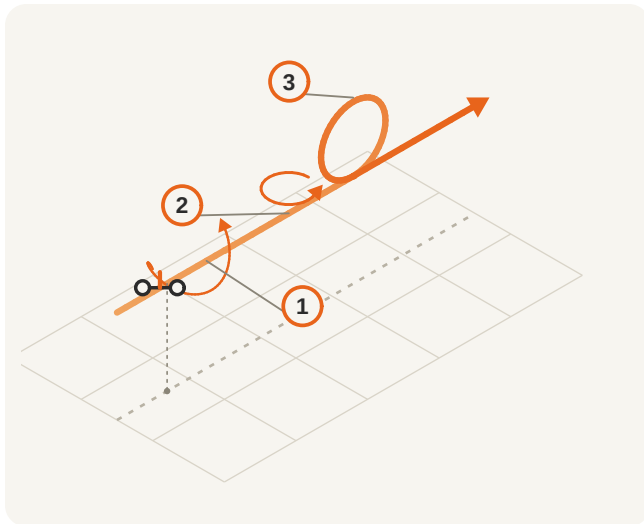
Barrel Roll

The coordinated roll at its core.

Cubik's Rube

Intermediate

Open Air



STEP BY STEP CONTROL INPUTS



Roll out first



Pin a flat yaw



Flip to finish

PREREQUISITES & WHERE TO PRACTICE

- Clean rolls, yaw spins, and flips
- A feel for chaining axes

Have each axis solid, then chain them roll-yaw-flip on one line, keeping it straight through all three. Smooth the hand-offs between the turns.

COMMON MISTAKES

- ✗ Drifting off line between the turns
- ✗ Rushing one axis into the next
- ✗ An uneven turn on one of the axes

VARIATIONS

Rubik's Cube

The same axes flip-first.

Rubik's Cube Pendulum

The combo on a pendulum swing.

RELATED TRICKS

Rubik's Cube

The flip-first order.

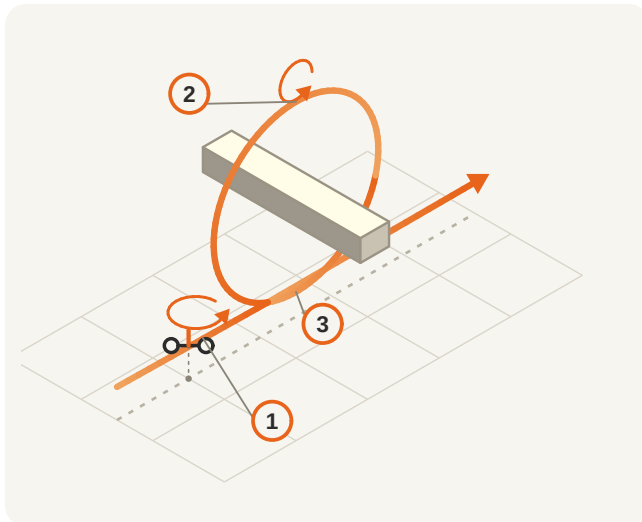
Rubik's Cube Pendulum

On a pendulum.

Double Dutch

Intermediate

Jump Ropes



HOW TO

A jump rope with a reverse 360 roll thrown in at the top. Yaw sideways and enter beneath the object as for a jump rope, throttle up and, once at the top of the loop, snap a reverse 360 roll, then continue looping back down and under the object. It is a jump rope dressed up with a roll over the top.

MASTERY LEVELS

- ★☆☆ Loop the object and land a reverse 360 roll up top.
- ★★★ Smooth jump rope with a clean reverse roll over the top.
- ★★★★ Tight loop and a crisp reverse 360, back down clean.

STEP BY STEP CONTROL INPUTS



90° flat yaw spin, enter under the object sideways



Throttle up and, at the top, perform a reverse 360 roll



Continue to loop back down and under the object

PREREQUISITES & WHERE TO PRACTICE

- A solid jump rope
- A quick, controllable roll

Get the jump rope dialed first, then add a reverse 360 roll right at the top of the loop, staying over the object. Keep the loop tight so the roll has room to complete before you come back down.

COMMON MISTAKES

- ✗ Rushing the roll before the top
- ✗ Losing the loop line during the roll
- ✗ Not completing the 360 before coming down

VARIATIONS

Jump Rope

The same loop without the roll.

Cinnamon Roll

A roll folded into the swing instead.

RELATED TRICKS

Jump Rope

The loop it builds on.

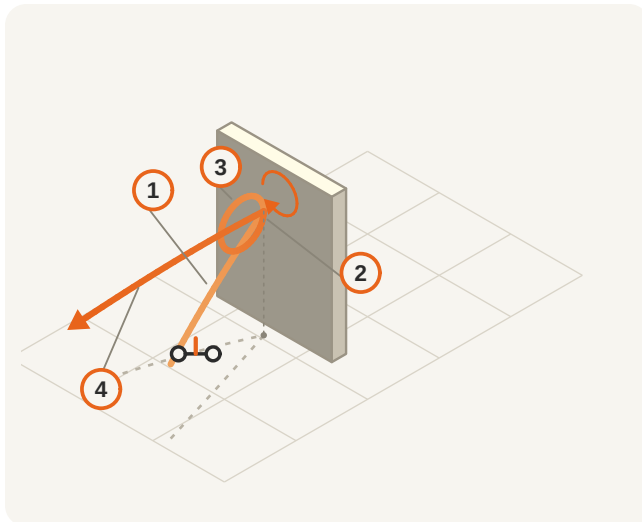
Cinnamon Roll

A jump rope with a roll in the swing.

Downtown Tap

Intermediate

Wall Tricks



HOW TO

Tapping an object while flipped away from it. Fly at the object and, as you reach it, roll 180 and pitch back 90 at the same time so you end up looking down and away from it, and tap it in that orientation. Straight after the tap, roll into a smooth front flip, then fly swiftly away to finish.

MASTERY LEVELS

- ★☆☆ Roll-and-pitch into the tap, front flip, and away.
- ★★★ Clean reorientation, tidy tap, smooth front flip out.
- ★★★★ Committed roll-pitch tap and a crisp front flip, gone.

STEP BY STEP CONTROL INPUTS



Fly toward the object



Roll 180 and pitch back 90 to face down and away, and tap



Perform a front flip



Swiftly fly away

PREREQUISITES & WHERE TO PRACTICE

- Comfortable multi-axis rotations
- A confident front flip

Break it down: practise the combined 180 roll and 90 pitch-back on its own first, then add a light tap, then the front flip out. Give yourself room off the object and build the pieces up slowly.

COMMON MISTAKES

- ✗ Not committing the roll and pitch together
- ✗ Tapping too hard and knocking off line
- ✗ Rushing the front flip before you are clear

VARIATIONS

Wall Tap

A plain tap, no reorientation.

Matty Flip

The flip on its own.

RELATED TRICKS

Wall Tap

The basic tap this dresses up.

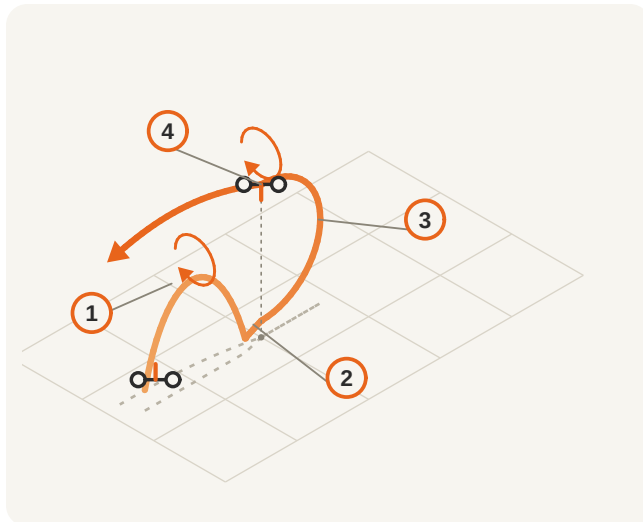
Flip

The front flip in the exit.

Flat Bounce Rewind

Intermediate

Rewinds



HOW TO

A roll and a loop stitched together low to the ground. Punch a roll and let it dish down toward the deck, correct the roll just before the ground, then punch a forward loop back up and, once you come around to upside down, roll out to finish. It drops to the deck and rewinds up and away.

MASTERY LEVELS

- ★☆☆ Roll to the deck, correct, loop up, and roll out.
- ★★★ Clean roll and correction, smooth loop and roll-out.
- ★★★★ Tight roll to the deck and a crisp loop, rolled out inverted.

STEP BY STEP CONTROL INPUTS



Punch a roll



Correct it near the ground



Punch a forward loop



Once upside down, roll out

PREREQUISITES & WHERE TO PRACTICE

- A confident roll and correction
- A punchy forward loop

Break it into pieces: a roll that dishes toward the deck and corrects near the ground, then a punched forward loop with a roll-out once you hit inverted. Start high and bring the bounce point down as it cleans up.

COMMON MISTAKES

- ✗ Not correcting the roll before the ground
- ✗ A weak loop punch that stalls out
- ✗ Rolling out before you are fully upside down

VARIATIONS

Stall Rewind

A rewind off a stall instead of the ground.

Barrel Roll

The roll on its own.

RELATED TRICKS

Stall Rewind

Another way to rewind.

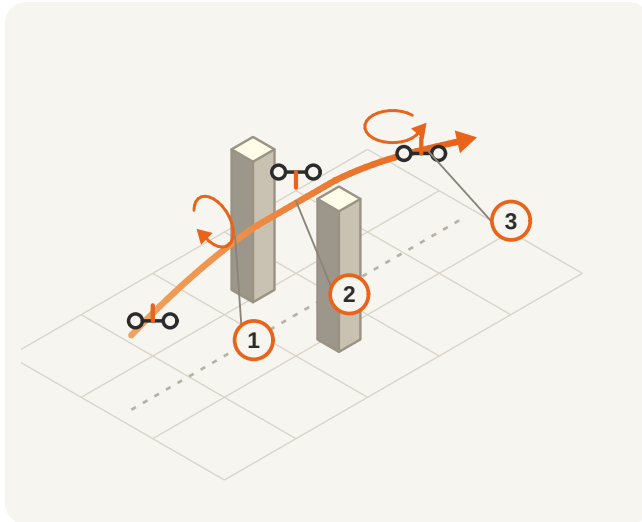
Flip

The forward loop it punches.

Half Matty

Intermediate

Matty Flips



HOW TO

A half matty flown through a gap. Approach the gap between two objects and start a cross-coordinated half roll to inverted. Once you come around upside down, pitch forward and feed in throttle to bring the quad back upright but now facing backwards, then keep going backwards through the gap and yaw out once you are past it. It flips your heading without turning around, and the line keeps going rather than doubling back.

MASTERY LEVELS

- ★☆☆ Half roll through the gap and out backwards.
- ★★★☆☆ Clean roll to inverted, smooth reorient, tidy yaw out.
- ★★★★★ Crisp cross-coordinated half roll, backwards through, dialed yaw out.

STEP BY STEP CONTROL INPUTS



Approach the gap and start a cross-coordinated half roll



Upside down, pitch forward and throttle up to face backwards



Continue backwards through the gap and yaw out past it

PREREQUISITES & WHERE TO PRACTICE

- A clean, controllable half roll
- Comfort inverted and flying backwards

Find a gap you can fly through. Get the cross-coordinated half roll to inverted smooth first, then the pitch-and-throttle reorient to upright-backwards, and finish with a clean yaw out once you are past the gap. Build it up slowly.

COMMON MISTAKES

- ✗ Rushing the half roll before the gap
- ✗ Not reorienting fully before you reach the gap
- ✗ Yawing out too early, before you are past it

VARIATIONS

Stall Half Matty

Hold a longer stall at the top before descending.

Half Matty Rewind

Rewind back over the top instead of passing under.

RELATED TRICKS

Matty Flip

The full version with a forward flip over the top.

Split S

The same reverse-and-descend, without the object.

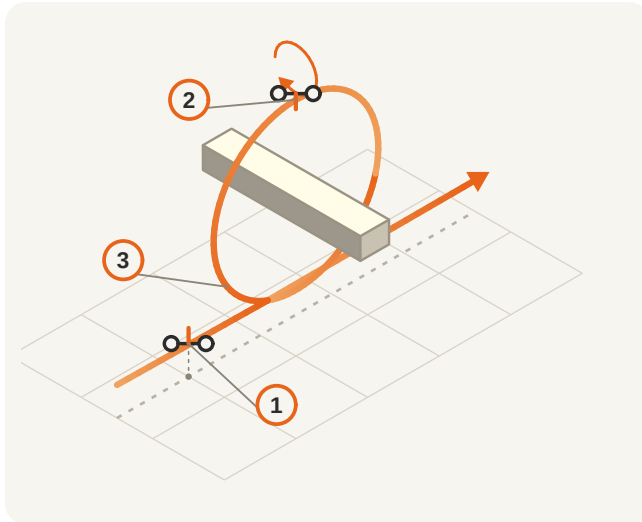
Snapback

Another half-rotation reversal.

Immelloop

Intermediate

Powerloops



STEP BY STEP CONTROL INPUTS



Powerloop up and over the object



At the top upside down, 180 roll to upright, facing where you started



Keep backwards, pitch back slightly for momentum through the bottom

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A clean 180 roll

Have the powerloop dialed, then add a 180 roll right at the inverted apex so you come upright but facing backwards. Keep the loop shape and let your momentum carry you backwards through the bottom. Give yourself room.

COMMON MISTAKES

- ✗ Rolling before you reach the inverted apex
- ✗ Losing the loop shape during the roll
- ✗ Not keeping the backwards momentum through the bottom

VARIATIONS

Power Switch

A yaw switch at the top instead of a roll.

Powerloop

The same loop without the apex roll.

RELATED TRICKS

Powerloop

The loop it is built on.

Power Switch

A powerloop that exits turned around, via yaw.

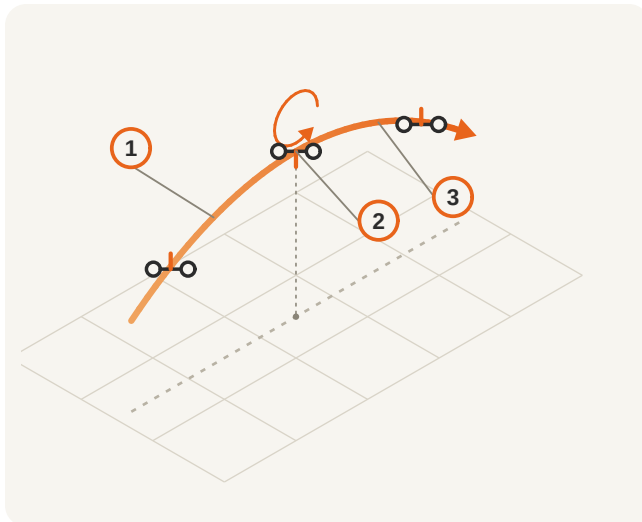
Matty Flip

Another over-the-top move.

Inverted Lookback

Intermediate

Open Air



HOW TO

An inverted hang built from a front roll, with no yaw at all. Gain a little forward speed, then blip the throttle and pitch into a front roll to bring the quad inverted, and pause there to hang upside down, looking back the way you came. When you are ready, finish the roll to come back upright and carry on forward. It is a front roll with a hang stalled into the middle of it.

MASTERY LEVELS

- ★☆☆ Flip to inverted, hang, and roll out forward.
- ★★★ Smooth flip into a clean inverted hang, tidy exit.
- ★★★★ Controlled front roll with a held inverted hang, dead level out.

STEP BY STEP CONTROL INPUTS



Gain forward momentum, then blip throttle and invert



Once upside down and looking back, hang for a bit



Finish the forward roll to continue moving forward

PREREQUISITES & WHERE TO PRACTICE

- A controllable front roll
- A feel for the inverted hang

Get a smooth front roll first, then learn to pause it at the inverted point and hold the hang before completing the rotation. Start high with plenty of room and build up the hang time as it gets steadier.

COMMON MISTAKES

- ✗ Rushing through the inverted point instead of hanging
- ✗ Losing the forward line during the hang
- ✗ Not completing the roll cleanly on the exit

VARIATIONS

Juicy Flick

A pop to inverted and a hang.

Matty Flip

The full front flip without the hang.

RELATED TRICKS

Juicy Flick

A blip to inverted with a brief hang.

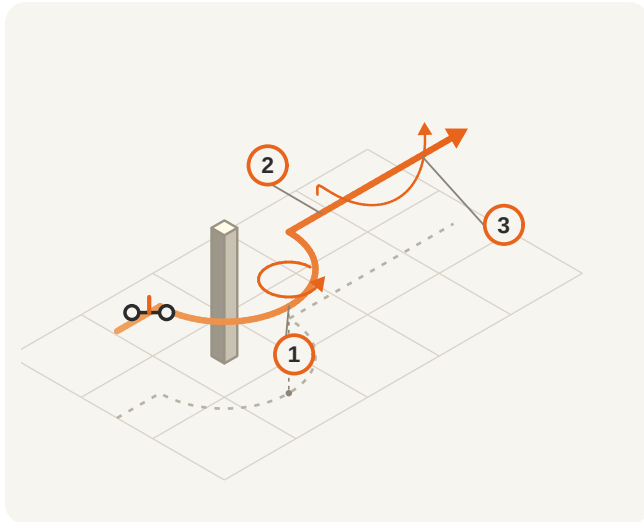
Matty Flip

A full front flip.

Juicy Cartwheel

Intermediate

Open Air



HOW TO

Yaw-spin around an object, then blip and roll out of it. Approach the object and yaw-spin all the way around it, and once you have come back around, blip the throttle and throw a roll. Balance the throttle through the roll so you complete it cleanly on the line. Marrying the yaw-spin into a coordinated roll is what gives it the juicy, flowing feel.

MASTERY LEVELS

- ★☆☆ Yaw-spin an object and roll out of it.
- ★★★ Smooth yaw-spin into a clean, floaty roll.
- ★★★★ Dialed both directions, on line, object framed.

STEP BY STEP CONTROL INPUTS



1 Approach object and yaw spin around it



2 Once around the object, blip throttle and roll



3 Slowly balance speed to complete the roll

PREREQUISITES & WHERE TO PRACTICE

- A clean yaw spin around an object
- A clean, coordinated roll

Get the yaw-spin around an object dialed first, then practice blipping and rolling out of it. Coordinate a little throttle through the roll so it floats and finishes on the line rather than dropping. Both directions.

COMMON MISTAKES

- ✗ Losing the object out of frame through the yaw spin
- ✗ Rushing the roll instead of balancing the throttle
- ✗ Dropping altitude or drifting off the line

VARIATIONS

Side Loop

A sideways loop around an object.

Barrel Roll

A corkscrew roll along the line.

RELATED TRICKS

Coin Drop

The same yaw-and-roll path, rewound.

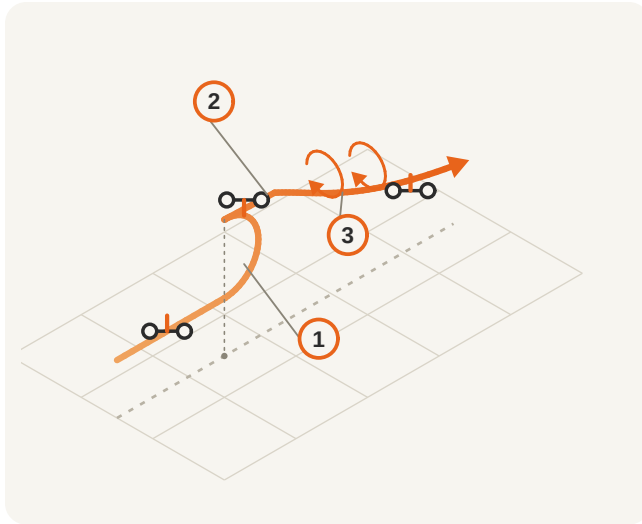
Roll

The axis it finishes on.

Juicy Double Roll

Intermediate

Open Air



HOW TO

A Juicy Flick finished with a quick double roll instead of a slow roll-out. From a cruising line, blip the throttle to unload the quad and snap a 180 pitch up to inverted, hang there floaty for a beat, then rip a quick 540 roll to recover forward on the same line. The contrast between the lazy hang and the snappy double roll is what sells it.

MASTERY LEVELS

- ★☆☆ Pop to inverted and double roll out on the same line.
- ★★★☆☆ Floaty flick into a crisp, even double roll.
- ★★★★ Effortless flick and a snappy 540 that lands dead on line.

STEP BY STEP CONTROL INPUTS



Blip the throttle and snap a 180 pitch to inverted



Hang inverted, floaty, for a beat



Do a quick "double" (540) roll

PREREQUISITES & WHERE TO PRACTICE

- A clean Juicy Flick
- An even, controlled double roll

Have the flick and the double roll each solid on their own, then join them: keep the hang lazy and let the 540 be the snappy part. Give yourself height and check the exit stays on the original line.

COMMON MISTAKES

- ✗ Starting the roll before the hang has settled
- ✗ An uneven 540 that stops off-axis
- ✗ Losing the line, so you recover off-track

VARIATIONS

Juicy Flick

The same flick with a slow single roll-out.

Double Roll

The double roll on its own.

RELATED TRICKS

Juicy Flick

The flick it is built on.

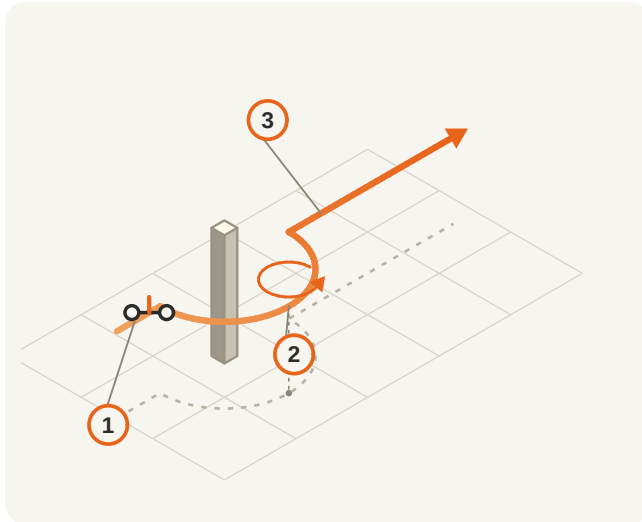
Double Roll

The 540 that finishes it.

Juicy Tracking-Lookback

Intermediate

Open Air



HOW TO

A tracking lookback with a flick through the middle.

Approach the object yawing past it at an angle so it stays centered, and as you pass, flick-loop the sticks and carry the yaw through with the opposite angle. Then level out, still looking back at the object. Holding the frame through the flick and the yaw reversal is what gives it the juicy feel.

MASTERY LEVELS

- ★☆☆ Yaw-track past an object and look back.
- ★★★ Smooth angled track with a clean flick and lookback.
- ★★★★ Buttery tracking lookback, object framed the whole way.

STEP BY STEP CONTROL INPUTS



1 Approach object, yawing past it at an angle, keeping it centered



2 Passing the object, flick loop clockwise, yaw the opposite angle



3 Level out looking back at the object

PREREQUISITES & WHERE TO PRACTICE

- A smooth yaw-track around an object
- A clean flick loop

Track an object with yaw as you pass at an angle, keeping it centered, then work the flick and the yaw reversal, leveling out with the object still framed behind you. Keep it floaty and give the object room.

COMMON MISTAKES

- ✗ Losing the object out of frame
- ✗ A jerky flick that breaks the frame
- ✗ Not carrying the yaw through the reversal

VARIATIONS

Juicy Cartwheel

The same path, rolled out.

Inverted Lookback

A lookback flown inverted.

RELATED TRICKS

Juicy Cartwheel

The same yaw-spin path.

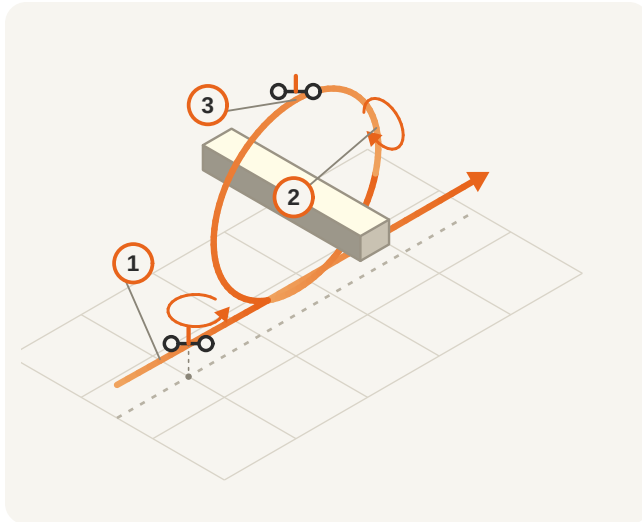
Inverted Lookback

A lookback flown inverted.

Jump Rope

Intermediate

Jump Ropes



HOW TO

A loop around a horizontal object flown flat and sideways: the quad yaws 90 degrees to face sideways and rolls its way around, staying level the whole time instead of pitching inverted like a powerloop. Yaw and enter beneath the object facing sideways, throttle up and roll to carry it up and around, and cut throttle and correct the roll as you pass back over the top. Because it rolls rather than pitches, it never goes inverted.

MASTERY LEVELS

- ★☆☆ Loop around the object flat and sideways.
- ★★★ Smooth, level loop, sideways the whole way round.
- ★★★★ Tight, level jump rope, dead sideways, clean over the top.

STEP BY STEP CONTROL INPUTS



1 90° flat yaw spin, enter under the obstacle sideways



2 Throttle up and roll right to loop around



3 Pass over the top, cut throttle and correct roll

PREREQUISITES & WHERE TO PRACTICE

- Coordinated throttle and roll
- Comfortable flying sideways (yawed 90°)

Find a horizontal object, a branch, bar, or gap edge. Yaw sideways, drop beneath it, and roll up and around it, keeping the quad level rather than letting it pitch inverted. Cut throttle as you clear the top. Build the loop up slowly.

COMMON MISTAKES

- ✗ Letting it pitch inverted like a powerloop
- ✗ Drifting off the sideways heading
- ✗ Carrying throttle over the top instead of cutting it

VARIATIONS

Burrito Roll

Add a roll and go under on the finish.

Double Dutch

A jump rope with a reverse 360 roll at the top.

RELATED TRICKS

Powerloop

The inverted, pitching cousin.

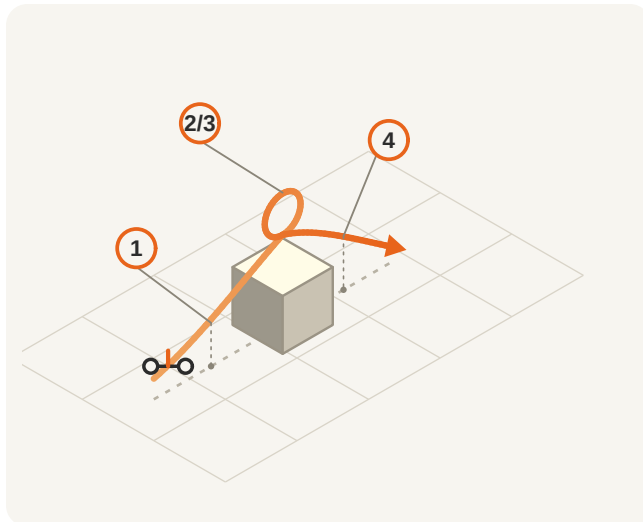
Burrito Roll

A jump rope with a roll and an under-pass.

Laser Flip

Intermediate

Open Air



HOW TO

A knife-edge yaw spin over an object. Climb over the object gaining height, and as you reach the top, twist to point the nose straight down at it. On that knife edge, feed in a coordinated yaw and spin around the vertical, then roll away and descend out. Pointing dead at the object and holding the knife edge through the spin is what makes it hard.

MASTERY LEVELS

- ★☆☆ Spin on a knife edge over an object.
- ★★★ Clean knife-edge yaw spin, nose on the object.
- ★★★★ Dialed laser flip, pinned on the object, smooth out.

STEP BY STEP CONTROL INPUTS



Approach object gaining height to pass over it



Approaching the top, twist into facing the object



Facing down at it, coordinated yaw spin on a knife edge



Completing the vertical yaw spin, roll away

PREREQUISITES & WHERE TO PRACTICE

- A confident knife edge
- A clean, coordinated yaw spin

Pick an object, climb over it, and practise twisting to point straight down at it. Feed in the yaw to spin around the vertical on the knife edge, keeping the nose on the object, then roll away. Start high and give yourself room.

COMMON MISTAKES

- ✗ Not getting the nose fully onto the object
- ✗ Falling off the knife edge mid-spin
- ✗ Losing too much height before you roll away

VARIATIONS

Knife Edge

Holding the vertical without the spin.

Inverted Yaw Spin

A yaw spin flown inverted.

RELATED TRICKS

Knife Edge

The vertical attitude it spins on.

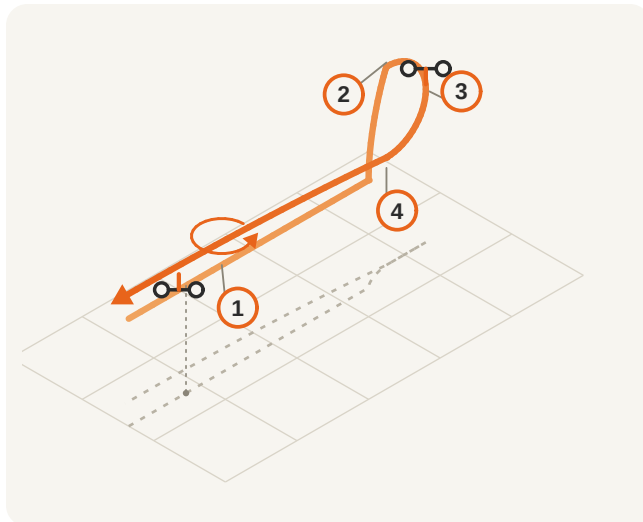
Inverted Yaw Spin

Spinning around the vertical.

Lookback & Loop

Intermediate

Open Air



HOW TO

A lookback that springs into a loop and rewinds. Carry some forward momentum into a 180 yaw so you are looking back the way you came, then punch the throttle and pitch up to spring to the top of a loop. Float through it upside down, around the back side, and at the bottom blip the throttle to carry back toward where you started. The abrupt spring into the loop is what sets it apart.

MASTERY LEVELS

- ★☆☆ Yaw back, spring into a loop, rewind out.
- ★★★☆☆ Clean 180 yaw sprung into a floaty back-side loop.
- ★★★★★ Seamless lookback, loop, and rewind back to start.

STEP BY STEP CONTROL INPUTS



1 Begin with forward momentum into a 180 yaw



2 Looking backwards, throttle punch and pitch up



3 Upside down, float through the loop



4 At the bottom, blip throttle back toward start

PREREQUISITES & WHERE TO PRACTICE

- A confident 180 lookback yaw
- A floaty loop

Carry momentum into a 180 yaw, then work the abrupt punch-and-pitch to spring up to the top of a loop. Float around the back side and blip back toward where you started. Keep the loop round and give yourself height.

COMMON MISTAKES

- ✗ A soft spring that never reaches the top of the loop
- ✗ Losing the line through the 180 yaw
- ✗ Not blipping to carry back toward the start

VARIATIONS

Inverted Lookback

A lookback flown inverted.

Lookback + Roll + Inverted Yaw Spin

A longer lookback combo.

RELATED TRICKS

Inverted Lookback

The inverted look.

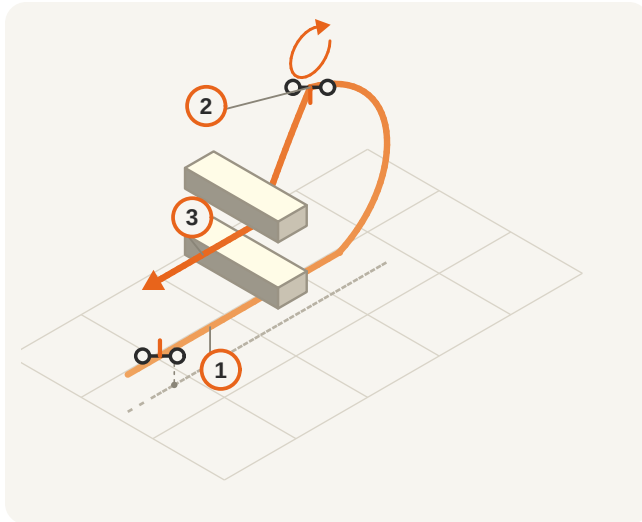
Loop

The loop it finishes with.

Loop to Matty

Intermediate

Powerloops



STEP BY STEP CONTROL INPUTS



1 Approach under an object, pitching drone up as if starting a powerloop



2 Facing backwards upside down above the objects, pitch back forward to upright facing forwards



3 Maintain throttle and alignment to pass through the gap backwards

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A Matty you can start from inverted

Fly the climb-and-peak on its own until you can arrive at the top with the nose coming over and speed in hand, because the Matty has nothing to work with off a dying climb. Then trade the loop for the Matty at the peak and line up the gap for the drop back through. Keep it all on one line so it tracks straight from above.

COMMON MISTAKES

- ✗ Carrying the loop over instead of catching it at the peak
- ✗ Mattying before you are level and inverted at the top
- ✗ Missing the gap alignment on the way down
- ✗ Losing the line so it no longer tracks straight

VARIATIONS

Immelmatt

An Immelmann into the Matty instead.

Matty Flip

The Matty on its own.

RELATED TRICKS

Loop

The loop it starts with.

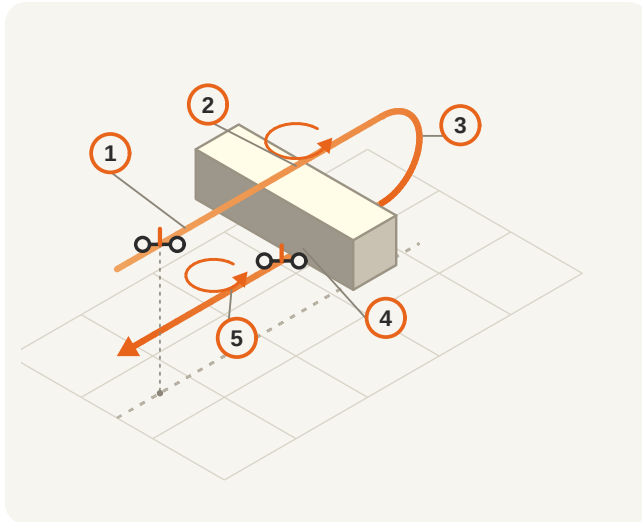
Matty Flip

The Matty it flows into.

Matty Twister

Intermediate

Matty Flips



HOW TO

A Matty with a 360 roll folded in while pointing straight down at the object. Approach over the object and pitch the nose down, and when you are facing down at it, throw a full 360 roll (which reads as a yaw from that attitude). Resume the pitch to finish the Matty over the far edge and back under, recovering at the bottom moving back the way you came but facing backwards. Optionally yaw out to face forward again.

MASTERY LEVELS

- ★☆☆ Fold a 360 roll into a Matty, facing down.
- ★★★☆☆ Clean face-down 360 roll, smooth Matty out.
- ★★★★★ Seamless Matty twister, roll pinned, tidy recovery.

STEP BY STEP CONTROL INPUTS



1 Approach over an object and pitch the drone down



2 Facing down at the object, do a 360 roll



3 Resume pitching to complete the Matty roll



4 Recover at the bottom, moving back but facing backwards



5 Optionally yaw out

PREREQUISITES & WHERE TO PRACTICE

- A clean Matty Flip
- A confident 360 roll

Get the Matty solid, then add a pause facing straight down at the object to fit a 360 roll before you finish the flip. Keep the roll centered on the object and recover clean under it. Yaw out to finish if you like.

COMMON MISTAKES

- ✗ Not getting the nose fully down before the roll
- ✗ An off-center roll that drifts off the object
- ✗ A messy recovery under the object

VARIATIONS

Matty Flip

Without the twist.

Matty 360

A full 360 yaw on the under instead.

RELATED TRICKS

Matty Flip

The base trick.

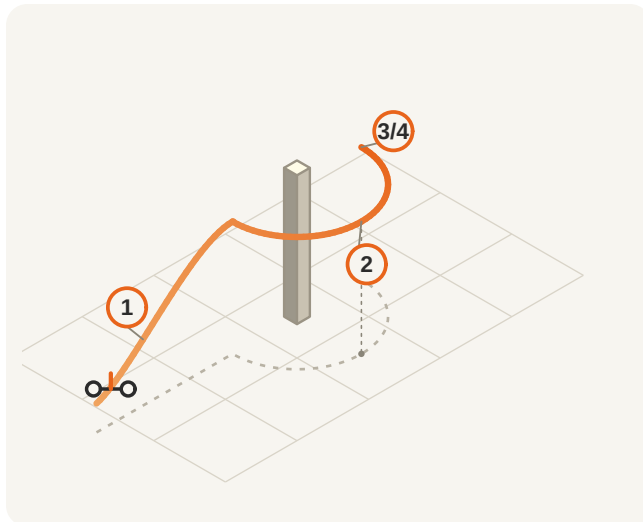
Matty 360

A full yaw 360 folded in.

Outward Side Lock Rewind

Intermediate

Rewinds



HOW TO

An outward side-lock rewind flown on an angle. Approach the side of the object banked about 30 degrees to the horizon, then swing around it holding that angle so it stays framed. Keep swinging until the object slides out of view, lock the quad in place, and blip the throttle to begin the rewind, retracing the swing back the way you came while holding the same angle.

MASTERY LEVELS

- ★☆☆ Swing around an object on an angle and rewind.
- ★★★ Clean angled swing, locked, smooth rewind back.
- ★★★★ Crisp outward side-lock rewind, angle held throughout.

STEP BY STEP CONTROL INPUTS



1 Approach the side of the object at a 30 degree angle



2 Swing around the object keeping the angle



3 As the object leaves view, lock and blip to rewind



4 Rewind back, holding the angle

PREREQUISITES & WHERE TO PRACTICE

- Confident flight on a held bank angle
- A feel for locking a subject and rewinding

Approach an object's side on a roughly 30 degree bank and swing around it holding the angle. When it leaves the frame, lock and blip back, retracing the swing on the same angle. Keep it smooth both ways.

COMMON MISTAKES

- ✗ Losing the angle through the swing
- ✗ Not locking cleanly before the rewind
- ✗ An uneven rewind that doesn't retrace the swing

VARIATIONS

Side-Lock Rewind

The inward version.

Sideways Rewind

Without locking a subject.

RELATED TRICKS

Side-Lock Rewind

The core side-lock rewind.

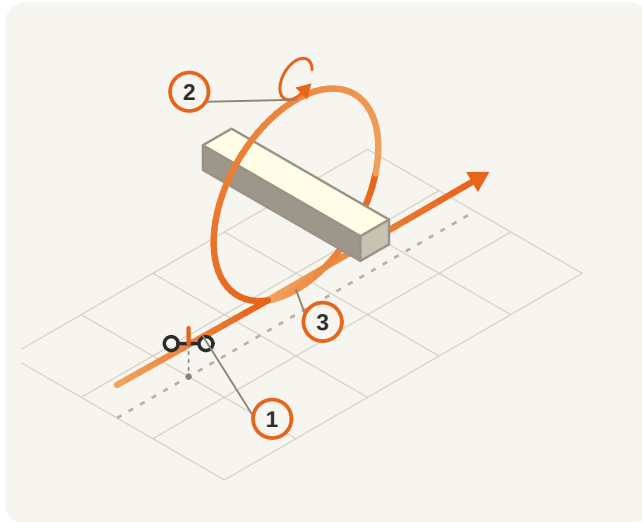
Sideways Rewind

A lateral rewind.

Power Flip

Intermediate

Powerloops



HOW TO

A powerloop with an extra in-place 360 loop thrown in at the top. Come in low and level with speed and pull into the loop, and when you reach the top inverted, snap a tight in-place 360 loop before continuing the main loop back down. Carry the throttle through and level out on the entry line. It is a powerloop with a flip folded into its apex.

MASTERY LEVELS

- ★☆☆ Powerloop with an in-place 360 at the top.
- ★★★☆☆ Clean powerloop, tidy 360 folded in at the apex.
- ★★★★★ Seamless power flip, both rotations crisp, level out.

STEP BY STEP CONTROL INPUTS



1 Approach low with speed; commit to the line



2 Loop around until upside down, and perform an in place 360 loop there, then continue the main loop back down



3 Carry throttle over the top, inverted



4 Relax pitch, catch level on the entry line

PREREQUISITES & WHERE TO PRACTICE

- A solid Powerloop
- A tight, quick flip

Get the powerloop dialed, then work the in-place 360 at the top: reach the apex inverted, snap the extra loop, and continue the powerloop down. Give yourself height over the obstacle.

COMMON MISTAKES

- ✗ Losing throttle at the top and dropping out
- ✗ A sloppy 360 that breaks the main loop
- ✗ Not catching level on the entry line

VARIATIONS

Powerloop

Without the in-place 360.

Reversed Power Flip

The other direction.

RELATED TRICKS

Powerloop

The base loop.

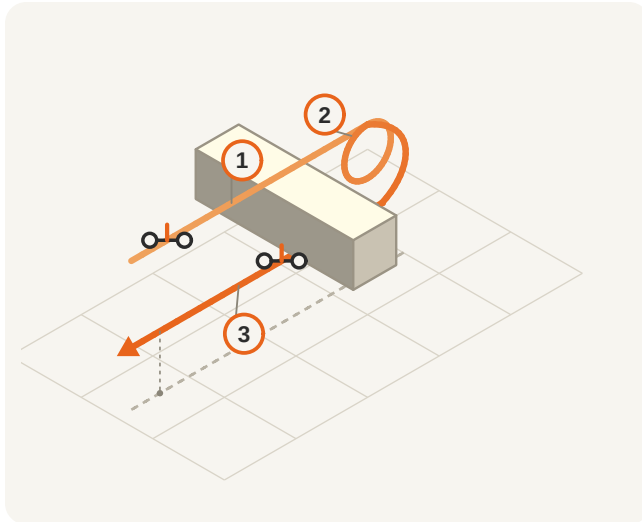
Double Dutch

A powerloop with a reverse 360 roll instead.

Power Matty

Intermediate

Matty Flips



HOW TO

A Matty with a forward loop folded in at the top. Fly over the object, and just before you drop into the matty, throw a forward loop. Then drop down behind the object and fly backward under it, feeding in throttle and pitching to level out as you pass beneath. The loop at the top is what sets it apart from a plain Matty.

MASTERY LEVELS

- ★☆☆ Forward-loop over the object, then Matty under.
- ★★★☆☆ Clean forward loop into a tidy Matty under-pass.
- ★★★★★ Crisp loop and Matty, aligned exit, smooth level-out.

STEP BY STEP CONTROL INPUTS



Fly over the object



Just before dropping into the matty, do a forward loop



Fly backward under, throttle up and level out

PREREQUISITES & WHERE TO PRACTICE

- A confident Matty Flip
- A clean forward loop

Find an object with clear air above and below. Fly over it, throw a forward loop just before you drop in, then drop down behind and fly backward under it, feeding in throttle to level out. Give yourself height.

COMMON MISTAKES

- ✗ A sloppy loop that doesn't line up the drop
- ✗ Not enough height, so you drop too low
- ✗ Losing the object out of frame

VARIATIONS

Matty Flip

The standard over-and-under, no loop.

Anti Matty

A full backflip instead of a forward loop.

RELATED TRICKS

Matty Flip

The base over-and-under.

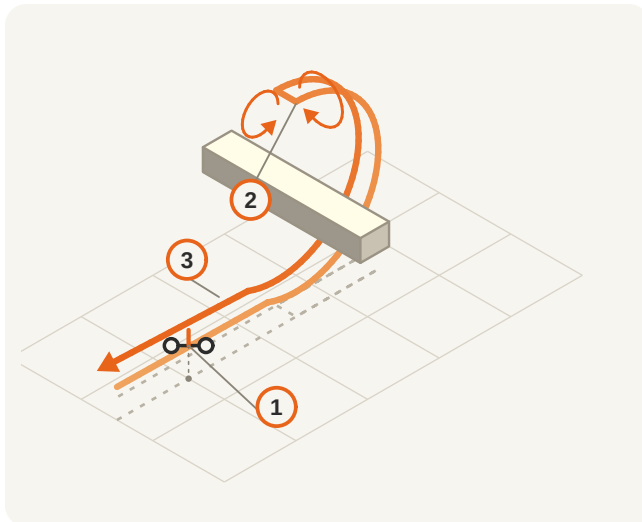
Anti Matty

The backflip version.

Power Split

Intermediate

Powerloops



STEP BY STEP CONTROL INPUTS

1



Approach under object, pitching drone up

2



Roll out at the top

3



Split-S down the far side

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A clean Split-S

Have both the powerloop and the Split-S dialed, then blend them: loop up, and at the top roll out and pull the split down the far side. Give yourself height for the descent.

COMMON MISTAKES

- ✗ Rolling out too early, before the top
- ✗ Not committing the throttle up the loop
- ✗ Diving out of the split instead of leveling

VARIATIONS

Powerloop

The full loop without the split exit.

Split S

The reversal without the loop entry.

RELATED TRICKS

Powerloop

The loop it starts with.

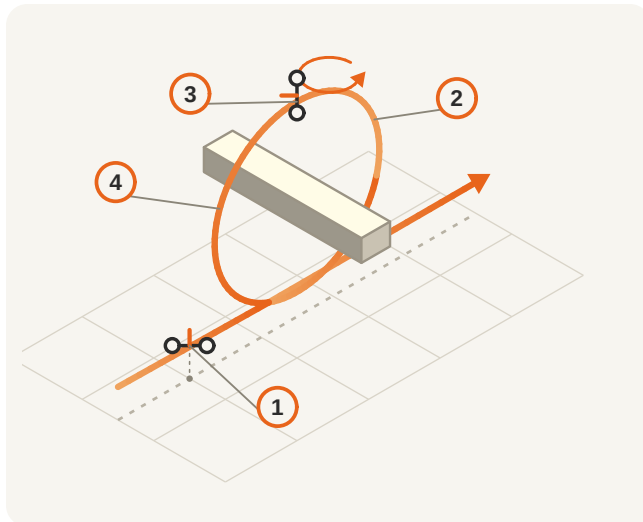
Split S

The reversal it finishes with.

Power Swap

Intermediate

Powerloops



HOW TO

A powerloop that swaps out into a backwards matty. Carry throttle up and over into the loop, and as you reach the top still facing up, roll 180 so your belly faces the sky. Let the momentum carry you around the other side of the roll and matty backwards under the object. It swaps the loop's exit for a rolled, backward under-pass.

MASTERY LEVELS

- ★☆☆ Loop up, roll belly-to-sky, matty out backwards.
- ★★★☆☆ Clean loop into a tidy top roll and backward under.
- ★★★★ Seamless power swap, roll pinned, smooth matty under.

STEP BY STEP CONTROL INPUTS



Approach low with speed; commit to the line



Full back-pitch + throttle to climb into the loop



At the top still facing up, 180 roll to face the sky



Let momentum carry around the roll and matty backwards under

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A clean 180 roll

Have the powerloop solid, then work the 180 roll at the top with your belly to the sky, letting momentum carry you around and back under the object. Give yourself height.

COMMON MISTAKES

- ✗ Rolling before you reach the top of the loop
- ✗ Not committing the throttle up the loop
- ✗ Losing the object as you come around backwards

VARIATIONS

Powerloop

Without the roll and swap.

Power Switch

A yaw switch at the top instead of a roll.

RELATED TRICKS

Powerloop

The loop it is built on.

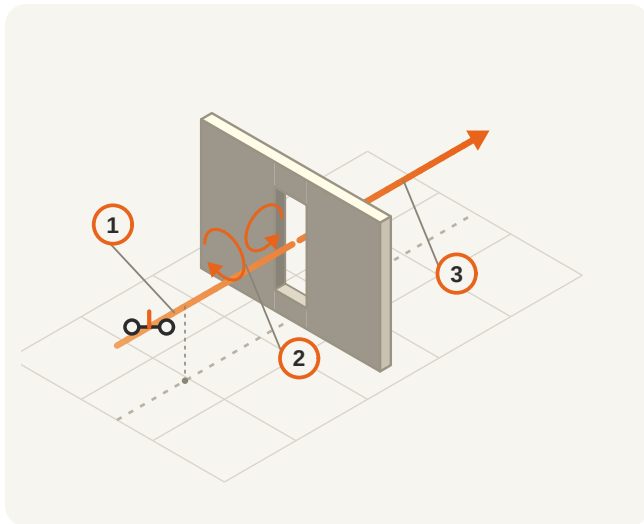
Matty Flip

The under-pass it finishes with.

Reverse Knife Edge

Intermediate

Gaps



HOW TO

A knife-edge taken backwards through a gap. Line up the narrow slot early and blip the throttle, then roll 90 onto the knife edge and pitch 180 so you slip through the gap backwards, narrow profile first. Float through and correct your orientation on the far side. It threads a tight slot backwards on the knife edge.

MASTERY LEVELS

- ★☆☆ Slip through a gap backwards on the knife edge.
- ★★★ Clean roll-and-pitch into a backward knife-edge gap.
- ★★★★ Tight slot threaded backwards, knife-edge, dead on line.

STEP BY STEP CONTROL INPUTS

1



Line up the narrow gap early, blip throttle

2



90 roll and 180 pitch to head backwards through the gap on knife edge

3



Float through the gap and correct orientation

PREREQUISITES & WHERE TO PRACTICE

- A solid knife-edge
- Comfortable rolling and pitching to face backwards

Line up a narrow gap, then practise rolling onto the knife edge and pitching 180 to slip through backwards. Commit the rotation before the gap and correct on the far side.

COMMON MISTAKES

- ✗ Rotating in the gap instead of before it
- ✗ Not enough bank to clear the sides
- ✗ Losing orientation flying through backwards

VARIATIONS

Knife Edge

Banked the usual way.

Ninja Star

A knife-edge gap in a combo.

RELATED TRICKS

Knife Edge

The other-way version.

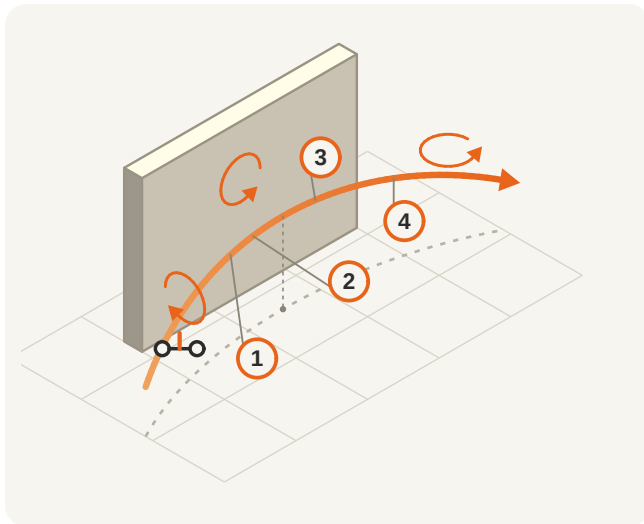
Gaps

Threading an opening flat.

Reverse Wall Ride

Intermediate

Wall Tricks



HOW TO

A wall ride flown backward. Approach the wall from the side and roll onto it as for a wall ride, and once you are moving parallel to the wall, throw a 180 yaw (which reads as a flip on its side) to face backward. Center the sticks and float backward down the wall, then blip away to recover the line, still facing backward. Optionally yaw forward again to finish. Riding it backward is the whole point.

MASTERY LEVELS

- ★☆☆ Roll onto the wall and ride it backward.
- ★★★ Clean 180 into a steady backward wall ride.
- ★★★★ Smooth reverse wall ride, dialed switch, tidy recovery.

STEP BY STEP CONTROL INPUTS



Approach wall from the side



Moving parallel to the wall, 180 yaw to face backwards



Facing backwards, center sticks to float through the wall ride



Blip away to recover the line, ending facing backwards



Optionally yaw forward facing again

PREREQUISITES & WHERE TO PRACTICE

- A solid forward wall ride
- A clean 180 yaw against the wall

Get the forward wall ride dialed, then practise the 180 to backward while riding, floating down the wall facing back before you blip away. Keep your gap steady once you are backward.

COMMON MISTAKES

- ✗ A slow 180 that drifts you off the wall
- ✗ Losing your gap once backward
- ✗ Not recovering the line cleanly off the wall

VARIATIONS

Reverse Wall Kiss

A backward tap instead of a full ride.

Wall Ride

The forward version.

RELATED TRICKS

Wall Ride

The forward version this builds on.

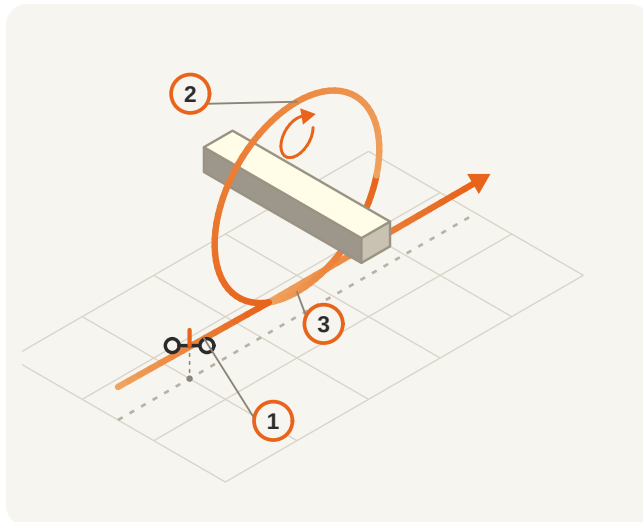
Snapback

Another quick reversal.

Reversed Power Flip

Intermediate

Powerloops



HOW TO

A powerloop with a reversed flip folded in at the top.

Approach under the object gaining throttle, and once you are under it with momentum, pitch up into the loop. Near the top, facing up, throw a forward flip, spun the opposite way to the loop, then complete it coming down the other side and pass back under the object.

MASTERY LEVELS

- ★☆☆ Powerloop with a reversed flip at the top.
- ★★★☆☆ Clean loop, tidy reversed flip at the apex.
- ★★★★★ Seamless reversed power flip, both rotations crisp.

STEP BY STEP CONTROL INPUTS



Approach under object gaining throttle; with momentum, pitch drone up



Facing up near the top of the loop, do a forward flip



Complete it coming down the other side and pass under the object

PREREQUISITES & WHERE TO PRACTICE

- A solid powerloop
- A clean flip both directions

Get the powerloop dialed, then add the reversed flip at the top: reach the apex facing up, throw the flip, and complete it coming down the far side. Give yourself height over the obstacle.

COMMON MISTAKES

- ✗ Flipping before you reach the top of the loop
- ✗ Not committing the throttle up the loop
- ✗ A sloppy flip that breaks the main loop

VARIATIONS

Power Flip

The flip spun the other way.

Powerloop

Without the flip at the top.

RELATED TRICKS

Power Flip

The mirror version.

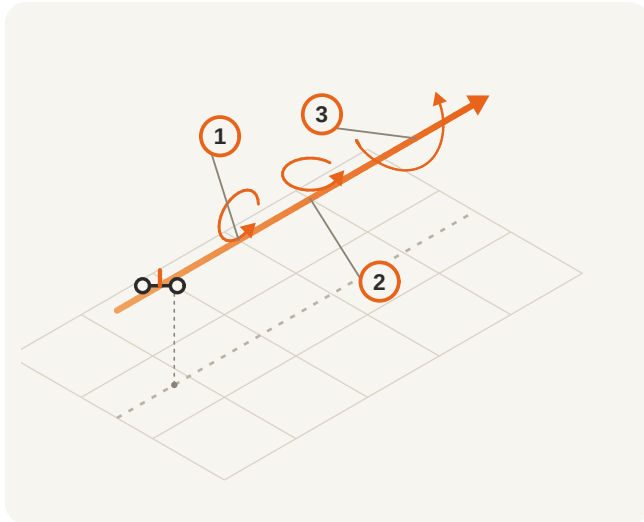
Powerloop

The loop it is built on.

Rubik's Cube

Intermediate

Open Air



HOW TO

A multi-axis combo that turns the quad on every axis in turn, like twisting a Rubik's Cube: a flip on the pitch axis, a flat yaw, and a roll to finish. Flip, pin the yaw, and roll out. Chaining all three axes cleanly on one line is the puzzle.

MASTERY LEVELS

- ★☆☆ Flip, yaw, and roll in one line.
- ★★★☆☆ Clean three-axis combo, each turn crisp.
- ★★★★ Seamless Rubik's Cube, every axis smooth and on line.

STEP BY STEP CONTROL INPUTS



Flip on the pitch axis



Pin a flat yaw



Roll out to finish

PREREQUISITES & WHERE TO PRACTICE

- Clean flips, rolls, and yaw spins
- A feel for chaining axes

Have each axis solid on its own, then chain them on one line: flip, yaw, roll, keeping the line straight through all three. Smooth the hand-offs between the turns.

COMMON MISTAKES

- ✗ Drifting off line between the turns
- ✗ Rushing one axis into the next
- ✗ An uneven turn on one of the axes

VARIATIONS

Cubik's Rube

The same axes in a different order.

Rubik's Cube Pendulum

A Rubik's Cube swung on a pendulum.

RELATED TRICKS

Roll

The roll axis.

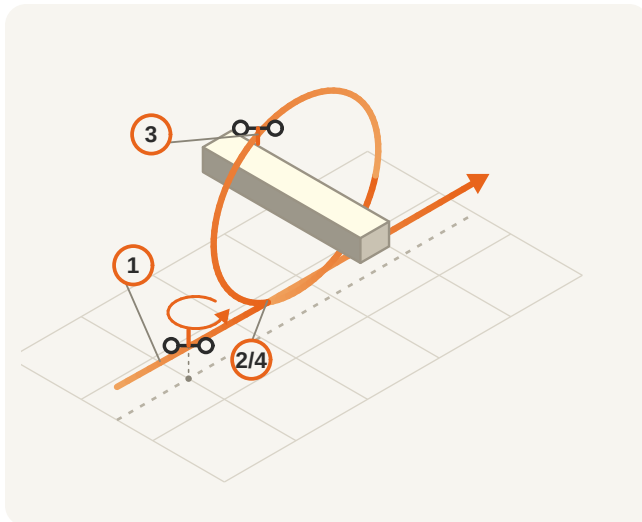
Loop

The pitch axis.

Sideways Powerloop

Intermediate

Powerloops



HOW TO

A powerloop flown yawed sideways, so the loop is driven by roll instead of pitch. Approach under the object with forward momentum and yaw so you are facing sideways, then once under it feed in throttle and roll to carry the loop up and over. Cut the throttle at the top while inverted, then catch it and continue back under the object. It is the powerloop rotated ninety degrees.

MASTERY LEVELS

- ★☆☆ Loop the bar sideways, driven by roll.
- ★★★ Round sideways loop with a clean catch.
- ★★★★ Tight, committed sideways powerloop, dead on line.

STEP BY STEP CONTROL INPUTS



Approach under object with forward momentum and yaw



Once under the object, begin the powerloop with throttle and roll



Cut throttle at the top of the loop while inverted



Catch and continue back under the object

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- Comfortable flying yawed sideways

Have the powerloop dialed, then yaw sideways and drive the same loop with roll instead of pitch. Give yourself height and commit to the roll.

COMMON MISTAKES

- ✗ Not committing the roll through the loop
- ✗ Clipping the bar mid-loop
- ✗ Losing the sideways yaw through the loop

VARIATIONS

Powerloop

The pitch-axis version.

Side Loop

A sideways loop without an obstacle.

RELATED TRICKS

Powerloop

The pitch-axis version.

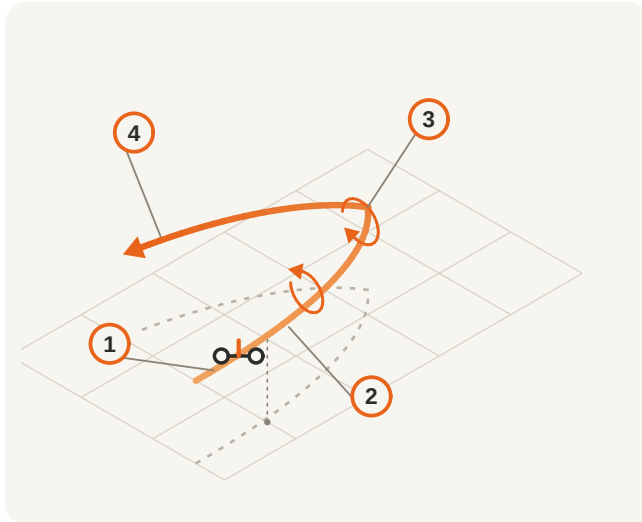
Side Loop

A roll-axis loop.

Sideways Rewind

Intermediate

Rewinds



HOW TO

A rewind flown sideways, like a backwards lookback. Roll about thirty degrees to bank the line, then feed in throttle and coordinate the roll and pitch to carry the turn round to a point. From that point begin the rewind, bringing it back the way you came and ending facing backwards.

MASTERY LEVELS

- ★☆☆ Bank out, point the turn, and rewind back.
- ★★★ Smooth banked slide into a clean pointed rewind.
- ★★★★ Buttery sideways rewind, ending dead backwards.

STEP BY STEP CONTROL INPUTS

1  
Roll slightly, around 30 degrees

2  
Throttle up and coordinate roll / pitch

3  
Begin the rewind

4  
End backwards facing

PREREQUISITES & WHERE TO PRACTICE

- A smooth coordinated banked turn
- A feel for rewinding a line

Start with a smooth banked turn, then tighten it to a point and rewind out of it, ending facing backwards. Keep the roll and pitch coordinated through the point.

COMMON MISTAKES

- ✗ A rounded turn with no point to rewind out of
- ✗ Uncoordinated roll and pitch through the turn
- ✗ Not ending cleanly backwards

VARIATIONS

Stall Rewind

A vertical rewind.

Side-Lock Rewind

A rewind past an object.

RELATED TRICKS

Stall Rewind

The vertical version.

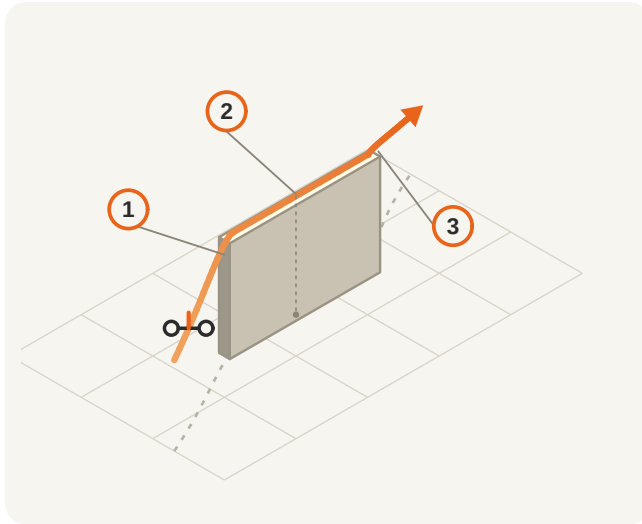
Side-Lock Rewind

A tracking rewind.

Slides / Grinds

Intermediate

Wall Tricks



HOW TO

Sliding or grinding the quad along the top edge of a rail, ledge, or wall. Ease onto the edge, slide along it keeping light, steady contact, and peel off the end. It is proximity flying taken to the limit, riding a line instead of just passing near it.

MASTERY LEVELS

- ★☆☆ Slide the rail and peel off clean.
- ★★★ Steady grind along the edge, controlled peel.
- ★★★★ Long, smooth grind held tight to the rail.

STEP BY STEP CONTROL INPUTS



Ease onto the rail



Slide along the top edge



Peel off the end

PREREQUISITES & WHERE TO PRACTICE

- Confident proximity flying near a surface
- A soft, steady touch

Find a clean, straight rail or ledge and practise easing on, holding a light slide, and peeling off. Keep the contact gentle so you do not snag. Scout the edge first.

COMMON MISTAKES

- ✗ Pressing too hard and snagging
- ✗ Drifting off the edge
- ✗ A rough peel-off that catches

VARIATIONS

Wall Ride

Riding a vertical face.

Perch

Settling on top instead of sliding.

RELATED TRICKS

Wall Ride

Riding a vertical face.

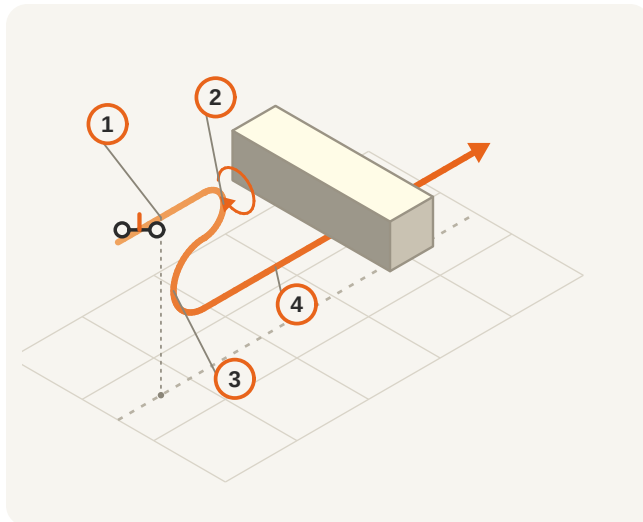
Perch

Resting on top of an object.

Split Stall Matty Rewind

Intermediate

Split-S



HOW TO

A high line dropped through a Split-S straight into a matty. Approach high above the object, and just above it cut the throttle into a Split-S with a 180 roll. Blip the throttle into a forward flip that carries you round the matty curve, then float down and catch it with throttle to continue under the object flying backwards. The two rotations trace a reverse S on the way down.

MASTERY LEVELS

- ★☆☆ Split-S into a matty and out under the object.
- ★★★☆☆ Clean Split-S, tidy matty, smooth catch underneath.
- ★★★★★ Seamless split-matty drop, backwards under, dead on line.

STEP BY STEP CONTROL INPUTS



Approach high above the object



Just above the object, Split-S: cut throttle with a 180 roll



Just above it, blip throttle into a forward flip into a matty



Float down, catch with throttle, continue under the object backwards

PREREQUISITES & WHERE TO PRACTICE

- A clean Split-S
- A solid Matty and a confident catch

Have the Split-S and the Matty solid on their own, then chain them above the object: cut throttle into the split, blip into the flip, and catch it underneath flying backwards. Give yourself height above the object.

COMMON MISTAKES

- ✗ A rough hand-off between the split and the matty
- ✗ Not blipping enough to carry the flip round
- ✗ Clipping the object on the way under

VARIATIONS

Stall Matty Rewind

Without the split entry.

Split Yaw

A split flown over and under with yaw.

RELATED TRICKS

Matty Flip

The over-and-under it is built on.

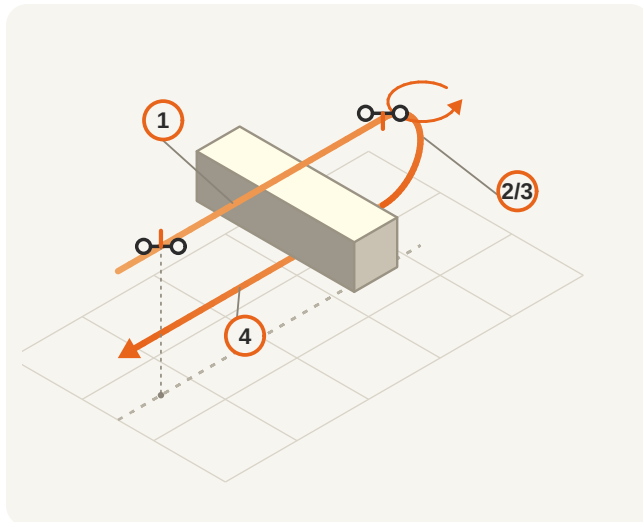
Stall Rewind

The rewind it finishes with.

Split Yaw

Intermediate

Split-S



HOW TO

Fly forward towards an object and perform a Split S as you fly over it. While descending, incorporate a 180° yaw spin and conclude by smoothly transitioning into backward flight, passing beneath the object.

MASTERY LEVELS

- ★☆☆ Reverse and drop under the object flying backward.
- ★★★☆☆ Smooth split with the yaw blended into the descent.
- ★★★★ Seamless roll, pull, and 180 yaw into a clean backward pass.

STEP BY STEP CONTROL INPUTS



1 Pass over the object



2 Once past the object, Split-S with a 180 roll to face down



3 Follow the roll quickly with a 180 yaw



4 Continue backwards under the object, toward the starting direction

PREREQUISITES & WHERE TO PRACTICE

- A solid Split-S
- A controllable flat yaw spin

Build it from a clean Split-S first, then layer the yaw in during the descent so the two blend into one motion. A bar or gap gives you the over-then-under line to aim for.

COMMON MISTAKES

- ✗ Yawing too early, before the half loop is established
- ✗ The yaw and pull fighting, so the exit wobbles
- ✗ Not enough altitude to finish the descent under the object

VARIATIONS

Split Yaw Rewind

Rewind back over the object out of the yaw.

Double Split Yaw

Carry a full 360 of yaw through the descent.

RELATED TRICKS

Split S

The base reverse-and-descend.

Yaw Spin

The flat spin folded into the descent.

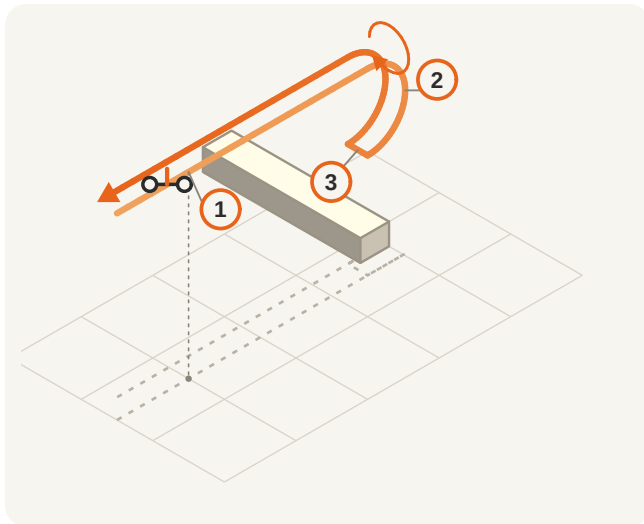
Half Matty

Another half-rotation reversal.

Split-S Rewind

Intermediate

Rewinds



HOW TO

A Split-S that only goes half way, then rewinds back out. Approach the object from above with some momentum, and as you pass it roll into a partial Split-S that curves down behind it. When you reach the object's own height, punch the throttle to rewind and pop back up, flying out the way you came. The line never drops below the object.

MASTERY LEVELS

- ★☆☆ Partial Split-S behind the object, then rewind out.
- ★★★☆☆ Clean roll into the drop, tidy rewind back up.
- ★★★★★ Precise partial Split-S, tight turn at height, smooth rewind.

STEP BY STEP CONTROL INPUTS



Approach the object from above with some momentum



Roll to perform a partial Split-S



At the object's mid-height, punch throttle and rewind out

PREREQUISITES & WHERE TO PRACTICE

- A solid Split-S
- A feel for the stall-and-rewind

Approach from above, roll into a partial Split-S behind the object, and punch the throttle to rewind before you drop below it. Keep the turn tight at the object's own height.

COMMON MISTAKES

- ✗ Dropping below the object instead of turning at its height
- ✗ Blowing through the turn
- ✗ A harsh punch that overshoots the rewind

VARIATIONS

Split S

Without the rewind.

Stall Rewind

A vertical stall rewind.

RELATED TRICKS

Split S

The reversal it starts with.

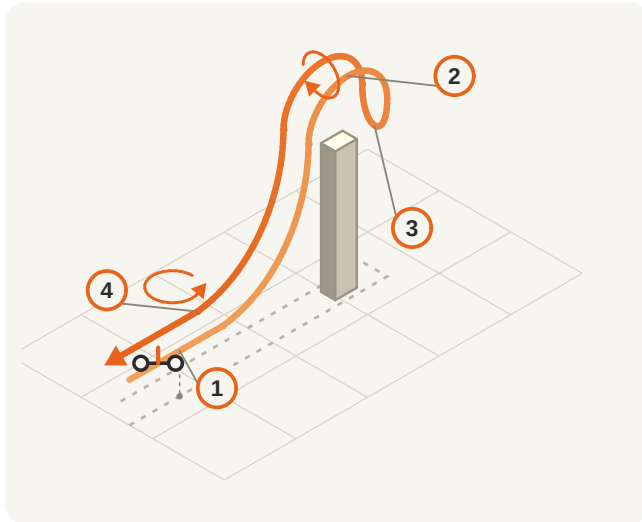
Stall Rewind

A vertical rewind.

Split-S to Stall Track

Intermediate

Split-S



HOW TO

A partial Split-S taken around an object and tracked back. Come in low with some momentum and climb before the object, rolling into a partial Split-S that carries you over the top and down the far side. At about the object's mid-height, punch the throttle to rewind out and track back close to the line you came in on, optionally yawing out at the end.

MASTERY LEVELS

- ★☆☆ Partial Split-S around the object, then track back.
- ★★★ Clean climb and roll, tidy turn behind the object.
- ★★★★ Precise partial Split-S, tight turn, smooth track home.

STEP BY STEP CONTROL INPUTS



1 Approach the object with some momentum



2 Roll to perform a partial Split-S



3 At the object's mid-height, punch throttle and rewind out



4 Optionally add a yaw out at the end

PREREQUISITES & WHERE TO PRACTICE

- A clean Split-S
- A confident stall and rewind

Climb before the object and roll into the partial Split-S, then punch the throttle at its mid-height to turn around and track back close to your entry line. Add the yaw out once the rest is smooth.

COMMON MISTAKES

- ✗ Carrying too much speed to turn cleanly
- ✗ Dropping the object out of frame
- ✗ A harsh, uneven pull out of the Split-S

VARIATIONS

Split-S

Without the stall.

Stall Rewind

A stall taken into a rewind.

RELATED TRICKS

Split-S

The base reversal.

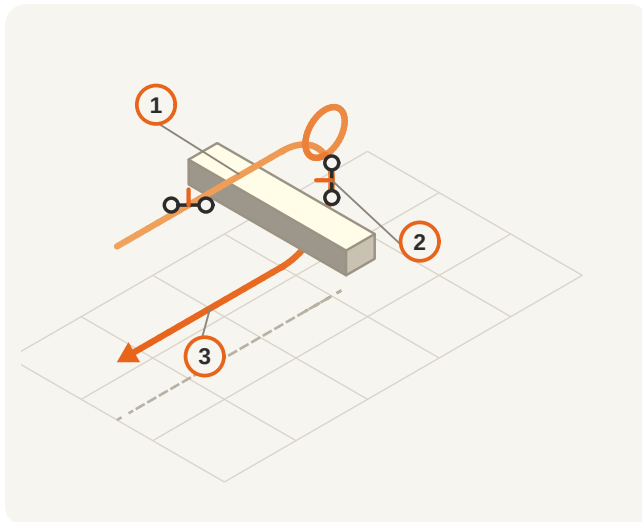
Stall Rewind

A stall taken into a rewind.

Splitback

Intermediate

Split-S



HOW TO

A Split-S with a backflip folded in at the top. Half roll to inverted, then snap a backflip right there before pulling through the descending half loop as in a Split-S. Catch it at the bottom and continue out. The split reverses you and the flip adds the style.

MASTERY LEVELS

- ★☆☆ Split-S with a backflip folded in at the top.
- ★★★ Smooth split with a clean backflip once inverted.
- ★★★★ Seamless backflip and Split-S as one flowing line.

STEP BY STEP CONTROL INPUTS



Half roll to inverted



Do a full backflip



Catch and continue out

PREREQUISITES & WHERE TO PRACTICE

- A solid Split-S
- A clean backflip

Build it from a clean Split-S, then add the backflip once you are inverted and before the descent. Keep the flip tight so it does not eat the altitude you need for the split.

COMMON MISTAKES

- ✗ Flipping before the half roll completes
- ✗ A loose flip that drifts off the line
- ✗ Losing altitude awareness through the combo

VARIATIONS

Split S

Without the backflip.

Splitback Rewind

Rewind back out of it.

RELATED TRICKS

Split S

The base reverse-and-descend.

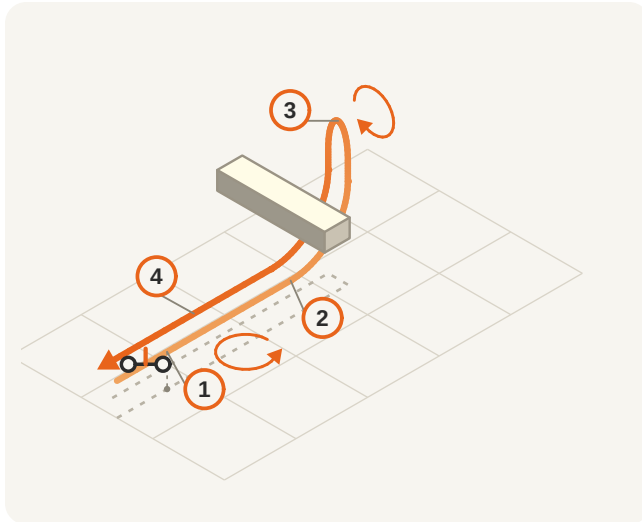
Double Roll

Shares the barrel-roll exit.

Stall Half Matty Rewind

Intermediate

Rewinds



HOW TO

Fly toward an object and execute a 180° yaw spin as you pass it. After clearing the object, add a small amount of throttle and perform a Half Matty to rewind back past the object.

MASTERY LEVELS

- ★☆☆ Yaw past the object and Half Matty back over it.
- ★★★☆☆ Clean 180 yaw into a smooth Half Matty rewind.
- ★★★★★ Seamless yaw, Half Matty, and rewind past the object.

STEP BY STEP CONTROL INPUTS



1 Approach the object, 180 yaw before passing under



2 Behind the object, throttle up to its height while pitching down



3 Brief stall at the object's height, then a 180 roll



4 Continue down and back under the object backwards

PREREQUISITES & WHERE TO PRACTICE

- A controllable 180 yaw
- A solid Half Matty

Get the 180 yaw and the Half Matty dialed on their own, then chain them: yaw before you pass under, climb only to the object's own height behind it, and roll off the stall to drop back under backwards.

COMMON MISTAKES

- ✗ Yawing too slow, drifting past the object
- ✗ Climbing past the object's height instead of stalling at it
- ✗ Losing the object on the way back under

VARIATIONS

Half Matty

Without the leading 180 yaw.

Stall Matty Rewind

A full Matty rewind instead.

RELATED TRICKS

Half Matty

The rewind half of the move.

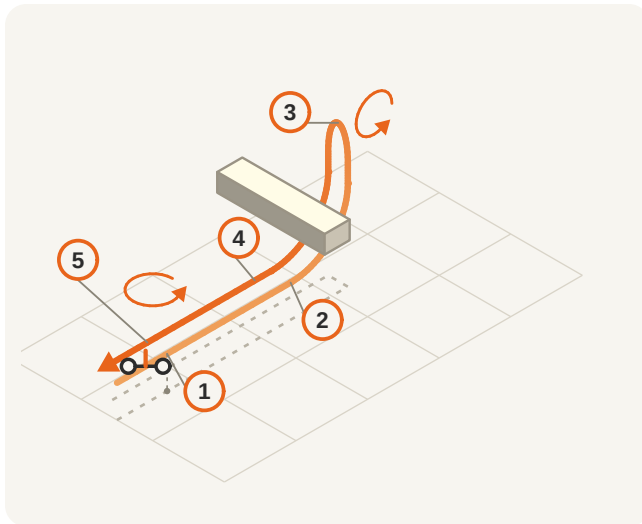
Split Yaw

Another yaw-and-reverse over an object.

Stall Matty Rewind

Intermediate

Rewinds



HOW TO

A Matty flown to a stall and rewound back. Pass under the object, then pitch back as if starting a powerloop so you sweep up the far side. Once you are facing up, throw a forward flip until you are inverted facing back toward the start, then ride it down and pass under the object backwards. Yaw back forwards to finish.

MASTERY LEVELS

- ★☆☆ Sweep up past the object and flip back under it.
- ★★★ Clean pitch-back into a smooth flip and rewind.
- ★★★★ Precise stall, tidy forward flip, exact pass back under.

STEP BY STEP CONTROL INPUTS



1 Approach the object, passing under it



2 Past the object, pitch back as if doing a powerloop



3 Facing up, forward flip to inverted facing back toward the start



4 Pass under the object backwards



5 Yaw back forwards

PREREQUISITES & WHERE TO PRACTICE

- A solid Matty Flip
- A feel for the stall at the top

Get the sweep up the far side smooth first, then work in the forward flip at the top before you ride it back down. Keep the object framed through the flip and the pass back under.

COMMON MISTAKES

- ✗ Flipping before the stall settles
- ✗ Losing the object on the way back under
- ✗ Not clearing the object underneath

VARIATIONS

Half Matty

Without the stall pause.

Stall Half Matty Rewind

Add a 180 yaw on the pass.

RELATED TRICKS

Half Matty

The rewind half of the move.

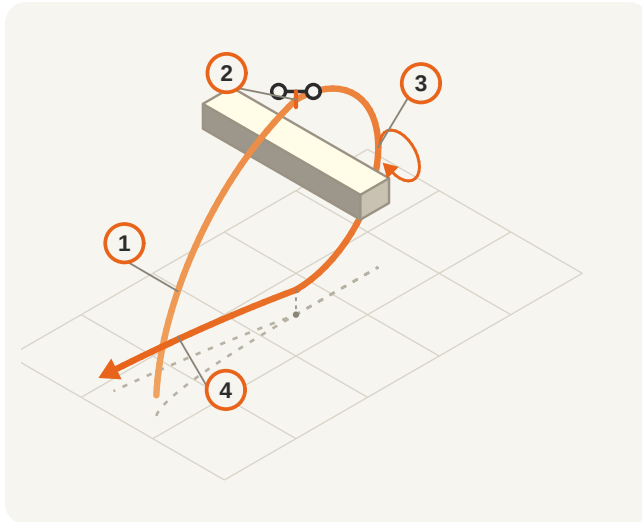
Stall Half Matty Rewind

A yaw-and-Matty rewind.

Tracking Front Flip

Intermediate

Matty Flips



HOW TO

A front flip flown around an object while keeping it framed the whole way round. Approach on the throttle, pitching up as you climb, and once you are level with the object and facing it, pitch back and cut the throttle so you begin turning around it, tracking it as you go. Float around until you are upside down, then cross-coordinate a roll and yaw back to upright and continue under it toward where you started.

MASTERY LEVELS

- ★☆☆ Flip while keeping an object framed.
- ★★★☆☆ Clean front flip, subject held in view.
- ★★★★ Tight tracking flip, object locked in frame throughout.

STEP BY STEP CONTROL INPUTS



Approach the object, throttling up and pitching up



Level with it, pitch back and cut throttle, turning around it



Float around until inverted, then cross-coordinate roll/yaw upright



Continue under the object back toward the start

PREREQUISITES & WHERE TO PRACTICE

- A clean front flip
- A feel for framing a subject

Climb to the object's height, then pitch back and let it carry you around it, keeping it framed. Work the roll-and-yaw back to upright once you come through inverted. Give the object room.

COMMON MISTAKES

- ✗ Losing the object as you come around
- ✗ Drifting toward or away from the subject
- ✗ An uncoordinated roll/yaw that leaves you off-axis

VARIATIONS

Flip

Without tracking a subject.

Juicy Tracking-Lookback

Tracking taken to a lookback.

RELATED TRICKS

Flip

The base rotation.

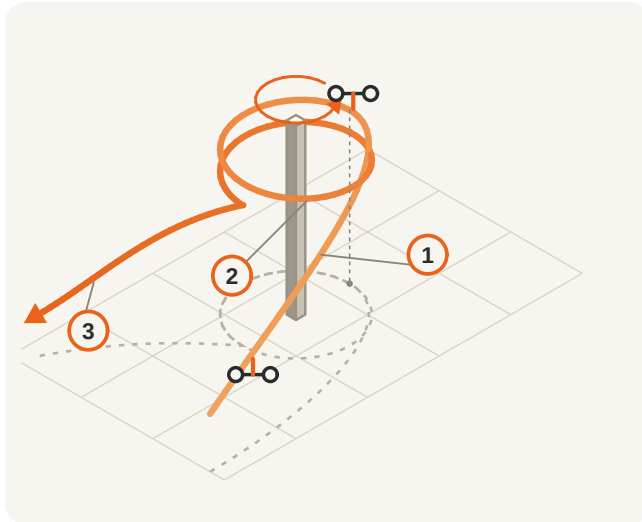
Matty Flip

A flip framed over an object.

Trippy Spin

Intermediate

Orbitals



HOW TO

Start by performing a Cradle. Once inverted with the object at the top of the screen, execute a coordinated roll and yaw to keep the object at the top of the screen as you orbit around it. From the right side of the object, roll right while yawing left (and vice versa from the left).

MASTERY LEVELS

- ★☆☆ Complete an inverted orbit keeping the object roughly framed.
- ★★★ Round orbit with the object held near the top of the frame.
- ★★★★ Smooth, constant inverted orbit, object locked top-center throughout.

STEP BY STEP CONTROL INPUTS

1



Climb inverted into position beside the pole top

2



Roll and yaw opposite ways to orbit, object held up top

3



Unwind and descend away

PREREQUISITES & WHERE TO PRACTICE

- Comfortable inverted, holding a hang
- Coordinated roll-and-yaw (orbits)

Use an isolated pole or post with room around it. Get comfortable inverted first, then add the orbit slowly. Keeping the object pinned at the top of the frame is the whole game.

COMMON MISTAKES

- ✗ Roll and yaw out of sync, so the orbit wobbles
- ✗ Losing the object out of frame
- ✗ Dropping the inverted hang mid-orbit

VARIATIONS

Double Trippy Spin

Two inverted orbits before unwinding.

Trippy Switch

Reverse the spin direction mid-orbit.

RELATED TRICKS

Orbits

The upright, ground-level version.

Yaw Spin

Spinning in place rather than orbiting.

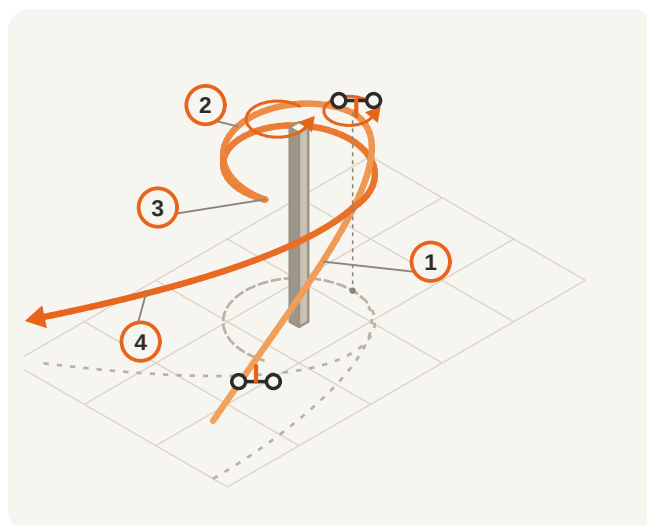
Inverted Yaw Spin

The inverted flat spin without the orbit.

Trippy Switch

Intermediate

Orbitals



HOW TO

A Trippy Spin that reverses direction mid-orbit. Get inverted and orbit the top of an object as in a Trippy Spin, then switch the spin the other way, unwinding back around before you descend away. Keeping the object framed through the reversal is the hard part.

MASTERY LEVELS

- ★☆☆ Orbit inverted, switch direction, and unwind.
- ★★★ Clean switch with the object held framed.
- ★★★★ Smooth inverted orbit and reversal, object locked in view.

STEP BY STEP CONTROL INPUTS



1 Climb inverted into position beside the pole top



2 Roll and yaw opposite ways to orbit, object held up top



3 Switch spin directions



4 Unwind and descend away

PREREQUISITES & WHERE TO PRACTICE

- A solid Trippy Spin
- Reversing a coordinated roll-and-yaw cleanly

Have the Trippy Spin dialed first, then practise switching the spin direction smoothly without losing the object. It is a control test, so keep the reversal gentle and framed.

COMMON MISTAKES

- ✗ Losing the object at the switch
- ✗ A jerky reversal that wobbles the orbit
- ✗ Dropping the inverted hang mid-switch

VARIATIONS

Trippy Spin

Without the switch.

Double Rolling Trippy Spin

Add rolls to the orbit.

RELATED TRICKS

Trippy Spin

The base inverted orbit.

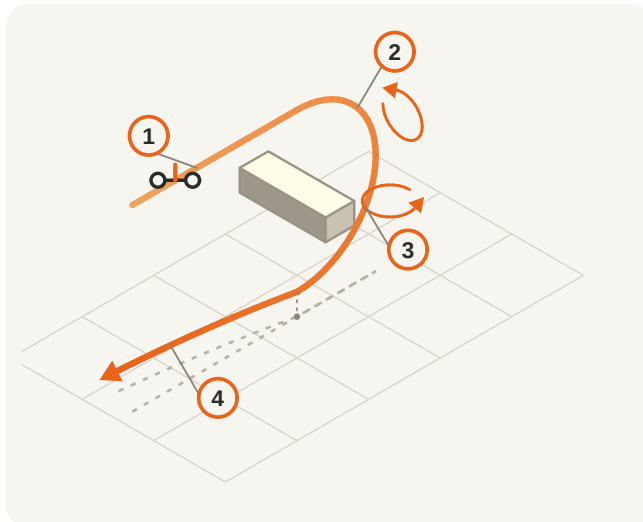
Orbits

The upright, single-direction version.

True Barani

Intermediate

Miscellaneous



HOW TO

A split-s with the second half of the rotation flown on the yaw axis instead of the pitch axis. Come in above an object and start what looks like a split-s, coordinating roll and pitch until you are pointed straight down. From there, rather than pulling the nose through, hang on the descent and yaw slowly around to face back up, then carry on underneath the object flying backwards. The slow, deliberate yaw out of the vertical is what makes it a barani and not a split-s.

MASTERY LEVELS

- ★☆☆ Stop a split-s nose-down and yaw out of it.
- ★★★ Smooth split-s entry into a controlled yaw around.
- ★★★★ Slow, hanging barani flown right over the object.

STEP BY STEP CONTROL INPUTS



Approach above object, setting up for a split-s



Just past it, coordinate roll/pitch to face down



Right after facing down, hanging juicy yaw to face up



Continue path backwards under the object

PREREQUISITES & WHERE TO PRACTICE

- A clean split-s
- Comfortable holding a nose-down hang

Fly plain split-s's over an object until the entry is automatic, then start stopping the rotation at nose-down and holding it. Once you can hang there, add the yaw and slow it right down. The whole trick reads on how unhurried that yaw looks.

COMMON MISTAKES

- ✗ Rushing the yaw so it reads as a split-s
- ✗ Starting the rotation before you are past the object
- ✗ Dropping too much altitude in the hang

VARIATIONS

Barani

The looser, less deliberate version.

Split S

The same entry finished on pitch instead of yaw.

RELATED TRICKS

Split S

The entry it is built on.

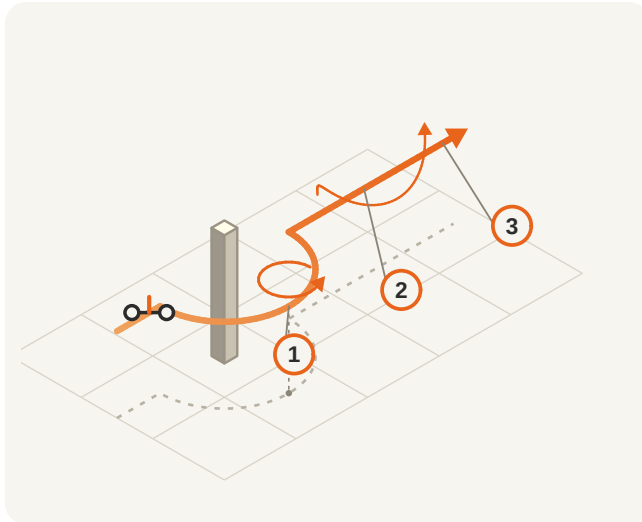
Barani

The same idea flown less deliberately.

Vanny Roll

Intermediate

Open Air



HOW TO

Approach an obstacle like a tree or pole and yaw-spin around it, tracking it through 180 degrees. Blip the throttle and cross past the obstacle with a coordinated 360 roll, keeping the line smooth, then blip once more and yaw out to fly away clean. It blends a tracking yaw, a rolling cross, and a yaw exit into one flowing pass.

MASTERY LEVELS

- ★☆☆ Track the obstacle 180 and cross with a full roll.
- ★★★ Smooth yaw-track, a clean crossing roll, and a tidy yaw out.
- ★★★★ Seamless track, coordinated roll, and yaw exit as one flowing line.

STEP BY STEP CONTROL INPUTS



1 Approach the obstacle, yaw-spin tracking it 180



2 Throttle blip and cross with a coordinated 360 roll



3 Another throttle blip and yaw out

PREREQUISITES & WHERE TO PRACTICE

- A controllable flat yaw spin and object tracking
- A clean coordinated 360 roll

Pick an obstacle with room around it, a tree or a pole. Get the 180 yaw-track smooth first, then add the rolling cross and the yaw-out, blending the three into one continuous pass. Keep the throttle blips light.

COMMON MISTAKES

- ✗ Losing the obstacle out of frame during the yaw-track
- ✗ A stiff or off-axis crossing roll
- ✗ Heavy throttle blips that break the flow

VARIATIONS

Double Vanny

Two rolling crosses linked around the obstacle.

Vanny Flick

Blend a juicy flick into the crossing roll.

RELATED TRICKS

Yaw Spin

The tracking yaw at the start.

Roll

The coordinated 360 across the middle.

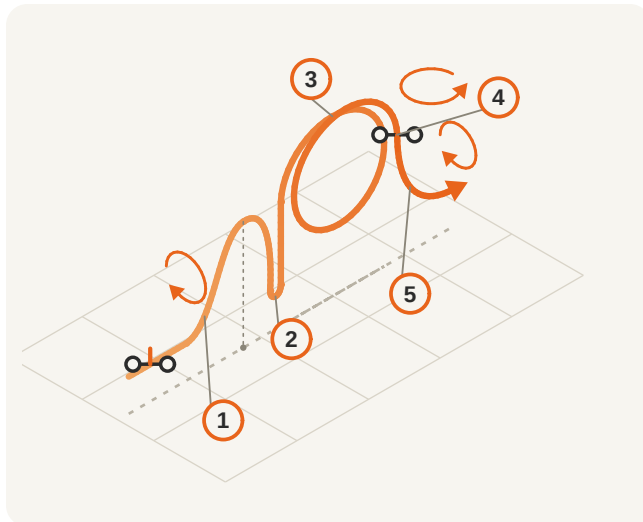
Orbits

Another obstacle-tracking maneuver.

540 Delayed Front Flip

Advanced

Open Air



HOW TO

A one and a half rotation flip flown off a blip, with the yaw hanging on the end of it. Blip the throttle and roll into a tight arc, let it drop off the back and catch it facing backwards, then blip hard again straight up and pitch into the flip. Pump the throttle through the rotation, up as the sky comes into view and down as the ground does, and carry it all the way round to inverted. Hang there and yaw 180, then catch on a last blip and roll out. Flown properly it tracks one straight line the whole way.

MASTERY LEVELS

- ★☆☆ Get through a 540 and catch it.
- ★★★ Clean blip-and-catch into a paced 540.
- ★★★★ Seamless 540 with a hanging yaw, dead on line.

STEP BY STEP CONTROL INPUTS



Start with a throttle blip and right roll



Catch at the bottom facing backwards with another blip, then pitch up for a backflip



Flip 540: throttle up as you see the sky, down when facing the ground, through to inverted



When upside down, do a 180 yaw spin



Catch with a throttle blip and roll out

PREREQUISITES & WHERE TO PRACTICE

- A reliable throttle blip and catch
- A committed backflip past inverted

Get the blip-and-catch on its own first, backwards and level, because everything after it depends on that catch. Then add the flip and work the throttle pumping until the rotation holds its height. The hanging yaw at the end comes last, once the 540 stops costing you altitude.

COMMON MISTAKES

- ✗ Missing the backwards catch and dropping out the bottom
- ✗ Flat throttle through the flip, so it sinks
- ✗ Stopping the rotation at 360 instead of carrying to inverted
- ✗ Rushing the yaw instead of letting it hang

VARIATIONS

540 Flip + Roll + Inverted Yaw Spin

A longer 540 combo.

Laser Flip

The snapped flip on its own.

RELATED TRICKS

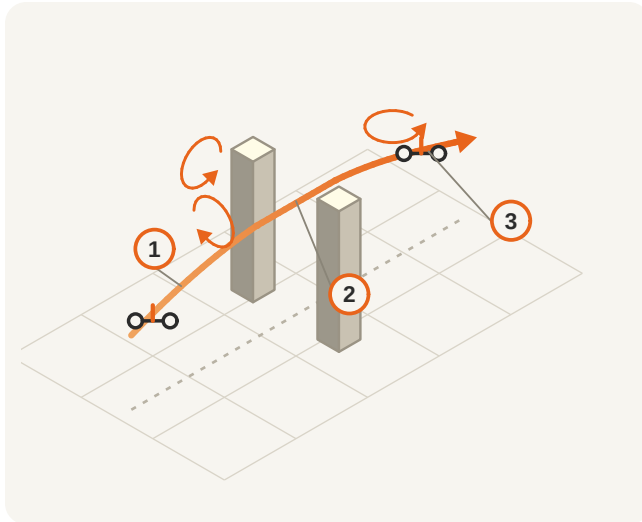
Laser Flip

A tight, snapped flip.

540 Half Matty

Advanced

Matty Flips



HOW TO

Execute a precise 540° roll (1.5 full rotations) and immediately follow it with the second half of a Matty flip.

MASTERY LEVELS

- ☆☆☆ Get the 540 round and come out backwards.
- ☆☆☆ Clean 1.5 rolls into a crisp flip upright.
- ☆☆☆ Precise 540, tight flip, exact backward line through the gap.

STEP BY STEP CONTROL INPUTS



Approach gap with a 540 roll



Just after the 540 when upside down, 180 flip to upright facing backwards



Continue backwards through the gap, optionally yaw out

PREREQUISITES & WHERE TO PRACTICE

- A precise, countable 540 roll
- A clean Half Matty

Get the 540 stopping dead inverted on its own, out in the open with no gap to worry about, because the flip has nowhere to go if the roll is off. Then chain the flip on immediately and only add the gap once the pair runs together without a pause.

COMMON MISTAKES

- ✗ Over or under rotating the 540 so the flip starts off-axis
- ✗ A gap between the roll and the flip
- ✗ Coming out of the flip still drifting sideways

VARIATIONS

540 Matty

Add the full flip instead of the half.

Half Matty

Without the leading 540 roll.

RELATED TRICKS

Half Matty

The half-roll reversal it ends with.

Matty Flip

The full Matty.

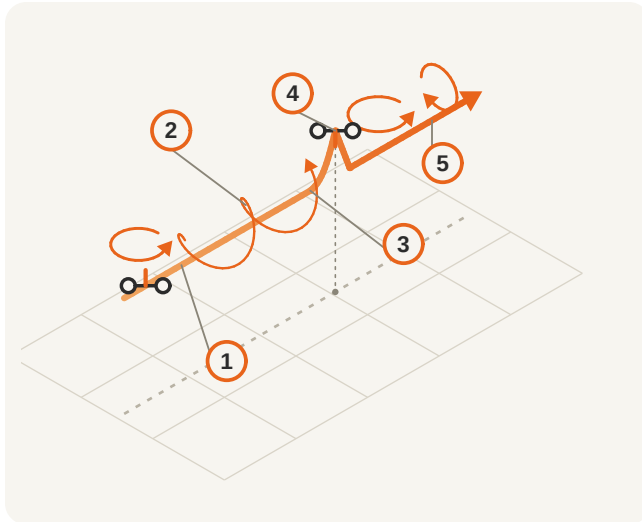
Double Roll

Shares the multi-rotation roll.

Double Side Roll + 180 Inverted Yaw Spin

Advanced

Open Air



HOW TO

Two side rolls chained into a 180 inverted yaw spin, all on one line. Build forward momentum and yaw ninety degrees so you are travelling sideways, which is what puts the rolls on their side. Blip the throttle into the first roll and blip again into the second, then a third blip lifts you to a point where you pitch down to inverted. Yaw 180 while you are upside down and roll out. Even, matched rolls and a line that never wanders are what make it read.

MASTERY LEVELS

- ★☆☆ Two side rolls and an inverted 180.
- ★★★ Clean, even rolls flowed into a pinned inverted 180.
- ★★★★ Seamless double side roll and inverted 180, dead on line.

STEP BY STEP CONTROL INPUTS

- 1 Gain forward momentum then do 90 degree yaw
- 2 Throttle blip and first roll, blip again and second roll
- 3 Throttle blip and pitch drone down inverted
- 4 Once inverted, 180 yaw
- 5 Roll out to finish

PREREQUISITES & WHERE TO PRACTICE

- A clean side roll
- A pinned inverted yaw

Get one side roll landing on heading before doubling it, because a first roll that finishes off-axis leaves nowhere to start the second. Then work the blip timing so the two rolls sit at the same height, and only add the inverted 180 once the pair runs without a correction between them.

COMMON MISTAKES

- ✗ Uneven rolls that wander off line
- ✗ Skipping the blip so the second roll sinks
- ✗ Rushing into the inverted 180
- ✗ Losing orientation inverted

VARIATIONS

Double Roll

Without the inverted 180.

Rapid Split-S + Inverted Yaw Spin Combo

A Split-S version.

RELATED TRICKS

Double Roll

The rolls it starts with.

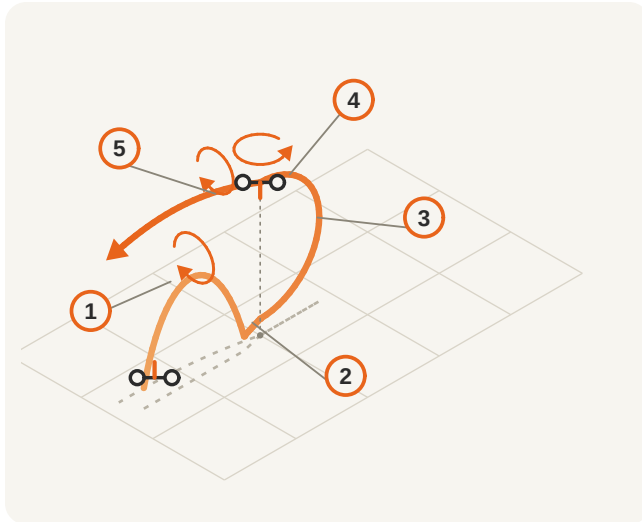
Inverted Yaw Spin

The inverted pivot.

Flip Rewind to Inverted Yaw Spin

Advanced

Rewinds



HOW TO

A flat bounce rewind with an inverted yaw spin folded in at the top. Punch a roll and let it dish down toward the deck, correct the roll just before the ground, then punch a forward loop back up and, once you come around to upside down, pin an inverted yaw spin. When you are ready, roll out to finish and fly away.

MASTERY LEVELS

- ★☆☆ Flip-rewind to inverted, spin, and roll out clean.
- ★★★☆☆ Smooth loop into a pinned inverted spin, tidy roll-out.
- ★★★★★ Crisp flip-rewind, dialed inverted 360, dead-level roll-out.

STEP BY STEP CONTROL INPUTS

1	2	3	4	5
Punch a roll	Correct it near the ground	Punch a forward loop	Once inverted, do a yaw spin	Now, roll out

PREREQUISITES & WHERE TO PRACTICE

- A confident flat bounce rewind
- A clean inverted yaw spin

Get the flat bounce rewind dialed first (roll to the deck, forward loop up), then add an inverted yaw spin at the top before you roll out. Keep it over one spot and give yourself room.

COMMON MISTAKES

- ✗ Rushing into the spin before you are inverted
- ✗ Losing the spin off-axis
- ✗ A messy roll-out after the spin

VARIATIONS

Flat Bounce Rewind

Without the inverted yaw spin.

Inverted Yaw Spin

The spin on its own.

RELATED TRICKS

Flat Bounce Rewind

The rewind it is built on.

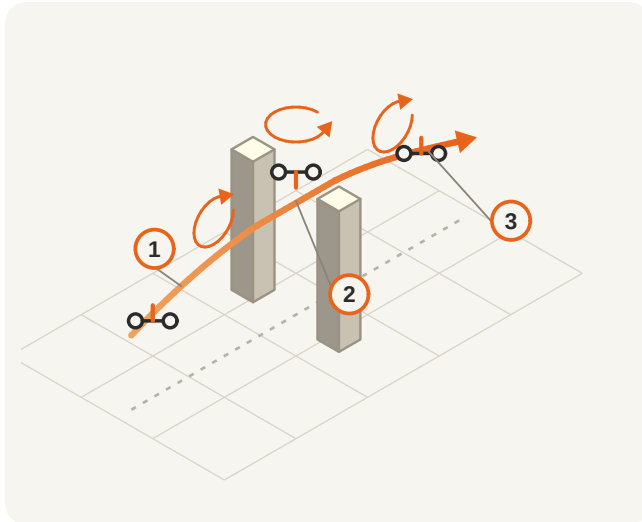
Inverted Yaw Spin

The spin pinned at the top.

Forani

Advanced

Matty Flips



HOW TO

A forward barani: a front flip with a half twist folded into it, so you reverse direction without ever rolling. Approach the object and pitch forward through 180 until you are upside down and facing back the way you came. The moment you are there, yaw 180. Then pitch down, slowly, until you are upright, carrying on backwards through the gap as you do it. Doing the whole reversal on pitch and yaw is what separates it from a Half Matty.

MASTERY LEVELS

- ★☆☆ Flip, yaw, and pitch back upright without a stall.
- ★★★☆☆ Clean forward barani through a gap.
- ★★★★★ Seamless Forani, every rotation slow and deliberate.

STEP BY STEP CONTROL INPUTS



Approach object, forward 180 flip



As soon as upside down facing backwards, 180 yaw



Slowly pitch the drone down until upright while moving backwards through the gap

PREREQUISITES & WHERE TO PRACTICE

- A controllable front flip you can stop inverted
- A pinned 180 yaw

Work the half flip on its own until you can stop it dead inverted, out in the open with no gap involved, because the yaw has nothing to sit on if the flip overshoots. Then add the yaw, and only last the slow pitch back upright. The slower that final pitch looks, the better the trick reads.

COMMON MISTAKES

- ✗ Over-rotating the first flip past inverted
- ✗ Yawing before you are properly upside down
- ✗ Snapping the final pitch instead of drawing it out
- ✗ Drifting off the line through the gap

VARIATIONS

Half Matty

The same reversal flown on a roll.

Barani

The backward version.

RELATED TRICKS

Half Matty

The roll-based reversal.

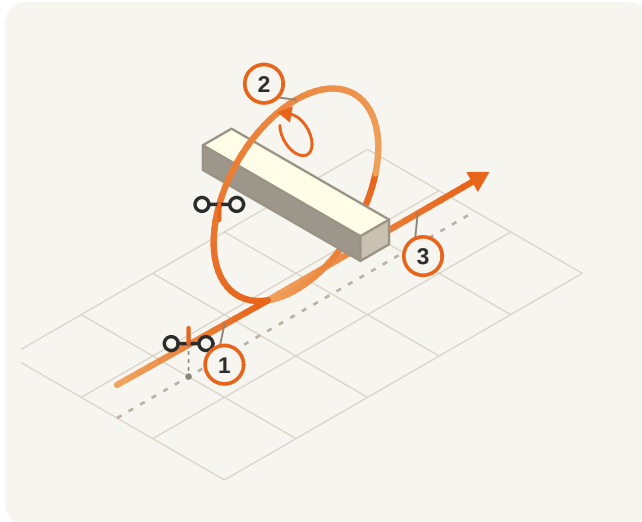
Barani

The same idea the other way round.

Half Mavvy

Advanced

Maverick Loops



HOW TO

A Maverick loop taken only half way round, the reversal made with a roll instead of finishing the arc. Come in level underneath the object and pitch back as you pass beneath it, so you climb out the far side. Once you are round and pointing up, blip the throttle and roll 180, which leaves you inverted rather than carrying on over the top. From there pitch down and let the momentum you already have take you back under. Timing the roll to the moment you are facing up is the whole trick.

MASTERY LEVELS

- ★☆☆ Get round the far side and roll to inverted.
- ★★★ Clean climb into a pinned 180 roll.
- ★★★★ Seamless Half Mavvy, roll timed to the top of the arc.

STEP BY STEP CONTROL INPUTS



Approach level under the object, pitching the drone back once passing under it



Once facing up on the other side of the object, throttle blip, then 180 roll



Once upside down, pitch drone down but continue momentum back under the gap

PREREQUISITES & WHERE TO PRACTICE

- A confident Maverick loop
- A pinned 180 roll

Fly the climb out on its own first and get used to arriving at the top with the nose up and something left in reserve, because a roll started from a dying climb will just drop you. Then add the blip and the roll together, and only worry about the exit line once inverted feels settled.

COMMON MISTAKES

- ✗ Rolling before you are properly facing up
- ✗ No blip, so the roll costs all your height
- ✗ Carrying on over the top instead of stopping at inverted
- ✗ Losing the object on the way back under

VARIATIONS

Maverick Loop

The full loop, no roll.

Mavvy Roll

A Maverick loop with a roll folded in.

RELATED TRICKS

Maverick Loop

The full version.

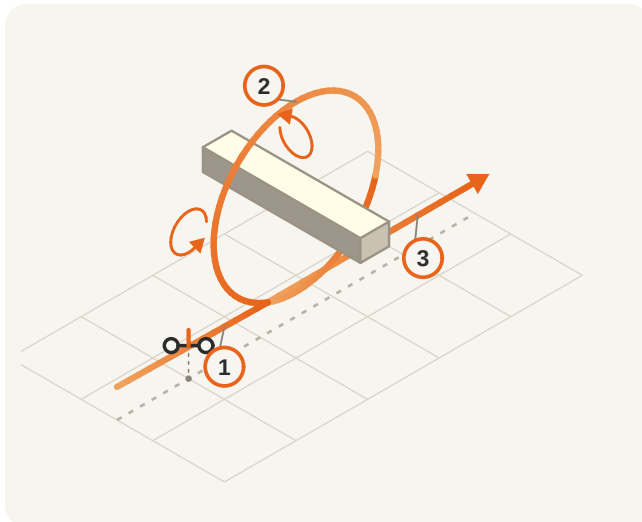
Reversed Power Flip

The same arc, reversed at the top.

Immelmann

Advanced

Powerloops



HOW TO

An Immelmann reversal flown around an object and finished like a Matty. Come in level underneath the object and pitch back as you pass beneath it, so you climb out the far side. Once you are round and pointing up, roll 180, which is the Immelmann half of it and leaves you inverted. From there pitch forward and let the momentum you already carry take you back under the object. Getting the roll to land exactly as the nose comes up is what keeps the two halves reading as one figure.

MASTERY LEVELS

- ★☆☆ Climbed out the far side and roll to inverted.
- ★★★ Clean Immelmann roll flowed into the return.
- ★★★★ Seamless Immelmann, roll timed to the top of the climb.

STEP BY STEP CONTROL INPUTS



Approach level under the object, pitching the drone back once passing under it



Once facing up on the other side of the object, 180 roll



Once upside down, pitch drone down but continue momentum back under the gap

PREREQUISITES & WHERE TO PRACTICE

- A clean Immelmann
- A pinned 180 roll

Fly the climb out on its own until you reliably arrive at the top with the nose up and speed in hand, because a roll started off a dying climb just drops you. Then put the roll on the end of it, and only worry about the line back under once inverted feels settled.

COMMON MISTAKES

- ✗ Rolling before the nose is properly up
- ✗ Losing height through the roll
- ✗ Pitching forward too hard and killing the momentum
- ✗ Losing the object on the way back under

VARIATIONS

Immelmann

The climbing reversal on its own.

Half Mavvy

The same arc with a blip into the roll.

RELATED TRICKS

Immelmann

The reversal it is built on.

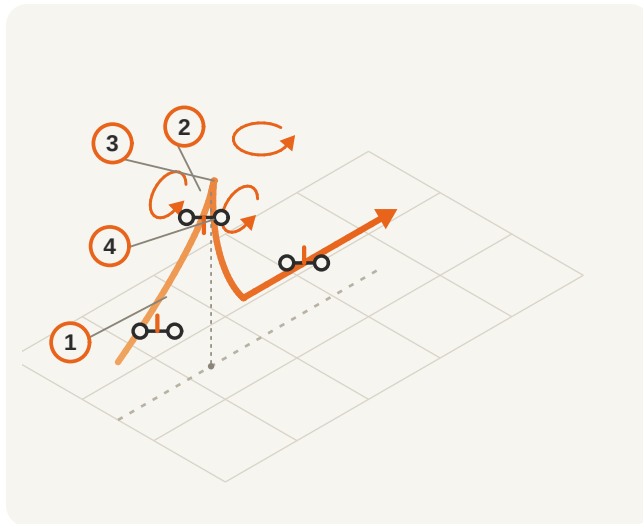
Half Mavvy

The same shape, blipped into the roll.

Inverted Uh-uh

Advanced

Open Air



HOW TO

The 'no-no' head shake flown at an inverted stall. Throttle punch up and invert, then at the top, while upside down, wag the yaw: 90 to the left, back 90 to the right, and 90 to the left again, like shaking your head no. Finish by rolling out of inverted and flying away. It is a cheeky shake held at the top of an inverted pop-up.

MASTERY LEVELS

- ★☆☆ Punch to inverted, wag the yaw no-no, and roll out.
- ★★★☆☆ Clean invert, even left-right-left yaw, tidy roll-out.
- ★★★★★ Crisp inverted stall with a sharp no-no shake and clean exit.

STEP BY STEP CONTROL INPUTS



Throttle punch and invert



When upside down, yaw left 90 degrees



Yaw back right 90 degrees



Yaw left 90 again, then roll out from inverted

PREREQUISITES & WHERE TO PRACTICE

- A confident punch to inverted
- Quick, controllable yaw

Get the punch-to-inverted stall dialed first, then add the yaw wag at the top, left-right-left, keeping it over one spot. Roll out cleanly to finish. Start high with plenty of room.

COMMON MISTAKES

- ✗ Drifting off the spot during the yaw wag
- ✗ Uneven or lazy yaw shakes
- ✗ A messy roll-out from inverted

VARIATIONS

Inverted Yaw Spin

A full inverted 360.

Yaw Spin

An upright flat spin.

RELATED TRICKS

Inverted Yaw Spin

A full inverted spin.

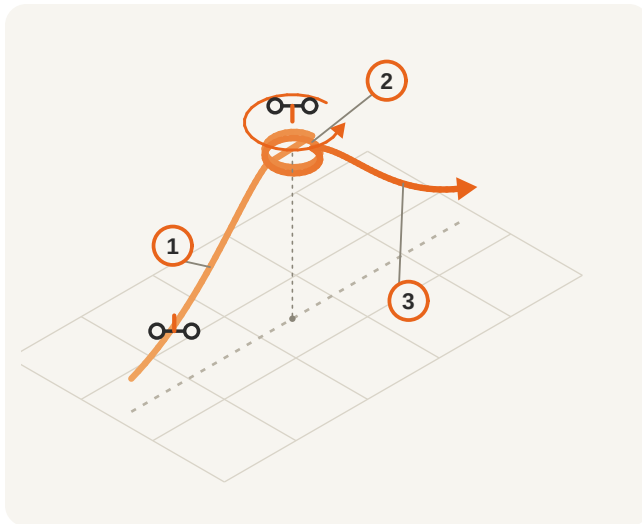
Yaw Spin

An upright flat spin.

Inverted Yaw Spin

Advanced

Open Air



HOW TO

Perform a 180° Pitch/Roll to invert, swiftly follow with an inverted 360° Yaw spin, and conclude with a Flip/Roll in the same direction.

MASTERY LEVELS

- ★☆☆ Invert, spin 360 on yaw, and recover upright.
- ★★★ Hold the inverted hang and pin a clean flat spin before recovery.
- ★★★★ Stable inverted hang, crisp 360 spin, smooth same-direction exit.

STEP BY STEP CONTROL INPUTS



1 Blip and flip to an inverted hover at the top of a climb



2 Hold the inverted hang and yaw a full 360



3 Flip back upright the same way and recover

PREREQUISITES & WHERE TO PRACTICE

- A solid yaw spin upright
- Confident flip to an inverted hover

This one wants altitude and commitment. Nail the flip-to-inverted and the upright yaw spin separately first, then combine them. The spin has to fire in the brief inverted float, so timing beats force.

COMMON MISTAKES

- ✗ Spinning before the hang settles, so it drifts
- ✗ Missing the inverted timing and dropping out
- ✗ Recovering the wrong direction, fighting the spin

VARIATIONS

Inverted Yaw Spin Rewind

Reverse back along the line out of the spin.

Double Inverted Spin

Two inverted rotations before recovery.

RELATED TRICKS

Yaw Spin

The upright version this builds on.

Trippy Spin

An inverted spin orbiting an object.

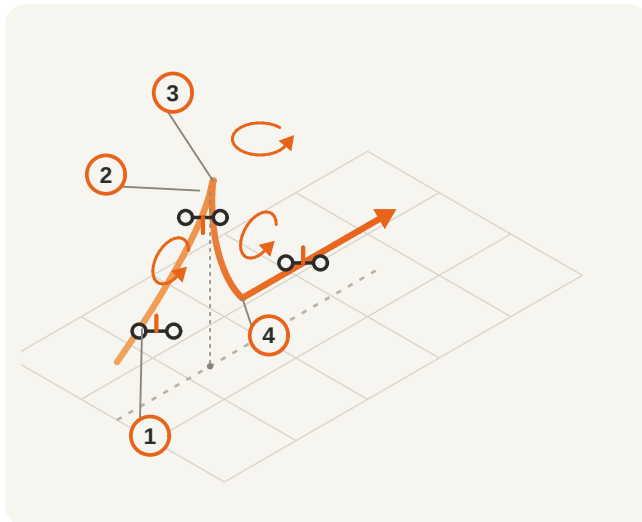
Juicy Flick

Shares the invert-and-recover feel.

Inverted Yaw Spin Rewind

Advanced

Rewinds



HOW TO

An inverted yaw spin rewind back to upright. Throttle punch up and invert, then at the top, while upside down, yaw left 90 and back right 90 as your inverted spin. Finally, pitch back to bring the quad upright again as you come down and level out, completing the move. It pops up inverted for the spin, then unwinds back the way it came.

MASTERY LEVELS

- ★☆☆ Invert, spin left-right, and rewind back upright.
- ★★★ Clean invert, tidy yaw spin, smooth rewind to level.
- ★★★★ Crisp inverted spin rewind dead-level on the exit.

STEP BY STEP CONTROL INPUTS



Throttle punch and invert



When upside down, yaw left 90 degrees



Yaw back right 90 degrees



Invert back to upright completing the move

PREREQUISITES & WHERE TO PRACTICE

- A confident punch to inverted
- A clean inverted yaw spin

Get the punch-to-inverted and the inverted yaw dialed first, then add the pitch back to upright as you come down, keeping it over one spot and leveling out clean. Start high with plenty of room.

COMMON MISTAKES

- ✗ Rushing the yaw before you are inverted
- ✗ Not fully inverting back to upright
- ✗ Leveling out off-heading

VARIATIONS

Inverted Uh-uh

A yaw wag at the top instead of the rewind.

Inverted Yaw Spin

The spin without the rewind.

RELATED TRICKS

Inverted Yaw Spin

The spin it builds on.

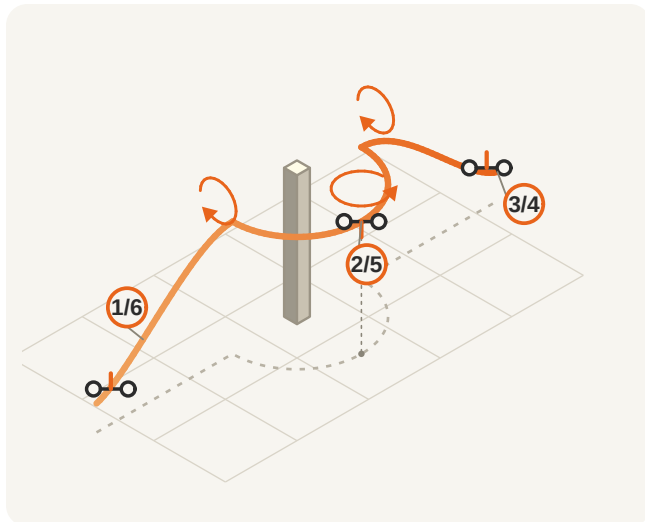
Inverted Uh-uh

The same inverted pop, with a yaw wag.

Inverted Yaw Track Rewind

Advanced

Rewinds



HOW TO

An Inverted Yaw Tracking pass, then rewind back the way you came. Roll to inverted as you approach and yaw all the way around the object, then invert back upright on the far side. To rewind, invert again and yaw back around the object the other way, inverting upright once more as you arrive back at the start. Keeping the object framed through both passes is the challenge.

MASTERY LEVELS

- ★☆☆ Yaw around an object inverted, then rewind it.
- ★★★☆☆ Clean inverted yaw-track flowed into a smooth rewind.
- ★★★★★ Crisp track and rewind, object framed throughout.

STEP BY STEP CONTROL INPUTS

1  	2  	3  	4  	5  	6  
Approach the object and roll inverted	Facing the object, yaw around it	Invert back upright	Begin the rewind, invert again in reverse	Rewind the yaw around the object	Invert back upright, back at the start

PREREQUISITES & WHERE TO PRACTICE

- A confident Inverted Yaw Tracking pass
- A feel for rewinding a line

Get the inverted yaw-track around an object dialed first, then practise reversing it: invert again and yaw back the other way, arriving upright where you started. Keep the object framed both ways and start slow.

COMMON MISTAKES

- ✗ Losing the object out of frame through either pass
- ✗ Reversed yaw sending it the wrong way inverted
- ✗ An uneven rewind that doesn't retrace the line

VARIATIONS

Inverted Yaw Tracking

Without the rewind.

Inverted Yaw Spin Rewind

A full spin instead of a track.

RELATED TRICKS

Inverted Yaw Tracking

The tracking pass.

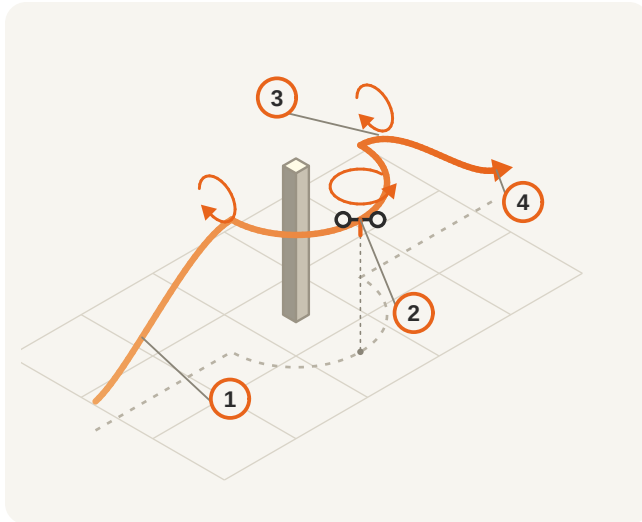
Inverted Yaw Spin Rewind

The spin-and-rewind.

Inverted Yaw Tracking

Advanced

Open Air



HOW TO

Tracking an object while inverted, with a clean roll in and out. Roll to inverted as you approach, and once you are upside down facing the object, work the yaw to swing around it and hold it framed. Roll back upright on the way out, then blip the throttle to catch the ground before you settle. The reversed controls make the yaw-tracking the hard part.

MASTERY LEVELS

- ★☆☆ Track an object while flying inverted.
- ★★★ Smooth inverted pass, object held in view.
- ★★★★ Object locked in frame through a full inverted pass.

STEP BY STEP CONTROL INPUTS



1 Approach the object and roll inverted



2 Facing the object, yaw around it



3 Invert back upright



4 Blip throttle to catch the ground

PREREQUISITES & WHERE TO PRACTICE

- Confident sustained inverted flight
- Tracking an object while yawing

Get comfortable holding inverted first, then add the yaw-tracking, keeping an object framed as you swing around it. Everything is reversed upside down, so start slow and wide.

COMMON MISTAKES

- ✗ Losing the object out of frame
- ✗ Reversed yaw sending it the wrong way
- ✗ Not catching the ground with a blip on the way out

VARIATIONS

Orbits

Tracking upright, in a circle.

Trippy Spin

An inverted orbit around an object.

RELATED TRICKS

Trippy Spin

An inverted orbit.

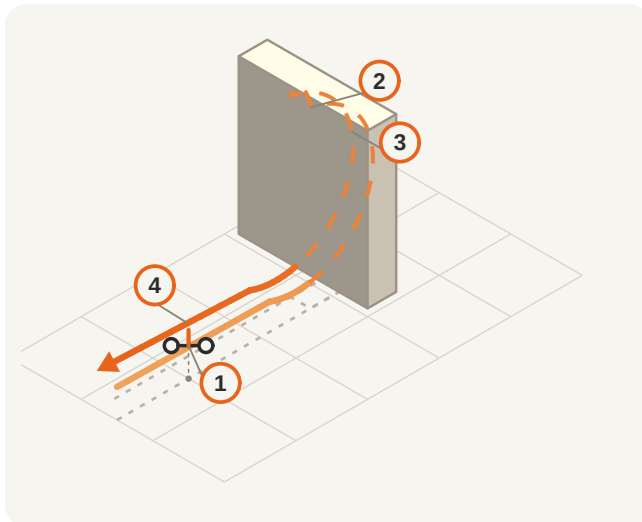
Orbits

Upright object tracking.

Jackie Chan

Advanced

Wall Tricks



HOW TO

A wall-runner: up the face of a wall, a bonk at the top, and a matty back the way you came. Pass underneath the wall low and, once you are out the far side, throttle up and pitch back so you climb its face. As the nose comes up, cut the throttle and keep pitching until you are looking at the ground, arriving into the wall for a light bonk rather than a hit. Blip the throttle to soften that contact, pitch down to reverse, and ride the matty back under the wall to where you started. The bonk should barely register; the climb and the reversal are what carry it.

MASTERY LEVELS

- ★☆☆ Get up the wall, touch it, and come back down.
- ★★★ Clean climb into a soft bonk and a tidy matty out.
- ★★★★ Seamless Jackie Chan, tap barely felt, exit dead on line.

STEP BY STEP CONTROL INPUTS



1 Pass under the object, throttle up on the opposite side pitching the drone up



2 When facing up, throttle down, continue pitching up until facing the ground, with a wall tap



3 Blip throttle to soften the wall tap, then pitch the drone down to reverse into a matty



4 Catch and backwards towards the start

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A soft, controlled wall bonk
- A solid matty

Get the climb up the face on its own first, without touching anything, so you know how much throttle carries you to the top with the nose coming over. Add the bonk only once you can arrive there slowly, because the tap is meant to be a touch, not a stop. The blip and the reversal come last.

COMMON MISTAKES

- ✗ Arriving at the wall too fast and hitting it properly
- ✗ No blip, so the tap kills all the momentum
- ✗ Not pitching far enough to be looking at the ground
- ✗ Drifting off line on the way back under

VARIATIONS

Power Switch Tap

The same idea on a horizontal object.

Wall Ride

Running the face rather than bonking it.

RELATED TRICKS

Power Switch Tap

The horizontal-object version.

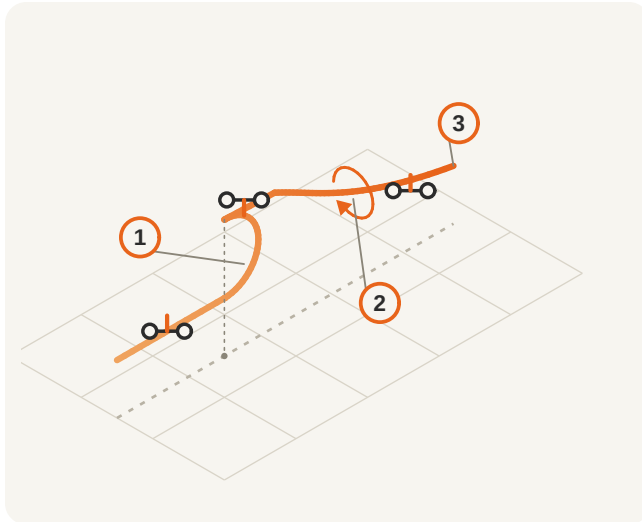
Matty Flip

The reversal it finishes with.

Juicy Flick Rewind

Advanced

Rewinds



HOW TO

A Juicy Flick flown out and then rewound straight back along the same line. Blip the throttle and drop it while pitching the nose down to pop up to inverted, then feed in the slow, floaty roll that gives the Juicy Flick its look. Rather than rolling all the way out and flying on, blip again at the far end and rewind the whole thing, running the roll back the other way so you retrace your path to where you started. The rewind is the flick played in reverse, so the smoother the first half, the cleaner the return.

MASTERY LEVELS

- ★☆☆ Flick to inverted, roll, and rewind back.
- ★★★☆☆ Clean juicy flick with a slow roll into a smooth rewind.
- ★★★★★ Seamless flick-roll-rewind, floaty and dead on line.

STEP BY STEP CONTROL INPUTS



1 Blip the throttle then drop it while pitching drone down



2 Do the slow roll of the Juicy Flick



3 Quick throttle blip and rewind the Juicy Flick with reverse roll

PREREQUISITES & WHERE TO PRACTICE

- A confident Juicy Flick
- A clean blip-and-catch to reverse on

Get the Juicy Flick itself floaty and repeatable first, because the rewind is only as good as the flick you are reversing. Then work stopping it at the far end and running the roll back the other way, keeping the line the same out and back. Film it: the two halves should mirror.

COMMON MISTAKES

- ✗ A harsh flick that leaves nothing to rewind cleanly
- ✗ Not blipping at the turnaround, so the rewind sinks
- ✗ Rolling the rewind the same way instead of reversing it
- ✗ Drifting off the line so out and back do not overlap

VARIATIONS

Juicy Flick

The flick without the rewind.

Double Juicy

Two flicks linked instead of reversed.

RELATED TRICKS

Juicy Flick

The flick it rewinds.

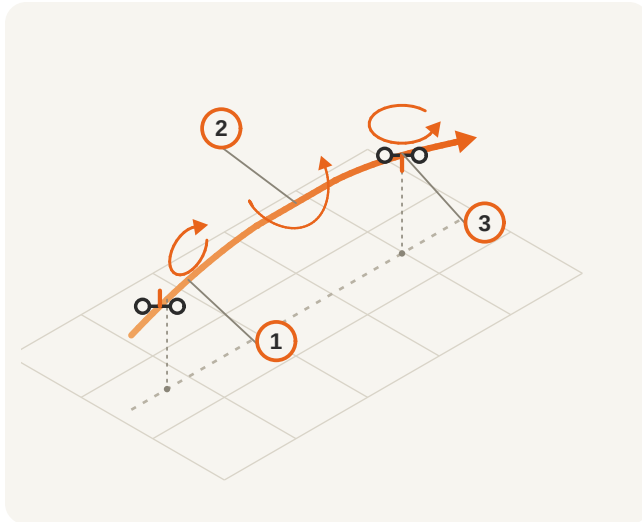
Juicy Eject

The same floaty pop, ejected out.

Lookback + Roll + Inverted Yaw Spin

Advanced

Open Air



HOW TO

A lookback, a roll, and an inverted yaw spin strung along one gently rising and falling line. Blip the throttle and pitch the nose down to flip to inverted, facing back the way you came, the lookback flown as a flip rather than a yaw. Level off on the horizon and roll a quick 360, then, still inverted and level, pin a full inverted yaw spin. Roll out to finish. It chains three distinct rotations, so smoothing the hand-offs between them is the whole game.

MASTERY LEVELS

- ★☆☆ Flip to inverted, roll, spin, and roll out.
- ★★★☆☆ Clean flip and 360 roll into a pinned inverted spin.
- ★★★★ Seamless flip-roll-spin-out, floaty and dead on line.

STEP BY STEP CONTROL INPUTS



Throttle blip, then pitch the drone down to flip facing backwards upside down



Once aligned with the horizon and inverted, do a quick 360 roll



Once back aligned with the horizon and inverted, perform inverted yaw spin



Roll out

PREREQUISITES & WHERE TO PRACTICE

- A flip to inverted you can stop on the horizon
- A pinned inverted yaw spin

Get the flip stopping clean and level inverted first, because the roll and the spin both need a settled horizon to start from. Then add the 360 roll, and last the inverted spin, chaining them on one line and keeping it straight through all three. Build the pace up as the hand-offs smooth out.

COMMON MISTAKES

- ✗ Not levelling on the horizon before the roll
- ✗ Drifting off line between elements
- ✗ A jerky hand-off into the spin
- ✗ Losing orientation on the inverted spin

VARIATIONS

Lookback & Loop

A lookback into a loop instead.

Side Lookback + Inverted Yaw Spin

A side-track version.

RELATED TRICKS

Lookback & Loop

A simpler lookback combo.

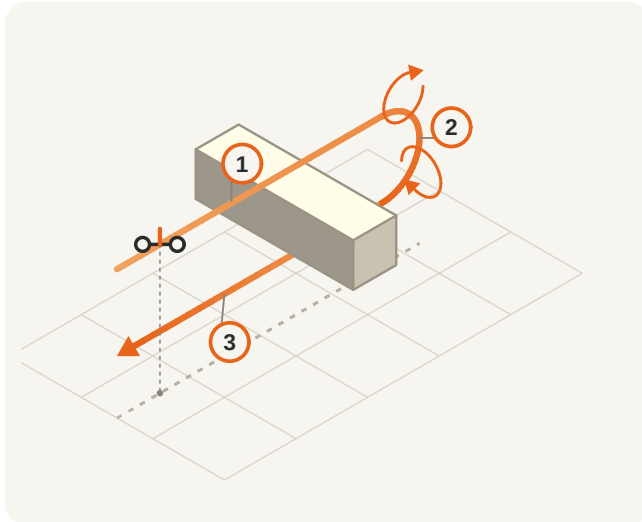
Inverted Yaw Spin

The spin it ends on.

Matty 360

Advanced

Matty Flips



HOW TO

A Matty with a full yaw spin folded in. Approach passing over the object and throw a front flip at the far edge. As you come over the top facing up at the sky, work in a complete 360 yaw so you spin once, then, still facing up, carry on backwards under the object. Blending the spin in without losing the object is the challenge.

MASTERY LEVELS

- ★☆☆ Add a full yaw 360 to the Matty pass-under.
- ★★★ Clean Matty with a smooth 360 folded in.
- ★★★★ Seamless Matty and a crisp 360, object framed throughout.

STEP BY STEP CONTROL INPUTS



1 Approach passing over object, then perform front flip



2 Once facing up to the sky, do a 360 yaw



3 Once facing back up to the sky, continue backwards under object

PREREQUISITES & WHERE TO PRACTICE

- A clean Matty Flip
- A controllable flat yaw 360

Have both the Matty and a flat 360 dialed first, then fold the spin into the under-pass. Keep the object framed as long as you can through the spin.

COMMON MISTAKES

- ✗ Rushing the 360 and losing the object
- ✗ The spin dragging you off the return line
- ✗ Under or over rotating the yaw

VARIATIONS

Matty Flip

Without the added 360.

Matty Twister

A yaw twist during the flip instead.

RELATED TRICKS

Matty Flip

The base trick.

Yaw Spin

The 360 folded in.

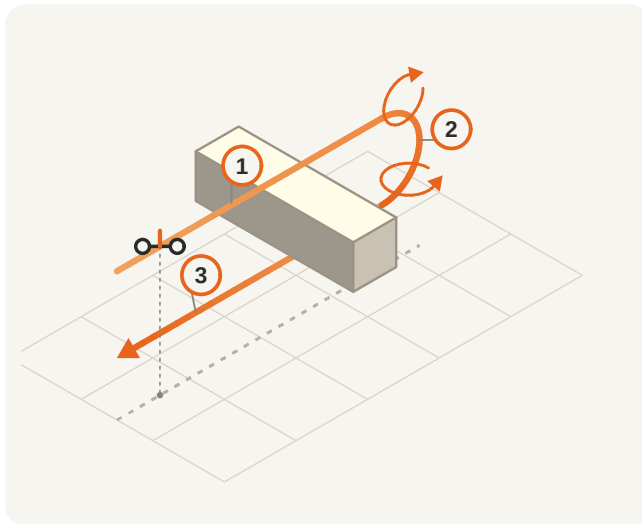
Split Yaw

Another Matty-family spin blend.

Matty Roll

Advanced

Matty Flips



HOW TO

A Matty with a full roll folded in. Approach passing over the object and throw a front flip at the far edge. As you come over the top facing up at the sky, roll a full 360, then, still facing up, carry on backwards under the object. The roll adds style without changing the line.

MASTERY LEVELS

- ★☆☆ Add a roll to the Matty pass-under.
- ★★★☆☆ Clean Matty with a smooth roll folded in.
- ★★★★★ Seamless Matty and roll, object framed throughout.

STEP BY STEP CONTROL INPUTS



1 Approach passing over object, then perform front flip



2 Once facing up to the sky, do a 360 roll



3 Once facing back up to the sky, continue backwards under object

PREREQUISITES & WHERE TO PRACTICE

- A clean Matty Flip
- A controllable roll

Have the Matty dialed first, then add the roll on the under-pass. Keep the roll rate matched to your speed so it flows, and keep the object framed.

COMMON MISTAKES

- ✗ Rolling too fast, off the line
- ✗ Losing the object through the roll
- ✗ Clipping the object under

VARIATIONS

Matty 360

A yaw 360 instead of a roll.

Matty Twister

A yaw twist during the flip.

RELATED TRICKS

Matty Flip

The base trick.

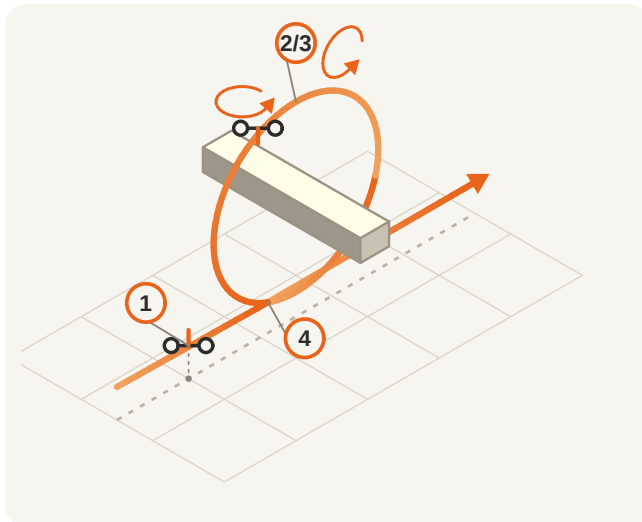
Matty 360

A yaw 360 folded in.

Mavani

Advanced

Maverick Loops



HOW TO

A powerloop with a 270 flip and a 180 inverted yaw folded in at the top. Come in under the object and pitch up as if starting a powerloop. When you are pointed straight up, throw a 270 forward flip so you end up inverted and facing backwards. There, at the peak, snap a 180 inverted yaw spin, then simply coast backwards to carry the loop the rest of the way round. Timing the flip and the yaw to the top of the loop is the whole trick.

MASTERY LEVELS

- ★☆☆ Fold a flip and a yaw into a powerloop.
- ★★★☆☆ Clean 270 flip into an inverted yaw at the top.
- ★★★★★ Seamless Mavani, flip and yaw timed to the peak.

STEP BY STEP CONTROL INPUTS



Approach under object, pitching drone up



When facing straight up, do a 270 forward flip



Once upside down facing backwards, do a 180 inverted yaw spin



Coast backwards to complete the loop

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A pinned inverted yaw spin

Get the powerloop climb solid so you arrive at the top with height to spare, because the flip and yaw both need room at the peak. Then add the 270 flip to stop yourself inverted and backwards, and only last the inverted yaw. Build it up with plenty of altitude.

COMMON MISTAKES

- ✗ Over or under rotating the 270 so you are not level inverted
- ✗ Rushing the yaw before the flip settles
- ✗ Losing orientation through the rotations at the top
- ✗ Not enough height to fit it all in

VARIATIONS

Mavik's Loop

The loop without the twist.

Donkey Loop

Another Maverick-loop variant.

RELATED TRICKS

Maverick Loop

The base loop.

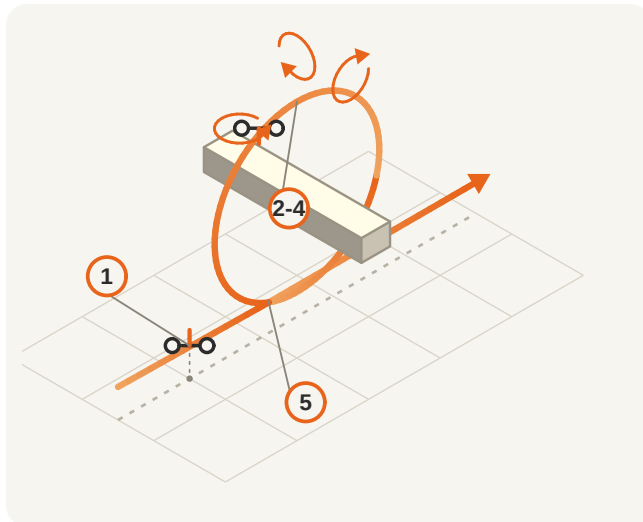
Forani

A barani on a Matty instead.

Mavik's Loop

Advanced

Maverick Loops



HOW TO

A powerloop with a quick pitch, roll and yaw stacked at the top. Come in under the object and pitch up as if starting a powerloop. Near the top of the loop, pitch the nose down so you end up inverted and facing backwards, then, in quick succession, snap a 180 roll to upright and a 180 yaw. With the three inputs done, simply coast the rest of the loop back under the object. Timing the pitch, roll and yaw tight to the peak is the whole trick.

MASTERY LEVELS

- ★☆☆ Get a pitch, roll and yaw in at the top of the loop.
- ★★★ Clean pitch-roll-yaw sequence timed to the peak.
- ★★★★ Seamless Mavik's Loop, the three inputs crisp and quick.

STEP BY STEP CONTROL INPUTS



Approach under object
pitching drone up



Near the top side of the
loop, pitch drone down to
face backwards upside
down



In quick succession,
perform a 180 roll to
upright



Next quickly 180 yaw



Finally coast under the
object

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- Quick, clean 180 roll and yaw

Get the powerloop climb solid so you arrive at the top with height to spare, because the three inputs all need room at the peak. Then work the pitch-to-inverted, and add the roll and yaw one at a time, keeping them quick and in order. Build it up with plenty of altitude.

COMMON MISTAKES

- ✗ Not enough height to fit the three inputs in
- ✗ Letting the roll and yaw run together sloppily
- ✗ Losing orientation through the sequence at the top
- ✗ Drifting off the loop line

VARIATIONS

Mavani

A Maverick loop with a barani twist.

Donkey Loop

Another Maverick-loop variant.

RELATED TRICKS

Mavani

A twisting Maverick loop.

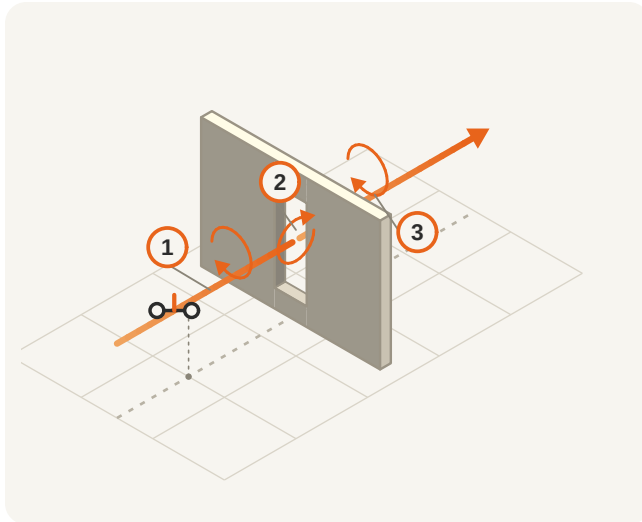
Powerloop

A loop over an object instead.

Ninja Star

Advanced

Gaps



HOW TO

Spinning through a narrow gap on the knife-edge, like a thrown ninja star. Blip the throttle and roll ninety degrees on the approach so you meet the gap sideways, knife-edge, to fit the slot. Yaw a full 360 as you thread it, which on the knife-edge reads as a flip through the opening, then roll back out level on the far side. It combines the precision of a knife-edge gap with a spin, so it is as much timing as line.

MASTERY LEVELS

- ★☆☆ Spin through the narrow gap knife-edge.
- ★★★☆☆ Clean spin threaded through the opening.
- ★★★★ Tight, fast spinning pass, dead through the slot.

STEP BY STEP CONTROL INPUTS



Throttle blip and approach gap with a 90 roll to face it sideways



360 yaw through the gap



Roll out on the other side of the gap

PREREQUISITES & WHERE TO PRACTICE

- A solid knife-edge gap
- A controllable spin

Get the knife-edge gap clean first, then add the spin, timing it so the narrow profile lines up with the slot as you pass. Start with a wider gap and tighten as your timing improves.

COMMON MISTAKES

- ✗ Mistiming the spin against the gap
- ✗ Not enough bank to fit the slot
- ✗ Clipping the sides on the way through

VARIATIONS

Knife Edge

A plain knife-edge gap.

Gaps

A flat gap thread.

RELATED TRICKS

Knife Edge

A plain knife-edge gap.

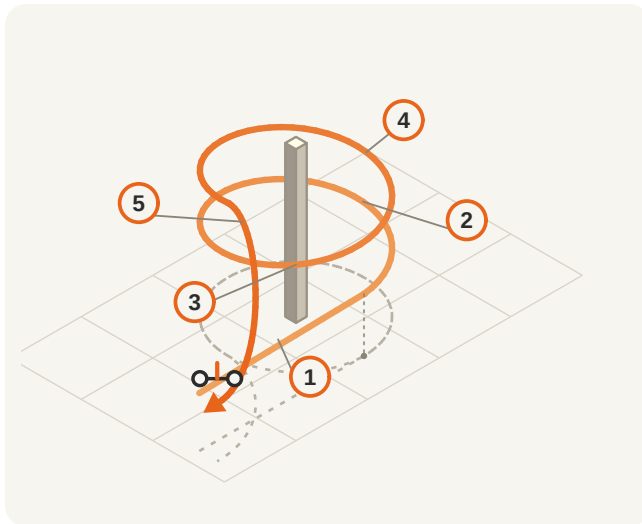
Gaps

Threading an opening flat.

Pole Dance

Advanced

Orbitals



HOW TO

Orbiting an object twice while keeping it framed, the way a dancer circles a pole. Fly toward the side of the object, then pitch back and yaw around it, holding it in view as you swing round. Level back out on the far side, still watching it, then pitch back and yaw around a second time. Level out once more and let the drone drift back away. Keeping the object framed through both orbits, coordinating the pitch-back with the yaw, is the whole dance.

MASTERY LEVELS

- ★☆☆ Spiral around the pole and peel off cleanly.
- ★★★☆☆ Tight, even spiral holding the pole close.
- ★★★★ Smooth, controlled spiral at close range, both directions.

STEP BY STEP CONTROL INPUTS



1 Fly toward side of the object



2 Pitch back and yaw around it, keeping object in view



3 Level back out, object in view



4 Pitch back and yaw around again keeping object in view



5 Level back out again object in view letting the drone back away

PREREQUISITES & WHERE TO PRACTICE

- Confident close-proximity orbits
- Coordinated pitch and yaw together

Pick a tall, clean pole with room around it. Start with a wide, gentle spiral and tighten the radius as your control improves. Both directions. Keeping the radius constant while climbing is the skill.

COMMON MISTAKES

- ✗ Losing the object out of frame through an orbit
- ✗ Pitch and yaw out of sync, so the orbit wobbles
- ✗ Clipping the object when you tighten in

VARIATIONS

Trippy Spin

An inverted orbit around the top.

Whiplash

A quick yaw whip around the pole.

RELATED TRICKS

Orbits

The flat, single-height version.

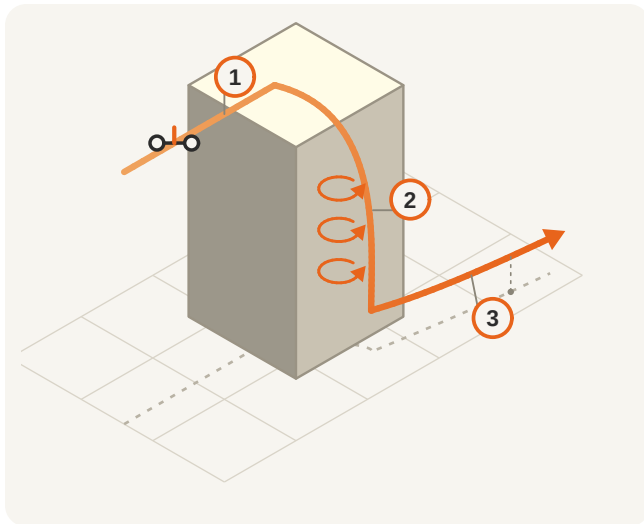
Trippy Spin

The inverted cousin.

Power Drill

Advanced

Powerloops



HOW TO

A dive down the face of a building or drop with full-rotation rolls all the way down, drilling like a bit. Approach the dive, then commit down the line rolling continuously, one full rotation after another, right to the bottom. Catch it there with a throttle blip to arrest the drop. It is a spectacular way to lose altitude, all controlled rotation.

MASTERY LEVELS

- ★☆☆ Corkscrew straight down and pull out.
- ★★★ Tight, even corkscrew descent, clean pull-out.
- ★★★★ Fast, drilling descent held dead-vertical, tidy exit.

STEP BY STEP CONTROL INPUTS



Approach dive



Dive the path down with multiple full rotation rolls to the bottom of the dive



Catch at the bottom with a throttle blip

PREREQUISITES & WHERE TO PRACTICE

- A confident roll and descent
- Orientation through continuous rotation

Give yourself height and practise a tight, even corkscrew straight down, keeping it vertical and controlled, then pull out cleanly at the bottom. The continuous rotation is disorienting, so build up slowly.

COMMON MISTAKES

- ✗ The spiral drifting off vertical
- ✗ Losing orientation in the rotation
- ✗ Pulling out too late

VARIATIONS

Barrel Roll

A horizontal corkscrew.

Sbang Tornado

A flat-spin descent instead of a corkscrew.

RELATED TRICKS

Barrel Roll

A horizontal corkscrew.

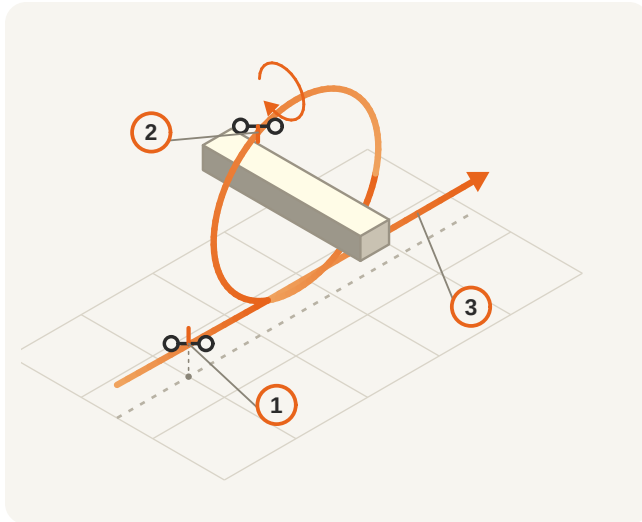
Powerloop

Another power descent.

Power Roll

Advanced

Powerloops



HOW TO

A powerloop with a full roll folded in at the top, the roll-axis cousin of the powerloop family. Pass under the object, throttle up and pitch back to climb the loop. At the peak, where you are inverted, snap a 360 roll, then carry the loop the rest of the way round and catch it at the bottom, continuing under the object. Timing the roll to the inverted peak is what makes it read.

MASTERY LEVELS

- ★☆☆ Powerloop with a 360 roll at the inverted top.
- ★★★ Clean roll pinned at the peak of the loop.
- ★★★★ Seamless power roll, the roll timed to inverted.

STEP BY STEP CONTROL INPUTS



1 Pass under the object, throttle up and begin pitching drone up



2 At the peak of the loop when inverted, 360 roll



3 Catch at the bottom of the loop continuing under

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A quick, clean 360 roll

Get the powerloop climb solid so you arrive inverted at the top with height to spare, because the roll needs room at the peak. Then add the 360 roll right at inverted and carry the loop through. Build it up with plenty of altitude.

COMMON MISTAKES

- ✗ Rolling before you are level inverted at the top
- ✗ A slow roll that costs the loop its shape
- ✗ Losing orientation through the roll

VARIATIONS

Mavvy Roll

The same loop kept upright over the top.

Powerloop

The loop without the roll.

RELATED TRICKS

Mavvy Roll

The upright-over-the-top version.

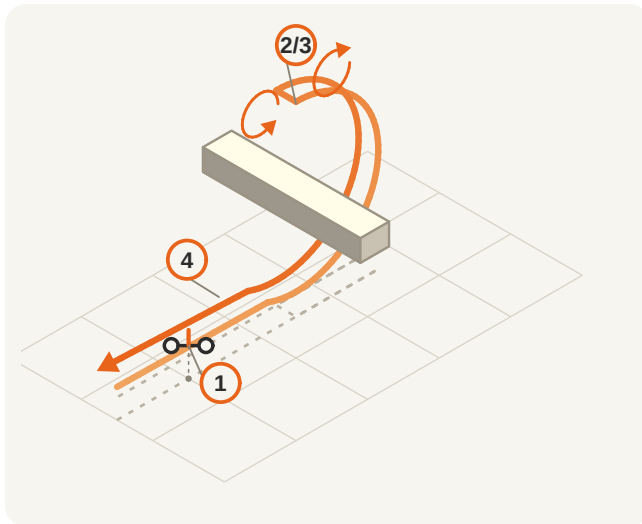
Powerloop

The base loop.

Power Switch

Advanced

Powerloops



HOW TO

A powerloop where the flip direction switches at the top. Come in under the object and pitch up to climb the loop. At the top, complete the flip to upright and hold a brief pause there, then blip the throttle and throw a forward flip the other way, switching direction. Ride the loop back down and carry on backwards under the object, on the same line you started on. The pause between the two flips is the signature.

MASTERY LEVELS

- ★☆☆ Loop up and switch the flip direction at the top.
- ★★★ Clean flip-to-upright, pause, then a forward flip.
- ★★★★ Seamless power switch, both flips crisp at the peak.

STEP BY STEP CONTROL INPUTS



Approach under object, pitching drone up



At the top of the loop complete the flip to upright with a pause



Throttle blip and forward flip



Continue backwards under the object on the path you started on

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- Control to pause and reverse a flip

Get the powerloop climb solid so you arrive at the top with height to spare, then work stopping the flip upright and holding it before the second flip. The pause and the switch of direction are the hard part, so build it up with plenty of altitude.

COMMON MISTAKES

- ✗ No pause, so the two flips blur together
- ✗ Not committing the throttle up the climb
- ✗ Losing the line so the exit misses the start path

VARIATIONS

Powerloop

Without the switch.

Power Swap

A swap with a different exit.

RELATED TRICKS

Powerloop

The loop it is built on.

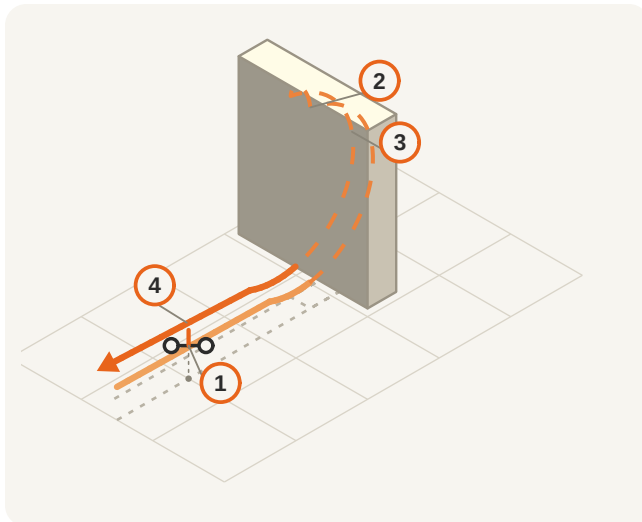
Power Swap

A related loop-and-swap.

Power Switch Tap

Advanced

Wall Tricks



HOW TO

Fly beneath an object and initiate a powerloop. At the peak of the loop, pitch back to look downward and tap the object with the bottom of your whoop. Without delay, perform a Matty flip to rewind back under the object.

MASTERY LEVELS

- ★☆☆ Powerloop, tap the top, and Matty out.
- ★★★☆☆ Clean powerloop with a soft tap into a tidy Matty rewind.
- ★★★★ Seamless Power Switch Tap, object framed and barely touched.

STEP BY STEP CONTROL INPUTS



1 Pass under the object, throttle up on the opposite side pitching the drone up



2 When facing up, throttle down, continue pitching up until facing the ground, with a wall tap



3 Blip throttle to soften the wall tap, then pitch the drone down to reverse into a matty



4 Catch and backwards towards the start

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A soft, precise tap and a solid Matty

Have the powerloop and the Matty solid, then work the tap at the peak: pitch back to look down and touch the object lightly before flipping back under. Keep the tap gentle and the object framed. Scout the object.

COMMON MISTAKES

- ✗ Tapping too hard and knocking off line
- ✗ Delaying between the tap and the Matty
- ✗ Losing the object over the top

VARIATIONS

Ceiling Tap

A tap on an overhead surface.

Wall Tap

A tap on a vertical face.

RELATED TRICKS

Powerloop

The loop it is built on.

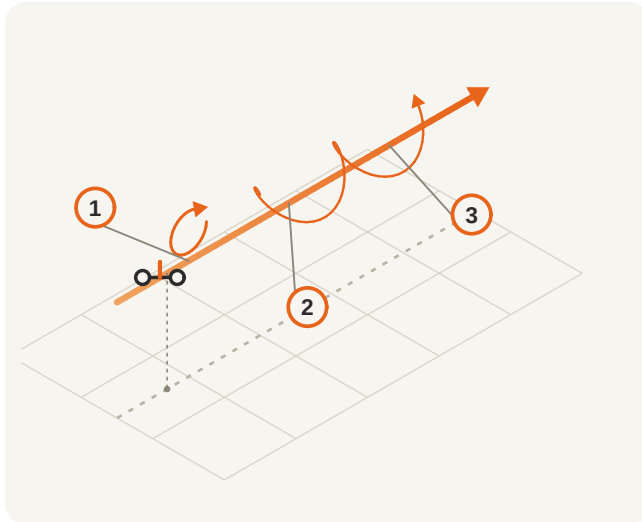
Matty Flip

The rewind it finishes with.

Rapid Look Back + Slow Double Roll

Advanced

Open Air



HOW TO

A study in contrast: a rapid, snapped lookback followed by a slow, deliberate double roll. Snap a rapid lookback, ease into a slow double roll, and catch level. The interplay between the sharp snap and the languid roll is the whole appeal.

MASTERY LEVELS

- ★☆☆ Snap a lookback, then a slow double roll.
- ★★★☆☆ Clean rapid lookback flowed into a smooth double roll.
- ★★★★★ Seamless snap-and-slow combo, the pace contrast crisp.

STEP BY STEP CONTROL INPUTS

1



Blip the throttle and snap a 180 forward flip, inverting while looking back

2



Double roll

3



Catch and level out

PREREQUISITES & WHERE TO PRACTICE

- A snappy lookback
- A controlled, even double roll

Snap the lookback sharp, then ease straight into a slow, even double roll, holding the line through both. Work the pace contrast so the snap and the slow roll each read clearly.

COMMON MISTAKES

- ✗ A soft lookback that does not snap
- ✗ Rushing the double roll
- ✗ Drifting off line between the two

VARIATIONS

Look Back + Double Roll

Capped with a flip.

Double Roll

The rolls on their own.

RELATED TRICKS

Double Roll

The rolls it uses.

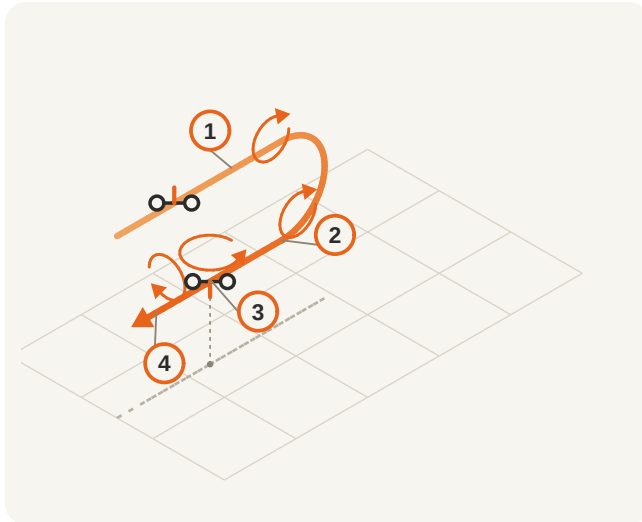
Lookback & Loop

Another lookback combo.

Rapid Split-S + Inverted Yaw Spin Combo

Advanced

Split-S



HOW TO

A fast Split-S dropped straight into an inverted yaw spin. Rapid Split-S over and down, level off inverted, and pin an inverted 360. The speed of the Split-S sets up the spin, so carrying it cleanly into the inverted pivot is the crux.

MASTERY LEVELS

- ★☆☆ Rapid Split-S into an inverted spin.
- ★★★☆☆ Clean fast Split-S flowed to a pinned inverted 360.
- ★★★★ Seamless rapid Split-S and inverted spin, dead on line.

STEP BY STEP CONTROL INPUTS

1



Blip throttle and do a rapid Split-S

2



Level out upright, then blip throttle and pitch into inverted

3



Inverted 360 yaw

4



180 roll out to level

PREREQUISITES & WHERE TO PRACTICE

- A fast, clean Split-S
- A pinned inverted yaw spin

Have the Split-S and the inverted spin each solid, then chain them: come out of the fast Split-S level and inverted, and pin the spin. Keep the line straight through the transition.

COMMON MISTAKES

- ✗ Carrying the Split-S off line into the spin
- ✗ Not leveling inverted before the spin
- ✗ Losing orientation upside-down

VARIATIONS

Split-S

The reversal on its own.

RELATED TRICKS

Split-S

The reversal it starts with.

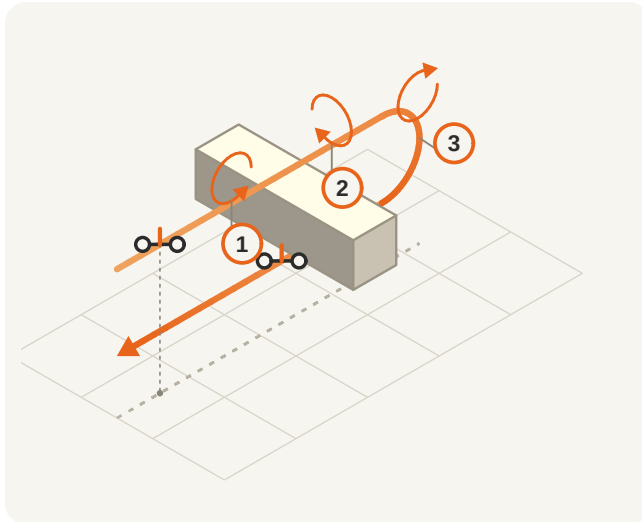
Inverted Yaw Spin

The spin it ends on.

Rubik's Cube Pendulum

Advanced

Open Air



HOW TO

A Rubik's Cube flown on a pendulum: swing down into a pendulum arc and turn the quad's axes at the bottom before swinging back up. Swing down, turn the axes at the bottom, and swing back up and out. It marries a multi-axis combo to the momentum of a pendulum.

MASTERY LEVELS

- ★☆☆ Turn the axes while swinging on a pendulum.
- ★★☆ Clean pendulum swing with the axes turned at the bottom.
- ★★★ Seamless Rubik's Cube Pendulum, every turn on the swing.

STEP BY STEP CONTROL INPUTS



1 Approach high, throttle blip and 180 back flip passing above the object



2 Cross coordinate a 360 roll, with a throttle blip in the middle of the roll



3 Half matty: once the roll is complete facing the sky, pitch the drone down and raise throttle to continue through the objects

PREREQUISITES & WHERE TO PRACTICE

- A confident pendulum
- A clean multi-axis combo

Have the pendulum and the Rubik's Cube each solid, then work the axis turns into the bottom of the swing where you have the most room. Keep the swing smooth and the turns crisp.

COMMON MISTAKES

- ✗ Rushing the turns and killing the swing
- ✗ Losing the line at the bottom
- ✗ An uneven, lopsided pendulum

VARIATIONS

Rubik's Cube

Without the pendulum.

Pendulum

The swing on its own.

RELATED TRICKS

Rubik's Cube

The multi-axis combo.

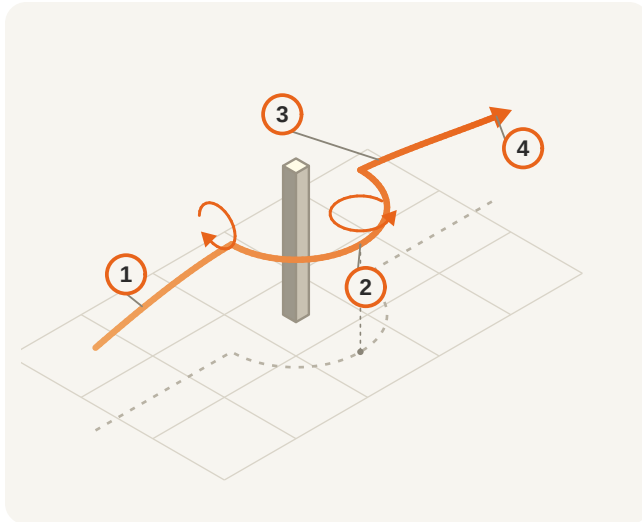
Pendulum

The swing it rides.

Rufio

Advanced

Open Air



HOW TO

A lively flourish: a pop up, a flip at the apex, and a roll on the way down. Pop it up, flip at the top, and roll away as you descend. It is a showy, expressive move, all about flow and attitude rather than any single hard element.

MASTERY LEVELS

- ★☆☆ Pop, flip, and roll away.
- ★★★☆☆ Clean pop into a flip and a rolling exit.
- ★★★★ Seamless Rufio, lively and smooth, dead on line.

STEP BY STEP CONTROL INPUTS

1  

Fly next to the object with a throttle punch

2  

Cross coordinate roll while tracking the object

3  

Blip the throttle and correct the roll

4  

Track the object while continuing backwards

PREREQUISITES & WHERE TO PRACTICE

- A clean flip and roll
- A feel for a floaty pop

Pop up into a flip and flow a roll as you come down, keeping it loose and lively rather than mechanical. Work the timing so the three parts blend into one flourish.

COMMON MISTAKES

- ✗ A stiff, mechanical feel
- ✗ Losing the line through the pop
- ✗ Dropping too fast to flow the roll

VARIATIONS

Juicy Flick

A floaty pop to inverted.

Rubik's Cube

A tighter multi-axis combo.

RELATED TRICKS

Juicy Flick

Another expressive pop.

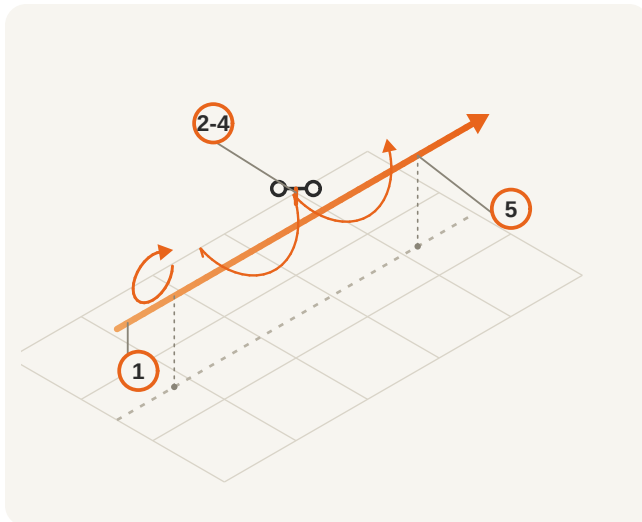
Loop

The flip at its core.

Sbang Tornado

Advanced

Open Air



HOW TO

A tornado: a flat yaw spin carried straight down into a fast, spinning descent. Set up over your spot, flat-spin straight down through a tight vertical column, and pull out to fly away at the bottom. The continuous flat rotation while dropping is what makes it a tornado, and what makes it disorienting.

MASTERY LEVELS

- ★☆☆ Flat-spin straight down and pull out.
- ★★★ Tight, even tornado descent with a clean exit.
- ★★★★ Fast, dead-vertical tornado, tidy pull-out on line.

STEP BY STEP CONTROL INPUTS

1



Start low near the ground, throttle blip, then pitch forward until looking at the ground

2



Cross coordinate roll

3



Softly blip throttle as you see the sky, keeping the roll consistent

4



Repeat steps 2 and 3 for as many tornado rolls as you want

5



Level out to finish moving backwards

PREREQUISITES & WHERE TO PRACTICE

- A solid flat yaw spin
- Orientation through continuous rotation

Have the flat spin dialed, then add descent, keeping the column tight and vertical as you spin down, and pull out cleanly at the bottom. Give yourself height and build up slowly: the spinning drop is disorienting.

COMMON MISTAKES

- ✗ The column drifting off vertical
- ✗ Losing orientation in the spin
- ✗ Pulling out too late

VARIATIONS

Power Drill

A rolling corkscrew descent instead.

RELATED TRICKS

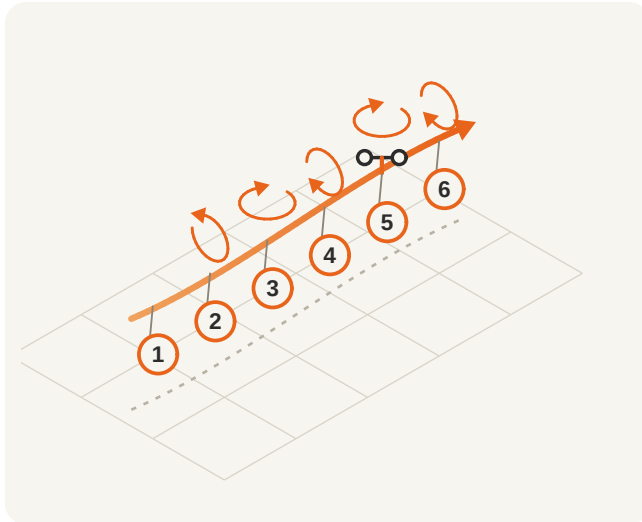
Power Drill

A corkscrew descent.

Side Lookback + Inverted Yaw Spin

Advanced

Open Air



HOW TO

An inverted sideways line dressed with a lookback and a full inverted yaw spin. Track sideways inverted, yaw back for a look, and pin an inverted 360. Holding a steady inverted side-track while swinging the view around is what makes it hard.

MASTERY LEVELS

- ★☆☆ Side-track inverted, look back, and spin.
- ★★★ Clean inverted side line into a lookback and 360.
- ★★★★ Seamless side lookback and inverted spin, dead on line.

STEP BY STEP CONTROL INPUTS

1



Roll slightly left so the horizon sits at a small angle

2



Start a side lookback with a throttle blip and coordinated roll/pitch, horizon almost level leaning left

3



Yaw and roll for the lookback

4



Throttle blip and cross coordinate roll

5



Once inverted, yaw spin

6



Roll out to upright

PREREQUISITES & WHERE TO PRACTICE

- Confident inverted side flight
- A pinned inverted yaw spin

Get a steady inverted sideways line first, then add the lookback, and finally the full spin, keeping the line tracking through all three. Build the speed of the spin up as it smooths out.

COMMON MISTAKES

- ✗ The inverted side-track sagging
- ✗ Drifting off line through the spin
- ✗ Losing orientation inverted

VARIATIONS

Inverted Lookback

The look without the spin.

Inverted Yaw Spin

The spin on its own.

RELATED TRICKS

Inverted Lookback

The lookback.

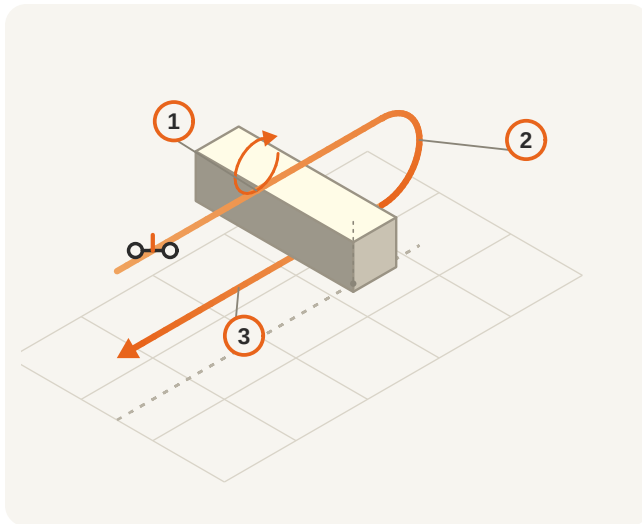
Inverted Yaw Spin

The spin.

SomerMatty

Advanced

Matty Flips



HOW TO

A Matty with a somersault folded in above the bar. Rise over the object, somersault above the bar, and finish the Matty flip and pass back under. It stacks an extra flip on top of the Matty, so keeping the object framed through both rotations is the hard part.

MASTERY LEVELS

- ★☆☆ Fold a somersault into the Matty.
- ★★★☆☆ Clean somersault and Matty flowed together.
- ★★★★★ Seamless SomerMatty, both crisp, object framed throughout.

STEP BY STEP CONTROL INPUTS



Rise over the object



Forward flip above object



Matty flip and under

PREREQUISITES & WHERE TO PRACTICE

- A solid Matty Flip
- A clean flip on demand

Have the Matty dialed, then add the somersault above the bar, flowing it into the Matty flip. Keep the object framed through both rotations. Give it height and build up slowly.

COMMON MISTAKES

- ✗ Losing the object through the somersault
- ✗ A rough hand-off between the two flips
- ✗ Dropping onto the bar between rotations

VARIATIONS

Matty Flip

Without the somersault.

Double Flip

Two flips without the object.

RELATED TRICKS

Matty Flip

The base trick.

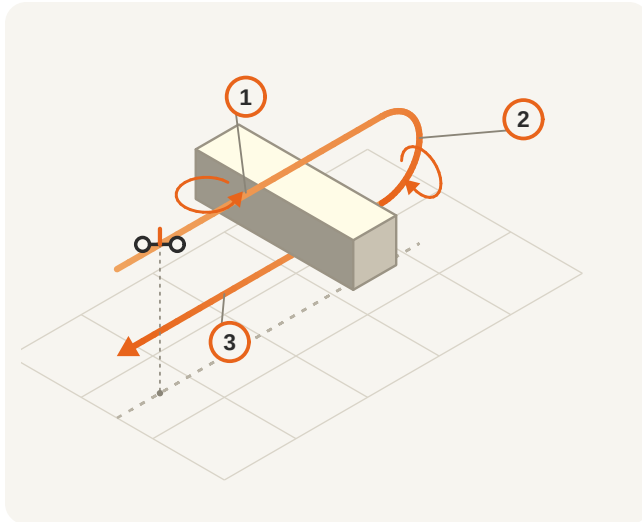
Loop to Matty

A loop into the Matty instead.

Stall 540 Half Matty Rewind

Advanced

Rewinds



HOW TO

A stall capped with a 540 spin, flowed through a half-Matty and reflowed back under. Climb to a stall, spin a 540 and carry a half-Matty over, then rewind back under. It threads a spin, an over-and-under, and a rewind into one demanding line.

MASTERY LEVELS

- ★☆☆ Stall, spin a 540, half-Matty and rewind.
- ★★★ Clean 540 into a half-Matty and a smooth rewind.
- ★★★★ Seamless stall-540-half-Matty-rewind, object framed throughout.

STEP BY STEP CONTROL INPUTS



Approach above the object with a 180 yaw spin passing the object



Once facing backwards, throttle blip and do a 540 roll



Once upside down, pitch down to upright and continue backwards under the object

PREREQUISITES & WHERE TO PRACTICE

- A confident stall and 540 spin
- A solid half-Matty and rewind

Have the 540, the half-Matty, and the rewind each solid, then chain them off a stall, keeping the object framed through the over-and-under. It is a long combo, so build it a piece at a time.

COMMON MISTAKES

- ✗ Rushing the 540 and drifting off line
- ✗ A rough hand-off into the half-Matty
- ✗ Losing the object through the rewind

VARIATIONS

Stall Half Matty Rewind

Without the 540 spin.

Five40 Half Matty

The 540 half-Matty without the rewind.

RELATED TRICKS

Stall Half Matty Rewind

The simpler version.

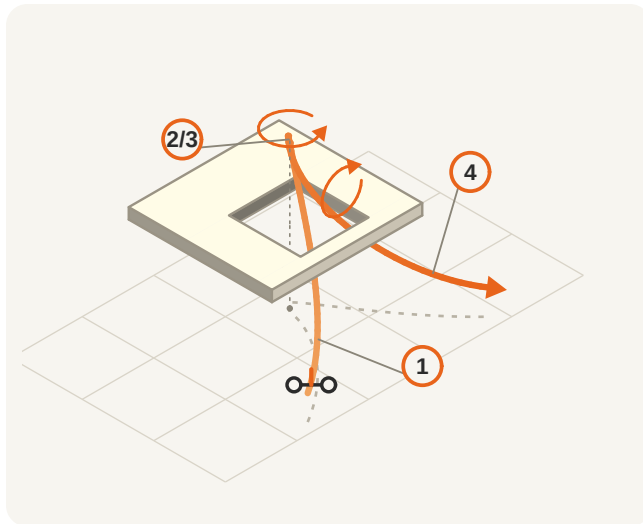
Stall Rewind

The rewind it finishes with.

Stellar Eject Roll

Advanced

Miscellaneous



HOW TO

A bigger, showier eject roll, tumbling through multiple rolls as you bail off the line. Cruise level, pick your out, then throw the quad into a string of rolls, ejecting away and down, and ride it out to a clean descent. It is the eject roll turned up to eleven.

MASTERY LEVELS

- ★☆☆ Tumble into rolls and eject away and down.
- ★★★ Multiple clean rolls into a controlled descent.
- ★★★★ A big, stylish eject, rolls tumbling out and clear.

STEP BY STEP CONTROL INPUTS



1 Eject: punch up through a gap, pitch to face down



2 Facing down through the gap from above, pin a 360 roll



3 As you begin to fall, quickly pitch down



4 Once facing the sky, continue through the gap backwards and slowly level out

PREREQUISITES & WHERE TO PRACTICE

- A confident eject roll
- Comfortable tumbling through multiple rolls

Build up from a single eject roll, adding rolls as your control improves. Give yourself plenty of height, since it eats altitude fast, and keep the descent controlled.

COMMON MISTAKES

- ✗ Losing orientation in the tumble
- ✗ Diving out of control
- ✗ Running out of height

VARIATIONS

Eject Roll

The single-roll version.

Juicy Eject

A floatier eject.

RELATED TRICKS

Eject Roll

The basic version.

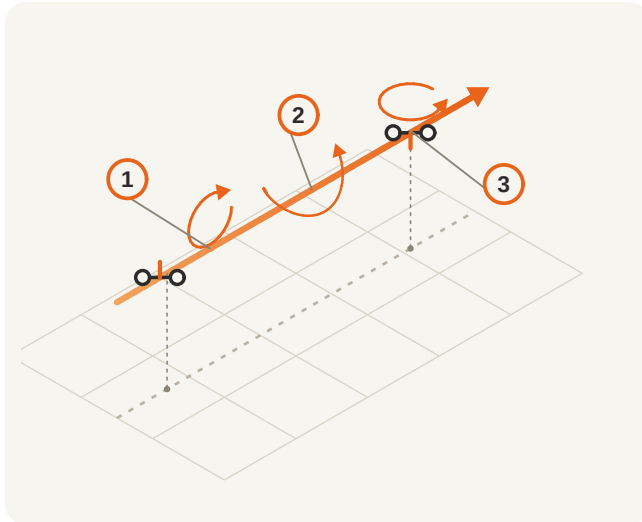
Roll

The rotation it multiplies.

540 Flip + Roll + Inverted Yaw Spin

Master

Open Air



HOW TO

A big three-part combo: a 540 flip, a roll, and a full inverted yaw spin, strung along one line. Snap the 540 flip, roll on the line, and pin an inverted 360. It packs three distinct rotations into one pass, so smoothing the hand-offs is everything.

MASTERY LEVELS

- ★☆☆ 540 flip, roll, and inverted spin in one line.
- ★★★☆☆ Clean 540 flip and roll into a pinned inverted 360.
- ★★★★★ Seamless 540-flip-roll-spin, every part crisp and on line.

STEP BY STEP CONTROL INPUTS



1 Blip throttle then drop throttle and snap a 540 flip



2 Once facing backwards, 360 roll, blipping throttle again when upright mid-roll



3 Once upside down, 180 yaw spin



4 Roll out to complete

PREREQUISITES & WHERE TO PRACTICE

- Clean flips, rolls, and inverted spins
- A feel for chaining rotations

Have each rotation solid on its own, then chain them along one line, keeping it straight through all three. Build the pace up only as the hand-offs clean up.

COMMON MISTAKES

- ✗ Drifting off line between rotations
- ✗ A rushed hand-off into the spin
- ✗ Losing orientation on the inverted 360

VARIATIONS

540 Delayed Front Flip

A 540 with a delayed flip.

Lookback + Roll + Inverted Yaw Spin

A lookback version.

RELATED TRICKS

540 Delayed Front Flip

Another 540 combo.

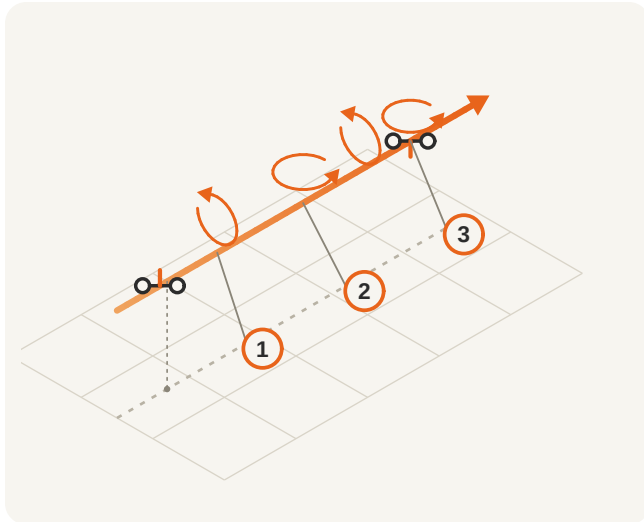
Inverted Yaw Spin

The spin it ends on.

Advanced Combo #3

Master

Open Air



HOW TO

A flowing open-air showcase: a flip in, a rolling middle, and a yaw out to finish. Flip in, roll through the middle, and yaw out. It is a freestyle sampler, chaining pitch, roll, and yaw into one clean line.

MASTERY LEVELS

- ★☆☆ Flip, roll, and yaw out in one line.
- ★★★ Clean flip and rolling middle into a tidy yaw out.
- ★★★★ Seamless advanced combo, every element crisp and on line.

STEP BY STEP CONTROL INPUTS



Slant sideways: blip throttle, pitch and roll left until the horizon is almost vertical (around 70 degrees)



Start a sideways rewind: yaw and roll right while pitching down



Blip throttle and roll



Once inverted, yaw spin

PREREQUISITES & WHERE TO PRACTICE

- Clean flips, rolls, and yaw spins
- A feel for stringing them together

Have each element solid, then chain them flip-roll-yaw on one line, keeping it straight and the hand-offs smooth. Vary the pace and make it your own.

COMMON MISTAKES

- ✗ Drifting off line between elements
- ✗ A jerky hand-off between rotations
- ✗ An uneven turn on one of the axes

VARIATIONS

Rubik's Cube

A tighter three-axis combo.

540 Flip + Roll + Inverted Yaw Spin

A bigger combo.

RELATED TRICKS

Rubik's Cube

A three-axis cousin.

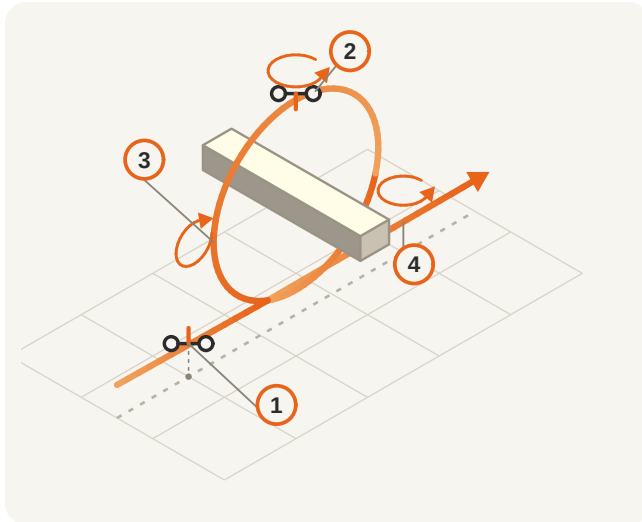
Loop

The flip at its core.

Barani

Master

Powerloops



HOW TO

Fly beneath the object and perform half a Powerloop so the horizon sits in the top half of your field of view, execute a precise 180° inverted yaw spin, and conclude with a Matty Flip back under the object.

MASTERY LEVELS

- ★☆☆ Half-loop, 180 inverted yaw, and Matty out.
- ★★★☆☆ Clean half-powerloop and inverted 180 into a tidy Matty.
- ★★★★ Seamless Barani, object framed through all three parts.

STEP BY STEP CONTROL INPUTS



1 Begin a powerloop under an object, pitching the drone up



2 Cut throttle as you invert, then 180 yaw



3 Half matty: follow the loop backwards, pitching the drone down as you complete the loop



4 Catch with a 180 yaw out

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A pinned inverted yaw and a solid Matty

Have the half-powerloop, the inverted 180, and the Matty each solid, then chain them beneath the object, keeping the horizon and the object where you want them through each part. It takes height and commitment.

COMMON MISTAKES

- ✗ Rushing the inverted 180 off line
- ✗ Losing the object between the parts
- ✗ A rough hand-off into the Matty

VARIATIONS

True Barani

The open-air twisting version.

Flipani

A flip dressed with a barani.

RELATED TRICKS

Powerloop

The half-loop it starts with.

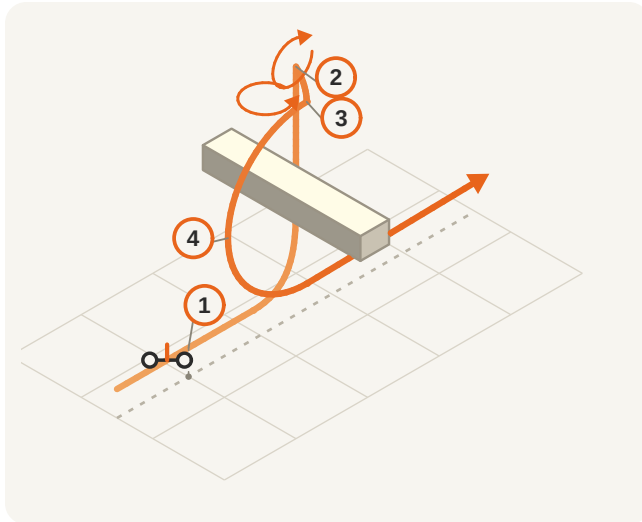
Matty Flip

The Matty it finishes with.

Donkey Loop

Master

Maverick Loops



HOW TO

A Maverick loop finished with a sharp donkey-kick: a hard pitch flick on the way out. Drop into the Maverick loop beneath the bar, loop up and around, and snap a donkey-kick as you exit. The kick is what gives it the mule-like snap and the name.

MASTERY LEVELS

- ★☆☆ Maverick loop with a donkey-kick out.
- ★★★ Clean Maverick loop finished with a sharp kick.
- ★★★★ Tight Donkey Loop, crisp kick, dead on line.

STEP BY STEP CONTROL INPUTS



1 Begin a powerloop: throttle up and pitch the drone up



2 Once facing up, 270 pitch the drone down to face backwards upside down



3 Once facing backwards, 360 inverted yaw spin



4 Once backwards again, begin pitch up to pass under the object completing the loop

PREREQUISITES & WHERE TO PRACTICE

- A confident Maverick loop
- A sharp, controlled pitch flick

Have the Maverick loop dialed, then add the kick on the exit: a hard, quick pitch flick as you come out. Keep it snappy and clear of the bar. Give yourself clearance and build up slowly.

COMMON MISTAKES

- ✗ A soft kick that does not read
- ✗ Clipping the bar on the exit
- ✗ Losing the line after the kick

VARIATIONS

Mavik's Loop

The loop without the kick.

Mavani

A Maverick loop with a twist instead.

RELATED TRICKS

Mavik's Loop

The base loop.

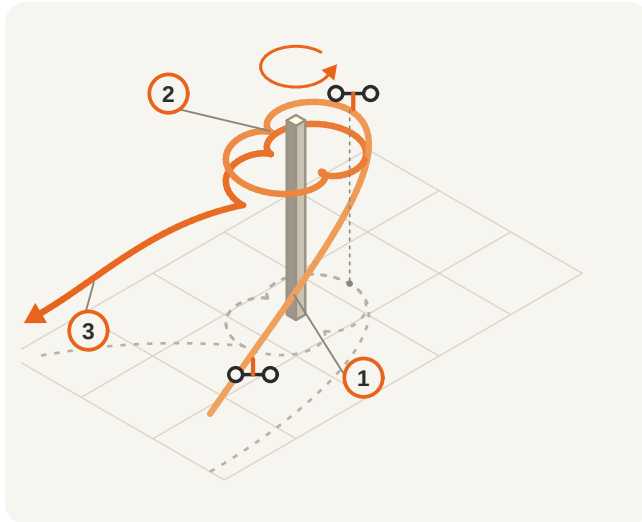
Mavani

A twisting Maverick loop.

Double Rolling Trippy Spin

Master

Orbitals



HOW TO

A Trippy Spin with rolls folded into the inverted orbit. Get inverted and orbit the top of an object as in a Trippy Spin, but roll the quad twice as you go around, layering rolls onto the spin. Keeping the object framed through the rolling orbit is brutally hard.

MASTERY LEVELS

- ★☆☆ Add rolls to the inverted trippy orbit.
- ★★★☆☆ Clean rolling orbit, object held framed.
- ★★★★★ Seamless double-rolling trippy spin, object locked in view.

STEP BY STEP CONTROL INPUTS



1 Approach the side of the object, pitch up to begin rotation once you are facing up



2 Fold rolls into the spin



3 Pitch down drone to exit

PREREQUISITES & WHERE TO PRACTICE

- A solid Trippy Spin
- Rolling while orbiting inverted

Have the Trippy Spin dialed cold, then add the rolls a little at a time, keeping the object framed. It is one of the hardest orbital tricks, so expect it to take a while.

COMMON MISTAKES

- ✗ Losing the object through the rolls
- ✗ The rolls wobbling the orbit
- ✗ Dropping the inverted hang

VARIATIONS

Trippy Spin

Without the rolls.

Trippy Switch

A trippy spin that reverses.

RELATED TRICKS

Trippy Spin

The base inverted orbit.

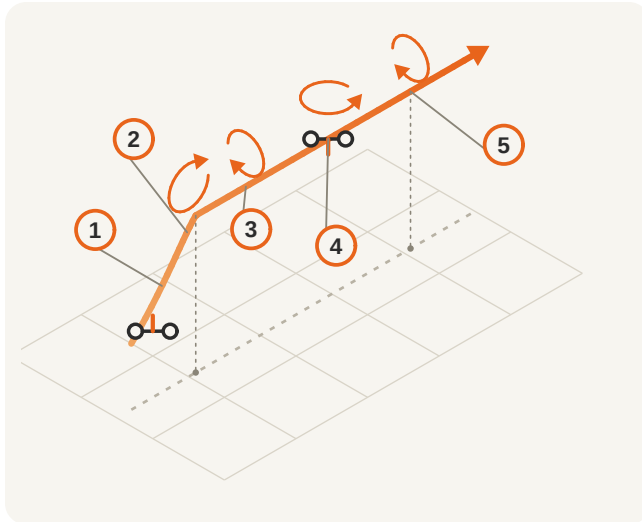
Trippy Switch

A reversing trippy spin.

Eject Lookback Roll + Inverted Yaw Spin

Master

Open Air



HOW TO

A hard eject to inverted flowed into a lookback roll and a full inverted yaw spin. Eject to inverted, lookback and roll, and pin an inverted 360. The eject launches the whole sequence, so committing to it sets up everything that follows.

MASTERY LEVELS

- ★☆☆ Eject, lookback-roll, and inverted spin.
- ★★★ Clean eject into a lookback roll and a pinned inverted 360.
- ★★★★ Seamless eject-lookback-roll-spin, dead on line.

STEP BY STEP CONTROL INPUTS

1  	2  	3  	4  	5  
Eject with a throttle punch	Forward flip	Once upside down, begin a cross coordinated 360 roll; throttle blip when the drone is upright	Continue rolling until upside down again, then begin the yaw spin	Roll out

PREREQUISITES & WHERE TO PRACTICE

- A confident eject
- A lookback roll and a pinned inverted spin

Have the eject, the lookback roll, and the inverted spin each solid, then chain them, committing to the eject and flowing straight into the roll and spin. Give it height and build up slowly.

COMMON MISTAKES

- ✗ A weak eject that stalls the combo
- ✗ Losing the line through the roll
- ✗ Losing orientation on the inverted spin

VARIATIONS

Eject Roll

Just the eject and roll.

Lookback + Roll + Inverted Yaw Spin

Without the eject.

RELATED TRICKS

Eject Roll

The eject-and-roll.

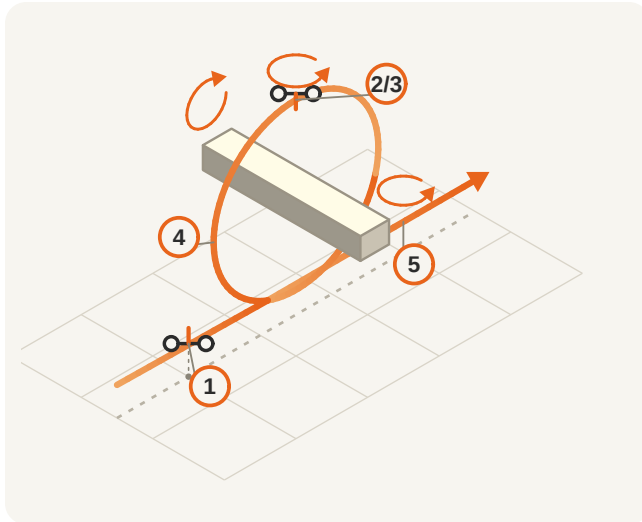
Lookback + Roll + Inverted Yaw Spin

The non-eject version.

Flipani

Master

Powerloops



HOW TO

A flip dressed with a barani, a twist folded in at the top of the loop. Pull into the flip, barani-twist at the top, and catch level as you fly out. It stacks a twist onto a clean loop, so coordinating the twist with the flip is the challenge.

MASTERY LEVELS

- ★☆☆ Add a barani twist to the flip.
- ★★★☆☆ Clean flip with a smooth twist at the top.
- ★★★★★ Seamless Flipani, flip and twist as one, dead on line.

STEP BY STEP CONTROL INPUTS



1 Begin a powerloop under an object, pitching the drone up



2 Cut throttle as you invert, then 180 yaw



3 Quickly follow with a 360 front flip



4 Half matty: follow the loop backwards, pitching the drone down as you complete the loop



5 Catch with a 180 yaw out

PREREQUISITES & WHERE TO PRACTICE

- A clean flip
- Comfortable twisting through a loop

Have the flip solid, then add the barani twist at the top, a little at a time. Keep the loop round and the line straight. Give it height and build up slowly.

COMMON MISTAKES

- ✗ Too much twist, throwing off the flip
- ✗ Losing the line through the twist
- ✗ An uneven loop out of the barani

VARIATIONS

Rollani

A barani off a roll instead of a flip.

Barani

The powerloop-context barani.

RELATED TRICKS

Loop

The flip it is built on.

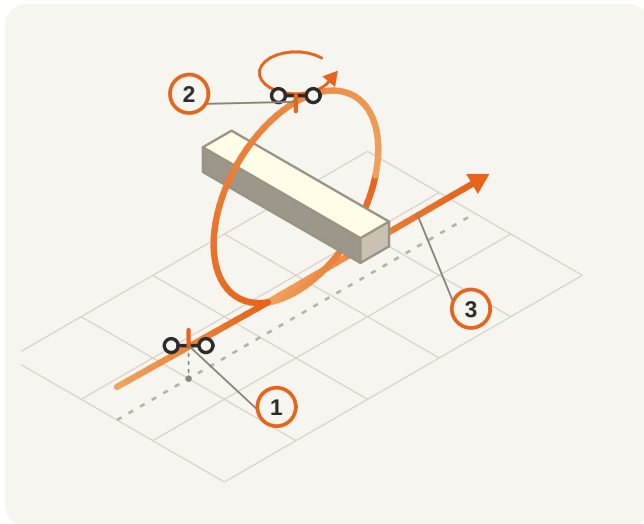
Rollani

The roll sibling.

Inverted 360 Powerloop

Master

Powerloops



HOW TO

A powerloop with a full flat 360 pinned at the inverted top. Carry throttle up and over the obstacle, and as you hang inverted at the peak, pin a complete 360 yaw before carrying the loop back down. Spinning while inverted at the top of a loop is as hard as it sounds.

MASTERY LEVELS

- ★☆☆ Add a full 360 yaw at the top of the powerloop.
- ★★★☆☆ Clean powerloop with a pinned 360 at the peak.
- ★★★★★ Seamless powerloop and inverted 360, dead on line.

STEP BY STEP CONTROL INPUTS



1 Approach under object, pitching drone up



2 At the peak of the loop, do a 360 yaw spin



3 Carry the loop back down

PREREQUISITES & WHERE TO PRACTICE

- A confident powerloop
- A pinned inverted yaw spin

Have the powerloop and the inverted spin dialed on their own, then combine them, pinning the 360 right at the top where you hang inverted. It takes height and commitment.

COMMON MISTAKES

- ✗ Rushing the 360 and drifting off the loop
- ✗ Losing the inverted timing
- ✗ Not committing the throttle

VARIATIONS

Powerloop

Without the spin.

Inverted Yaw Spin

The spin on its own.

RELATED TRICKS

Powerloop

The loop it is built on.

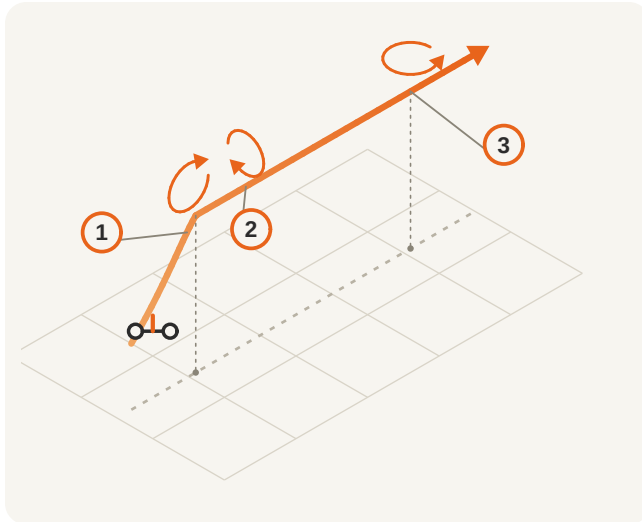
Inverted Yaw Spin

The inverted spin.

Look Back + Double Roll

Master

Open Air



HOW TO

The extended version: a rapid lookback, a slow double roll, and a flip to cap it. Snap a rapid lookback, flow a slow double roll, and cap it with a flip. The added flip turns the pace-contrast drill into a full showpiece.

MASTERY LEVELS

- ★☆☆ Lookback, slow double roll, capped with a flip.
- ★★★☆☆ Clean snap-and-slow combo finished with a tidy flip.
- ★★★★★ Seamless extended combo, every part crisp and on line.

STEP BY STEP CONTROL INPUTS



Throttle blip then lookback



When upside down looking backwards, cross coordinate a 540 roll



Once upright again still facing backwards, yaw out

PREREQUISITES & WHERE TO PRACTICE

- A snappy lookback and even double roll
- A clean flip

Have the rapid-lookback-slow-double-roll dialed, then add the flip on the end, flowing straight out of the roll into the loop. Keep the line true through all three.

COMMON MISTAKES

- ✗ A rough hand-off into the flip
- ✗ Rushing the double roll
- ✗ Losing the line through the combo

VARIATIONS

Rapid Look Back + Slow Double Roll

Without the flip.

Advanced Combo #3

Another open-air showcase.

RELATED TRICKS

Rapid Look Back + Slow Double Roll

The shorter version.

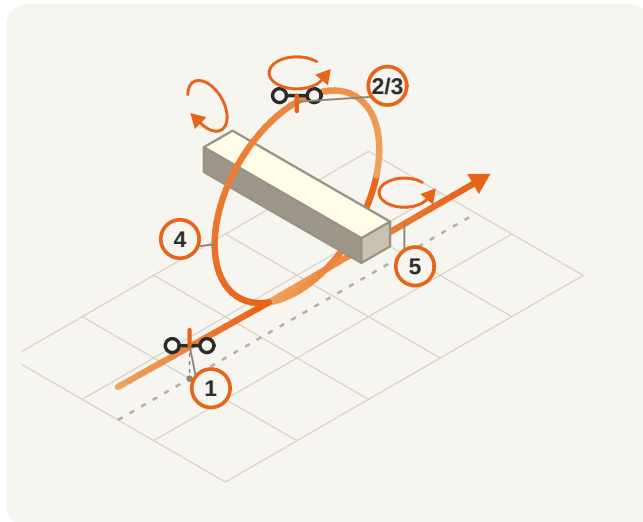
Loop

The flip it ends on.

Rollani

Master

Powerloops



HOW TO

A roll leading into a barani, a twisting flip. Roll on the line, barani-twist through a flip, and catch level as you fly out. It flows a roll into a twisting loop, so the hand-off between the roll and the barani is what to smooth.

MASTERY LEVELS

- ★☆☆ Roll into a barani twisting flip.
- ★★★☆☆ Clean roll flowed into a smooth barani.
- ★★★★★ Seamless Rollani, roll and barani as one, dead on line.

STEP BY STEP CONTROL INPUTS



1 Begin a powerloop under an object, pitching the drone up



2 Cut throttle as you invert, then 180 yaw



3 Quickly follow with a 360 roll



4 Half matty: follow the loop backwards, pitching the drone down as you complete the loop



5 Catch with a 180 yaw out

PREREQUISITES & WHERE TO PRACTICE

- A clean roll
- A confident barani twist

Have the roll and the barani each solid, then flow one into the other, keeping the line straight through both. Give it height and build up slowly.

COMMON MISTAKES

- ✗ A jerky hand-off between roll and barani
- ✗ Too much twist, throwing off the flip
- ✗ Drifting off the line

VARIATIONS

Flipani

A barani off a flip instead of a roll.

Barani

The powerloop-context barani.

RELATED TRICKS

Roll

The roll it starts with.

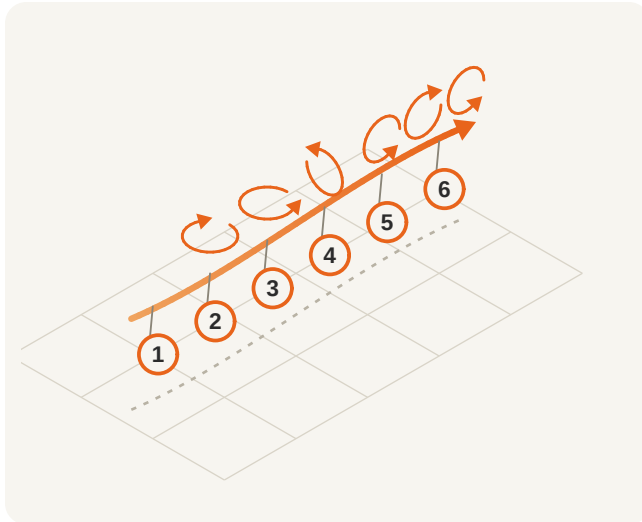
Flipani

The flip sibling.

Sbang Ultimate Combo

Master

Open Air



HOW TO

The grand finale of the Sbang family: a tornado spun straight down, rolled out at the bottom, and flown away on a climbing escape. Drop into the tornado, spin straight down, and roll out and climb away. It packs a spinning descent and a rolling recovery into one showstopper.

MASTERY LEVELS

- ★☆☆ Tornado down and roll out to escape.
- ★★★☆☆ Clean spinning descent rolling into a smooth climb-out.
- ★★★★★ Seamless Sbang Ultimate, dead-vertical tornado, crisp escape.

STEP BY STEP CONTROL INPUTS

- 1** Begin a sideways skate, blip sticks in quickly, then out
- 2** Skate sideways into a rewind, with coordinated yaw/pitch/roll; horizon should be diagonal
- 3** Rewind: blip the throttle and rewind the coordinated yaw/pitch/roll, holding the diagonal horizon
- 4** Throttle blip into a 180 roll
- 5** Once inverted, pitch the drone up
- 6** Once level and upright, flat bounce rewind: blip throttle followed by pitch, then reverse pitch to level

PREREQUISITES & WHERE TO PRACTICE

- A confident Sbang Tornado
- A clean roll and a strong climb-out

Have the tornado and the roll-out each solid, then chain them: spin the descent tight and vertical, roll out cleanly at the bottom, and climb away. Give yourself plenty of height and build up slowly.

COMMON MISTAKES

- ✗ The tornado drifting off vertical
- ✗ Losing orientation in the spinning descent
- ✗ A ragged roll-out at the bottom

VARIATIONS

Sbang Tornado

The tornado without the roll-out.

Power Drill

A rolling corkscrew descent.

RELATED TRICKS

Sbang Tornado

The tornado it builds on.

Power Drill

Another spinning descent.

Sources

SOURCES

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stick inputs, beginner how-to

authoritative per-trick how-to, point values...

prerequisites, practice progression, gear...

common mistakes, practice drills

authoritative video tutorials, cross-check

precise definitions, combo/competition context

stick-input notation, cross-reference

juicy double roll tutorial

Learning progression

A suggested order to learn tricks, grouped by difficulty and listed in order within each tier. Variations sit beneath their base trick. Tick each box as you land it.

Beginner

- ☐ Beginner Matty
- ☐ Beginner Switch
- ☐ Building Dive
- ☐ Cinnamon Roll
- ☐ Cradle
- ☐ Face Punch
- ☐ Flip
 - ☐ Double Flip
 - ☐ Segmented Flips/Rolls
- ☐ Gaps
- ☐ Inverted Rewind
- ☐ Jump Rope
- ☐ Knife Edge
 - ☐ Reverse Knife Edge
- ☐ Maverick Loop
- ☐ Orbits
- ☐ Roll
 - ☐ Double Roll
 - ☐ Vanny Roll
- ☐ Side-Lock Rewind
- ☐ Split S
 - ☐ 540 Split S
 - ☐ Split Stall Matty Rewind
- ☐ Stall Rewind
- ☐ Wall Ride
 - ☐ Reverse Wall Ride
- ☐ Wall Tap
- ☐ Whiplash
- ☐ Yaw Spin
 - ☐ Inverted Yaw Spin
- ☐ Barrel Roll
- ☐ Dive
- ☐ Land
- ☐ Loop
- ☐ Stall

Novice

- ☐ Anti Matty
- ☐ Burrito Roll
- ☐ Eject Roll
 - ☐ Stellar Eject Roll
- ☐ Immelmann
- ☐ Juicy Flick
- ☐ Matty Flip
- ☐ Mavvelmann
- ☐ Mavvy Roll
- ☐ Perch
- ☐ Powerloop
 - ☐ Inverted 360 Powerloop
- ☐ Roll Tap
- ☐ Side Loop
- ☐ Slide Disarm
- ☐ Snapback
- ☐ Stall 360 Rewind
- ☐ Stall Flip Rewind
- ☐ Backward Flying
- ☐ Boomerang
- ☐ Front Swing
- ☐ Juicy Eject
- ☐ Pendulum
- ☐ Slingshot
- ☐ Throwback
- ☐ Wall Bonk

Learning progression (cont.)

Intermediate

- ☐ Blind Loop
- ☐ Ceiling Tap
- ☐ Cubik's Rube
- ☐ Double Dutch
- ☐ Downtown Tap
- ☐ Half Matty
 - ☐ 540 Half Matty
- ☐ Immelloop
- ☐ Inverted Uh-uh
- ☐ Loop Tap
- ☐ Matty Twister
- ☐ Power Flip
 - ☐ Reversed Power Flip
- ☐ Power Matty
- ☐ Power Split
- ☐ Power Swap
- ☐ Rubik's Cube
- ☐ Split Yaw
- ☐ Splitback
- ☐ Stall Half Matty Rewind
- ☐ Stall Matty Rewind
- ☐ Trippy Spin
 - ☐ Double Rolling Trippy Spin
- ☐ Trippy Switch
- ☐ True Barani
- ☐ Alley-Oop
- ☐ Backwards Dive
- ☐ Coin Drop
- ☐ Flat Bounce Rewind
- ☐ Inverted Lookback
- ☐ Juicy Cartwheel
- ☐ Juicy Double Roll
- ☐ Juicy Tracking-Lookback
- ☐ Laser Flip
- ☐ Lookback & Loop
- ☐ Outward Side Lock Rewind
- ☐ Sideways Powerloop
- ☐ Sideways Rewind
- ☐ Slides / Grinds
- ☐ Split-S Rewind
- ☐ Split-S to Stall Track
- ☐ Tracking Front Flip

Advanced

- ☐ Forani
- ☐ Half Mavvy
- ☐ Immelmatt
- ☐ Inverted Yaw Tracking
- ☐ Loop to Matty
- ☐ Matty 360
- ☐ Matty Roll
- ☐ Mavani
- ☐ Mavik's Loop
- ☐ Ninja Star
- ☐ Pole Dance
- ☐ Power Roll
- ☐ Power Switch
- ☐ Power Switch Tap
- ☐ Stall 540 Half Matty Rewind
- ☐ 540 Delayed Front Flip
- ☐ Double Side Roll + 180 Inverted Yaw Spin
- ☐ Flip Rewind to Inverted Yaw Spin
- ☐ Inverted Yaw Spin Rewind
- ☐ Inverted Yaw Track Rewind
- ☐ Jackie Chan
- ☐ Juicy Flick Rewind
- ☐ Lookback + Roll + Inverted Yaw Spin
- ☐ Power Drill
- ☐ Rapid Look Back + Slow Double Roll
- ☐ Rapid Split-S + Inverted Yaw Spin Combo
- ☐ Rubik's Cube Pendulum
- ☐ Rufio
- ☐ Sbang Tornado
- ☐ Side Lookback + Inverted Yaw Spin
- ☐ SomerMatty

Learning progression (cont.)

Master

- ☐ Barani
- ☐ Donkey Loop
- ☐ Flipani
- ☐ Rollani
- ☐ 540 Flip + Roll + Inverted Yaw Spin
- ☐ Advanced Combo #3
- ☐ Eject Lookback Roll + Inverted Yaw Spin
- ☐ Look Back + Double Roll
- ☐ Sbang Ultimate Combo

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"If you're not crashing, you're not flying."